



**Numerical solutions of lattice quantum fields
with a hierarchy of Schroedinger-like equations.**

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Schrodinger-like equations**

Ludwig, Mark Allen, Ph.D.

The University of Arizona, 1990

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NUMERICAL SOLUTIONS OF LATTICE QUANTUM FIELDS
WITH A HIERARCHY OF SCHRÖDINGER-LIKE EQUATIONS

by

Mark Allen Ludwig

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A Dissertation Submitted to the Faculty of the

DEPARTMENT OF PHYSICS

In Partial Fulfillment of the Requirements
For the Degree of

DOCTOR OF PHILOSOPHY

In the Graduate College

THE UNIVERSITY OF ARIZONA

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the dissertation prepared by Mark A. Ludwig

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ABSTRACT

Systems of quantized fields can be described by an infinite hierarchy of coupled equations. Such a hierarchy is derived from first principles for a simple interacting field theory to illustrate this type of a representation. The perturbation series for the S matrix is derived from the hierarchy equations in order to show its equivalence to the usual expansion in Feynman amplitudes. An inquiry is then conducted to determine whether this type of representation is useful for solving problems. Truncations of the hierarchy which predict simple bound states are examined in the weak coupling limit, and equations describing a hydrogen-like atom are obtained. Next, the numerical approximation of a truncated hierarchy is studied, and a scattering / particle creation process is modeled in one dimension with a resulting accuracy of 1 to 2 percent. Finally, the mathematical questions of convergence which arise in connection with quantized fields are discussed within the context of the hierarchy equations.

CHAPTER 1

INTRODUCTION

Perturbation theory and the Bethe - Saltpeter¹ equation gave early researchers the techniques required to solve and understand a wide variety of quantized field problems and establish quantum electrodynamics on solid experimental ground. In the past two decades, however, the need to solve a wider variety of quantized field problems has become more pressing, in response to which many techniques have been developed, ranging from semi-classical approximations² to numerical techniques involving lattice gauge theories³ and lattice Hamiltonian field theories.⁴

While many of these techniques are valuable for solving one class of problems or another, no single

1. E. Saltpeter, H. Bethe, Phys. Rev. 84 (1951) 1232-1242.
2. N. Neveu, "Semiclassical Quantization Methods in Field Theory", Non-Linear and Collective Phenomena in Quantum Physics, (J. Gervais, M. Jacob, Editors), pp. 32-39, World Scientific, Singapore, 1983.
3. Claudio Rebbi, Lattice Gauge Theories and Monte Carlo Simulations, pp. 23-37, Interscience, New York, 1983.
4. T. Barnes, G. Daniell, Phys. Rev. D 28 (1983) 2045-2058.

method seems to be a panacea for the field theorist's tangles. As such any new technique which might be able to solve typical problems encountered in quantum field theory seems worthy of investigation.

At the suggestion of Prof. Richard Morse, I attempted to derive a hierarchy of equations which would describe a system of interacting quantized fields in a number representation, where states with different numbers of particles couple to each other. We supposed that such a system would be amenable to truncation and numerical solution. The results, presented here, center around a field theoretic version of the Schrodinger equation, which takes the form of a hierarchy with the desired structure.

It seems that many authors down through the years have suggested the possibility of using a Schrodinger equation to describe a system of quantized fields^{5 6 7 8 9}

5. F. Mandl, Introduction to Quantum Field Theory, pp. 1-24, Interscience, New York, 1959.

6. J. Bjorken, S. Drell, Relativistic Quantum Fields, p. 7, McGraw Hill, New York, 1965.

7. K. Nishijima, Fields and Particles, p. 102, Benjamin, New York, 1969.

and a few have developed the idea to some extent, albeit in an abstract fashion.^{10 11 12 13}

While the derivation of a Schrodinger - hierarchy may seem arcane and antiquated at first sight, it appears to be something which has been overlooked by physicists for the past sixty years.

When discussed in the literature, it seems that the Schrodinger equation for fields is either cast in the form of a functional equation,

$$i\frac{\partial}{\partial t}\Psi(\phi(x))=H(\phi(x),\pi(x))\Psi(\phi(x))$$

which immediately proves to be rather difficult to get

8. C. Itzykson, J. B. Zuber, Quantum Field Theory, pp 481,482, McGraw Hill, New York, 1980.

9. N. N. Bogoluibov, D. Shirkov, Introduction to the Theory of Quantized Fields, 3rd Edition, pp. 405-417, Wiley, New York, 1980.

10. K. Symanzik, "Schrodinger Representation in Renormalizable Quantum Field Theory" in Structural Elements in Particle Physics and Statistical Mechanics (J. Honerkamp, K. Pohlmeyer, H. Romer, Editors), pp. 287-299, Plenum, New York, 1981.

11. James Glimm, The Schrodinger Equation for Quantum Fields with Nonlinear Non-local Scattering, Aarhus Universitat, Matematisk Institut, 1966.

12. James Glimm, Comm. Math. Phys. 8 (1968) 123-25.

13. J. Ehlers, et. al., Eds., Constructive Quantum Field Theory, Lecture Notes in Physics, No. 25, Springer Verlag, New York, 1973.

any numbers from, or it is discussed only in the vaguest terms with no reference to the form it takes.

Of course, physicists have probably had good reason to stay away from such an infinite hierarchy of equations, as it seems that they would be totally inaccessible and even incomprehensible without numerical problem solving techniques and computers big enough to tackle the problems -both relatively recent developments.

In Chapter 2, the hierarchy of equations for a simple interacting system of fields is derived from basic principles. Although the derivation is for a specific system, the techniques are quite general and may be applied to a wide variety of field theories. Because of the unusual form of these equations, the rudiments of a mathematical framework within which to understand the system in the Schrodinger picture is also developed here.

Chapter 3 deals with the connection between the hierarchy equations and the ordinary perturbative approach to field theory, showing them to be equivalent. This result is non-trivial because the Schrodinger picture singles out a time axis, so manifest covariance is lost, even though the system is relativistically invariant.

In Chapter 4, the renormalization counterterms for the theory are introduced, and the lowest order terms are

derived in preparation for Chapter 5, where the hierarchy is reduced, in the weak coupling limit, to a Schrodinger equation for a system with a finite number of degrees of freedom. This type of reduction provides a connection between the full quantized field theory and the Schrodinger theory of a hydrogen - like atom, including unstable bound states and their decay modes. Though we will treat only the simplest example here, the possibility of incorporating relativistic corrections, corrections in the effective potential, or many-body effects is fairly straight forward, albeit messy.

In Chapter 6 we discuss the possibility of numerically approximating the field equations. It is found that the hierarchy representation is sufficiently economical to allow one to simulate dynamical experiments like scattering with reasonable computing requirements. Here, the renormalization parameters were computed numerically and a scattering problem was run, the results of which compare favorably with perturbative results in the regime where perturbation theory is valid.

Finally, in Chapter 7, we briefly discuss some of the sticky mathematical problems with the hierarchy which are inherent to all field theories. While it is beyond the scope of this work to provide a rigorous defense of the

mathematics, we can come to terms with it and understand the renormalization phenomenon in the context of this approach.

CHAPTER 2

A DERIVATION OF THE HIERARCHY EQUATIONS

The derivation of a hierarchy of equations for a quantized field is relatively independent of the form of the Lagrangian. Here, for the sake of simplifying the mathematics as much as possible, we will consider a simple interacting scalar boson system defined by the Lagrangian density

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \partial^\mu \psi - m^2 \psi^2 + \partial_\mu \partial^\mu \varphi - \mu^2 \varphi^2) + \lambda \psi^2 \varphi \quad (1)$$

The unphysical nature of a cubic boson theory, due to the lack of a negative energy bound on the Hamiltonian, presents some formal problems. A device of Glimm and Jaffee¹⁴ is sufficient to resolve such problems for the work presented here. The addition of a high, even order (e.g. $\lambda_2 \varphi^6 \psi^6$), weakly coupled term to (1) renders it formally sound¹⁵, but does not significantly affect any of the computations carried out here because of its high order and weakness.

14. J. Glimm, A. Jaffe, T. Spencer, The Particle Structure of the Weakly Coupled $P(\phi)^2$ Model, in "Constructive Quantum Field Theory" (G. Velo, A. Wightman, Eds.) pp. 169,170, Springer Verlag, New York, 1973.

15. J. Glimm, Commun. Math. Phys. 8 (1968) 12-25.

We consider such a theory because it has several advantages. The system has obvious similarities to quantum electrodynamics and the pion - nucleon system, making it possible to anticipate some of the results that it produces, including the bound states. Since the fields are real scalar fields, there are no negative energy anti-particles and no spin states, thereby simplifying numerical calculations. Since the interaction is only of third order, the number of terms in the hierarchy equations stays as small as possible, thereby facilitating computations.

The derivation of a hierarchy of equations describing this quantized system will follow the derivation of the connection between the Schrodinger and Heisenberg pictures in the case of a fixed number of particles.¹⁶ Dealing with variable numbers of particles, however, complicates both the derivation and the resulting equations and their interpretation.

The Hamiltonian for this system is

$$H_\lambda = \int \left(\frac{1}{2} [\pi_\psi^2 + \nabla \psi \nabla \psi + m^2 \psi^2 + \pi_\varphi^2 + \nabla \varphi \nabla \varphi + \mu^2 \varphi^2] - \lambda \psi^2 \varphi^2 \right) d^{\sigma}x \quad (2)$$

where σ is the dimension of space, $\pi_\psi = \dot{\psi}$, and $\pi_\varphi = \dot{\varphi}$. One

16. A. L. Fetter, J. D. Walecka, Quantum Theory of Many-Particle Systems, (McGraw Hill, New York) 1978, pp. 4-15.

direction has already been singled out for time, so the resulting equation will not be manifestly Lorentz invariant. This is unavoidable if we are to derive an equation for a wavefunction which describes an entire system of particles at a given time.

The first step in deriving a Schrodinger equation is to write H_λ in a number representation. To this end, we Fourier transform, using

$$\varphi_k = [(2\pi)^3 2\omega'_k]^{-1/2} \int \varphi(x) e^{ikx} d^3x$$

$$\varphi(x) = (2\pi)^3 \int \varphi_k \sqrt{2\omega'_k} e^{-ikx} d^3k$$

where $\omega_k = \sqrt{m^2 + k^2}$ and $\omega'_k = \sqrt{\mu^2 + k^2}$, and likewise for ψ , π_ψ , and π_φ . Substituting these into H_λ and performing the x integration,

$$H_\lambda = \int [\omega_k \{ \pi_{-k}^\psi \pi_k^\psi + (m^2 + k^2) \psi_{-k} \psi_k \} + \omega'_k \{ \pi_{-k}^\varphi \pi_k^\varphi + (\mu^2 + k^2) \varphi_{-k} \varphi_k \}] d^3k \quad (3)$$

$$- \lambda \int \int \int \frac{8\omega_k \omega_p \omega_q}{(2\pi)^3} \varphi_p \psi_{-k} \psi_k \delta(k+p+q) d^3k d^3p d^3q$$

Next, define creation and annihilation operators,

$$\varphi_k = \frac{1}{2\omega'_k} (b_k + b_{-k}^\dagger) \quad \pi_k^\varphi = \frac{-i}{2} (b_k - b_{-k}^\dagger)$$

$$\psi_k = \frac{1}{2\omega_k} (a_k + a_{-k}^\dagger) \quad \pi_k^\psi = \frac{-i}{2} (a_k - a_{-k}^\dagger)$$

where a_k annihilates the field of mass m (hereafter referred to as the m -particle) and b_k annihilates the field of mass μ (hereafter referred to as the μ -particle). Plugging these into (3) gives

$$\begin{aligned}
H_\lambda = & \int \frac{1}{2} [\omega_k (a_k a_k^\dagger + a_k^\dagger a_k) + \omega'_k (b_k b_k^\dagger + b_k^\dagger b_k)] d^3k \\
& - \lambda \int \int \int \frac{\delta^*(k+p+q)}{\sqrt{8(2\pi)^3 \omega_k \omega'_p \omega_q}} \left\{ b_p (a_k a_q + a_k a_q^\dagger + a_k^\dagger a_q + a_k^\dagger a_q^\dagger) \right. \\
& \left. + b_{-p}^\dagger (a_k a_q + a_k a_q^\dagger + a_k^\dagger a_q + a_k^\dagger a_q^\dagger) \right\} d^3k d^3q d^3p
\end{aligned} \tag{4}$$

In order to derive a hierarchy, we view H_λ as a finite sum over discrete space, and later take the limit in which the finite space becomes continuous. To go to a discrete space, define

$$a_j = a_{k_j} (\Delta k)^{1/2}$$

etc., where Δk is a finite volume element in σ -space and write

$$\int f(k) dk \longrightarrow \sum_{j=-p}^p f(k_j) \Delta k \tag{5}$$

where k_j is an element in the finite space numbered by the single index j , so that $k_{-j} = -k_j$. Finally, define

$$\delta_{1,i,j} = \begin{cases} 1 & \text{if } k_1 + k_j + k_i = 0 \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Then H_λ becomes

$$\begin{aligned}
H_\lambda = & \sum_{j=-p}^p \left\{ \frac{1}{2} \omega_j (a_j a_j^\dagger + a_j^\dagger a_j) + \frac{1}{2} \omega'_j (b_j b_j^\dagger + b_j^\dagger b_j) \right\} \\
& - \lambda \sum_{i,j,1=-p}^p \left[8(2\pi)^3 (\Delta k)^3 \omega'_1 \omega_i \omega_j \right]^{-1/2} \times \\
& \left\{ b_1 (a_i a_j \delta_{1,i,j} + a_i a_j^\dagger \delta_{1,i,-j} + a_i^\dagger a_j \delta_{1,-i,j} + a_i^\dagger a_j^\dagger \delta_{1,-i,-j}) \right. \\
& \left. + b_1^\dagger (a_i a_j \delta_{1,-i,-j} + a_i a_j^\dagger \delta_{1,i,-j} + a_i^\dagger a_j \delta_{1,i,j} + a_i^\dagger a_j^\dagger \delta_{1,i,j}) \right\}
\end{aligned} \tag{7}$$

In order to quantize this system, we impose the equal time commutation relations

$$[a_i, a_j^\dagger] = \delta_{ij}, \quad [b_i, b_j] = \delta_{ij}, \quad (8)$$

$$[a_i, a_j] = [b_i, b_j] = [a_i, b_j] = [a_i, b_j^\dagger] = 0$$

and treat H_λ as an operator, writing a Schrodinger equation

$$i \frac{\partial |\Psi\rangle}{\partial t} = H_\lambda |\Psi\rangle \quad (9)$$

for this system. To make the equation explicit, the state vector $|\Psi\rangle$ must be expanded in some basis. We choose the eigenstates of the free Hamiltonian H_0 with $\lambda=0$. Let

$$|n_{-p}, \dots, n_j, \dots, n_p; m_{-p}, \dots, m_1, \dots, m_p\rangle \quad (10)$$

be the state with n_j μ -particles in the k_j state and m_1 μ -particles in the k_1 state. This is an eigenstate of H_0 with

$$H_0 |n_{-p}, \dots, m_p\rangle = \sum_{j=-p}^p \left[\frac{1}{2} (\omega_j + \omega_j^*) + n_j \omega_j + m_j \omega_j' \right] |n_{-p}, \dots, m_p\rangle \quad (11)$$

A general state $|\Psi\rangle$ may be expanded as

$$|\Psi\rangle = \sum_{n_{-p}, \dots, n_p=0}^{\infty} C_{n_{-p}, \dots, n_p}^{n_{-p}, \dots, n_p} |n_{-p}, \dots, m_p\rangle \quad (12)$$

Substituting this into (9) gives

$$\begin{aligned}
\sum_{n_{-p}, \dots, n_p=0}^{\infty} i \frac{\partial C_{n_{-p}, \dots, n_p}^{n_{-p}, \dots, n_p}}{\partial t} |n_{-p}, \dots, n_p\rangle = & \quad (13) \\
\sum_{n, n=0}^{\infty} \left(\sum_{j=-p}^p \left[\omega_j \left(n_j + \frac{1}{2} \right) + \omega'_j \left(m_j + \frac{1}{2} \right) \right] C_{n_{-p}, \dots, n_p}^{n_{-p}, \dots, n_p} |n_{-p}, \dots, n_p\rangle \right. \\
- \lambda \sum_{n, n=0}^{\infty} C_{n_{-p}, \dots, n_p}^{n_{-p}, \dots, n_p} \sum_{i,j,l=-p}^p \left[\frac{(\Delta k)^2}{8(2\pi)^2 \omega'_i \omega'_j \omega_j} \right] \times \\
\left\{ \sqrt{m_i n_j} \delta_{1,i,j} |n_{-p}, \dots, n_i-1, \dots, n_j-1, \dots, m_i-1, \dots, m_p\rangle \right. \\
+ \sqrt{m_i n_i (n_j+1)} \delta_{1,i,-j} |n_{-p}, \dots, n_i-1, \dots, n_j+1, \dots, m_i-1, \dots, m_p\rangle \\
+ \sqrt{m_i n_j (n_i+1)} \delta_{1,-i,j} |n_{-p}, \dots, n_i+1, \dots, n_j-1, \dots, m_i-1, \dots, m_p\rangle \\
+ \sqrt{m_i (n_i+1) (n_j+1)} \delta_{1,-i,-j} |n_{-p}, \dots, n_i+1, \dots, n_j+1, \dots, m_i-1, \dots, m_p\rangle \\
+ \sqrt{(m_i+1) n_j} \delta_{1,-i,-j} |n_{-p}, \dots, n_i-1, \dots, n_j-1, \dots, m_i+1, \dots, m_p\rangle \\
+ \sqrt{(m_i+1) (n_i+1) n_j} \delta_{1,i,-j} |n_{-p}, \dots, n_i+1, \dots, n_j-1, \dots, m_i+1, \dots, m_p\rangle \\
+ \sqrt{(m_i+1) (n_j+1) n_i} \delta_{1,-i,j} |n_{-p}, \dots, n_i-1, \dots, n_j+1, \dots, m_i+1, \dots, m_p\rangle \\
+ \sqrt{(m_i+1) (n_i+1) (n_j+1)} \delta_{1,i,j} |n_{-p}, \dots, n_i+1, \dots, n_j+1, \dots, m_i+1, \dots, m_p\rangle
\end{aligned}$$

Now we transform to a uniform basis,

$$|n_{-p}, \dots, n_i, \dots, n_j, \dots, m_i, \dots, m_p\rangle \quad (14)$$

By orthogonality of these basis vectors we can then equate the coefficients of each basis vector to get an equation for the time derivative of each C as a function of other C's. The result is

$$\begin{aligned}
i \frac{\partial C_{n_{-p}, \dots, n_p}^{m-p, \dots, m_p}}{\partial t} = & \sum_{j=-p}^p \left[\frac{1}{2} (\omega_j + \omega'_j) + n_j \omega_j + m_j \omega'_j \right] C_{n_{-p}, \dots, n_p}^{m-p, \dots, m_p} \\
& - \lambda \sum_{i,j,l=-p}^p \left[\frac{(\Delta k)^3}{8(2\pi)^3 \omega_i \omega_j \omega'_l} \right]^{1/2} \{ \delta_{1,i,j} \sqrt{(m_i+1)(n_j+1)(n_l+1)} C_{n_{-p}, \dots, n_i+1, \dots, n_j+1, \dots, n_l+1, \dots, n_p}^{m-p, \dots, m_i+1, \dots, m_p} \\
& + \sqrt{(m_i+1)(n_l+1)n_j} \delta_{1,i,l,-j} C_{n_{-p}, \dots, n_i+1, \dots, n_l-1, \dots, n_j, \dots, n_p}^{m-p, \dots, m_i+1, \dots, m_p} \\
& + \sqrt{(m_i+1)(n_j+1)n_l} \delta_{1,-i,j} C_{n_{-p}, \dots, n_i-1, \dots, n_j+1, \dots, n_l, \dots, n_p}^{m-p, \dots, m_i+1, \dots, m_p} \\
& + \sqrt{(m_i+1)n_l n_j} \delta_{1,-i,-j} C_{n_{-p}, \dots, n_i-1, \dots, n_j-1, \dots, n_l, \dots, n_p}^{m-p, \dots, m_i+1, \dots, m_p} \\
& + \sqrt{m_i(n_l+1)(n_j+1)} \delta_{1,-i,-j} C_{n_{-p}, \dots, n_i+1, \dots, n_j+1, \dots, n_l, \dots, n_p}^{m-p, \dots, m_i-1, \dots, m_p} \\
& + \sqrt{m_i n_l (n_j+1)} \delta_{1,i,j-1} C_{n_{-p}, \dots, n_i-1, \dots, n_j+1, \dots, n_l, \dots, n_p}^{m-p, \dots, m_i-1, \dots, m_p} \\
& + \sqrt{m_i n_j (n_l+1)} \delta_{1,-i,j} C_{n_{-p}, \dots, n_i+1, \dots, n_j-1, \dots, n_l, \dots, n_p}^{m-p, \dots, m_i-1, \dots, m_p} \\
& + \sqrt{m_i n_l n_j} \delta_{1,i,j} C_{n_{-p}, \dots, n_i+1, \dots, n_j-1, \dots, n_l, \dots, n_p}^{m-p, \dots, m_i-1, \dots, m_p}
\end{aligned} \tag{15}$$

This is an equation linking states of different particle numbers, and it describes the complete system of quantized fields with creation and annihilation processes.

$C_{n_{-p}, \dots, n_p}^{m-p, \dots, m_p}$ is a complex number describing the probability of finding n_j m -particles with momentum k_j , etc. Assuming the total number of m -particles in such a state is N and the total number of μ -particles is M , the momenta of these particles can simply be labeled

$$(p_1, \dots, p_M; q_1, \dots, q_M) \tag{16}$$

where each of the p 's and q 's takes on the value k_j for some j , $-p > j > p$. There is a different C for each possible set of values that $(p_1, \dots, q_M)$ can take, each of which is a complex number. Thus, one may view C as a function of

$(p_1, \dots, q_M)$ and write

$$C_{n-p, \dots, n_p}^{m-p, \dots, m_p} = A \psi_{MN}(p_1, \dots, p_N; q_1, \dots, q_M) \quad (17)$$

where A is some constant depending on normalization. ψ_{MN} will be the Schrodinger wavefunction describing N m -particles and M μ -particles in momentum space. In general, there will be ψ_{MN} 's with all possible non-negative integral values of N and M , because any number of particles can exist in the system.

To find the normalization constant A in (17), note that the C 's are defined so that

$$1 = \langle \Psi | \Psi \rangle = \sum_{N, N=0}^{\infty} \sum_{\Sigma m=0, \Sigma n=0} |C_{n-p, \dots, n_p}^{m-p, \dots, m_p}|^2 \quad (18)$$

For the ψ_{MN} 's we want to write

$$1 = \langle \Psi | \Psi \rangle = \sum_{N, N=0}^{\infty} \sum_{p_1, q_1, \dots, k-p}^k |\psi_{MN}(p_1, \dots, p_N; q_1, \dots, q_M)|^2 \frac{(\Delta k)^{N+M}}{\prod_j (2\omega_{q_j}) \prod_j (2\omega_{p_j})} \quad (19)$$

in analogy to the usual Schrodinger theory, using the conventional relativistic measure for bosons. We thus take the view that $|\psi_{MN}|^2$ is a probability density for finding the system in a state with N m -particles and M μ -particles with specified momenta. Equation (19) is simply the statement that the probability of finding the system in some state is unity.

The sums in (18) and (19) give a different weight to each term. Two factors are involved. The first is from the measure, which contributes a factor

$$\sqrt{\frac{\Delta k}{2\omega_p}} \quad (20)$$

for each particle. The second is a combinatorial factor, due to the fact that the sum in (19) will include some C's more than once. Essentially, the problem at hand is how to put N particles into 2p boxes, with n_j particles in the j'th box, etc. This factor is

$$\sqrt{\frac{N!}{\prod_i n_i!}} \quad (21)$$

The resulting relationship is

$$C_{n_{-p}, \dots, n_p}^{m_{-p}, \dots, m_p} = \psi_{NM}(p_1, \dots, q_M) \frac{(\Delta k)^{N+M}}{\prod_{i=1}^N (2\omega_{p_i}) \prod_{j=1}^M (2\omega_{q_j})} \frac{N!M!}{\prod_{i=-p}^p (n_i! m_i!)} \quad (22)$$

for corresponding ψ 's and C's. Substituting this into (15) and taking the continuum limit, one arrives at the final result,

$$i \frac{\partial \psi_{NM}}{\partial t} = \left(\sum_{j=1}^N \omega_{p_j} + \sum_{j=1}^M \omega_{q_j} \right) \psi_{NM}(p_1, \dots, p_N; q_1, \dots, q_M) \quad (23)$$

$$\begin{aligned} & -\lambda (2\pi)^{\sigma/2} \sqrt{(N+2)(N+1)(M+1)} \times \\ & \quad \int \int \int \psi_{N+2, M+1}(p_1, \dots, p_N, r, u; q_1, \dots, q_M, s) \delta'(r+u+s) \frac{d^3 r d^3 u d^3 s}{8 \omega_r \omega_u \omega_s} \\ & - \frac{\lambda}{\sqrt{(2\pi)^\sigma M N (N-1)}} \sum_{i=1}^M \sum_{\substack{j=1 \\ i \neq j}}^N \delta'(q_i + p_i + p_j) \times \\ & \quad \psi_{N-2, M-1}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, \hat{q}_i, \dots, q_M) \\ & - 2\lambda (2\pi)^{-\sigma/2} \sqrt{M+1} \sum_{j=1}^N [\\ & \quad \int \int \psi_{N, M+1}(p_1, \dots, \hat{p}_j, \dots, p_N, s; q_1, \dots, q_M, r) \delta'(r+s-p_j) \frac{d^3 r d^3 s}{4 \omega_s \omega_r}] \\ & - 2\lambda (2\pi)^{-\sigma/2} M^{-1/2} \sum_{i=1}^M \sum_{j=1}^N [\\ & \quad \int \psi_{N, M-1}(p_1, \dots, \hat{p}_j, \dots, p_N, s; q_1, \dots, \hat{q}_i, \dots, q_M) \delta'(q_i + p_j + s) \frac{d^3 s}{2 \omega_s}] \\ & - \lambda \sqrt{\frac{(N+1)(N+2)}{M(2\pi)^\sigma}} \sum_{i=1}^M [\\ & \quad \int \psi_{N+2, M-1}(p_1, \dots, p_N, r, s; q_1, \dots, \hat{q}_i, \dots, q_M) \delta'(q_i - r - s) \frac{d^3 r d^3 s}{4 \omega_r \omega_s} \\ & - \lambda \sqrt{\frac{M+1}{(2\pi)^\sigma N(N-1)}} \sum_{\substack{i,j=1 \\ i \neq j}}^N [\\ & \quad \int \psi_{N-2, M+1}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, q_M, r) \delta'(p_i + p_j - r) \frac{d^3 r}{2 \omega_r}] \end{aligned}$$

where the vacuum energy has been absorbed by replacing ψ_{NM} with $\exp(-i\gamma t)\psi_{NM}$ and a " $\hat{}$ " over a variable signifies its omission, and any ψ_{NM} with a negative subscript is identically zero. Equation (23) is an infinite hierarchy of equations linking states with different numbers of particles. It is an exact description of the quantized fields represented by the Lagrange density of eq. (1). Notice that it is not based on perturbation theory,

although bare particles are used to describe states of the system. To find physical states, one must solve the eigenvalue problem for this hierarchy. The eigenvectors will describe physical states in terms of bare particles. The solution of this hierarchy will be discussed in detail in the following sections. First, however, let us examine the structure of the hierarchy to understand it a little better.

We denote the space on which ψ_{NM} is defined by V_{NM} . V_{NM} is a $\sigma(N+M)$ dimensional real vector space. One may think of the total wavefunction $|\Psi\rangle$ as a vector whose components are the ψ_{NM} 's,

$$|\Psi\rangle = (\psi_{00}, \psi_{10}(k), \psi_{01}(p), \psi_{20}(r, s), \dots) \quad (24)$$

The norm of $|\Psi\rangle$ is defined by

$$\langle\Psi|\Psi\rangle = \psi_{00}^* \psi_{00} + \int \psi_{10}^*(k) \psi_{10}(k) \frac{d^3k}{2\omega_k} + \int \psi_{01}^*(p) \psi_{01}(p) \frac{d^3p}{2\omega_p} + \dots \quad (25)$$

and the Hilbert space of states is defined as the closure of the space of all vectors $|\Psi\rangle$ with finite norm. The Hamiltonian described by (23) is a linear, hermitian operator in this space of states. Thus, this system is analogous to a standard quantum mechanical system, except that it is defined in a larger space of states. Probability and energy conservation follow directly from the hermiticity of H , as usual.

To make the connection with ordinary quantum mechanics, define the projection operator Π_{MN} by

$$\Pi|\Psi\rangle = (0, 0, \dots, \psi_{NM}, 0, 0, \dots) \quad (26)$$

Π_{MN} projects $|\Psi\rangle$ onto the space V_{MN} . Using this operator, one may write

$$\psi_{NM}(p_1, \dots, q_N) = \langle p_1, \dots, q_N | \Pi_{NM} | \Psi \rangle \quad (27)$$

and

$$i \frac{\partial \psi_{NM}}{\partial t}(p_1, \dots, q_N) = \langle p_1, \dots, q_N | \Pi_{NM} H_\lambda | \Psi \rangle \quad (28)$$

The probability of finding a certain number of particles

$$P_{NM} = \int \psi_{NM}^* \psi_{NM} \frac{dp dq}{\prod 2\omega_p \prod 2\omega_q} \quad (29)$$

is conserved only if H_λ is block diagonal with respect to the spaces V_{MN} , in which case one obtains a set of decoupled finite dimensional Schrodinger equations from (23).

The hierarchy (23) can be transformed to coordinate space in theory, but it is not very practical, because the equations contain relativistic terms of the form $\sqrt{m^2 + k^2}$, which, under Fourier transformation would naively result in something like $\sqrt{m^2 + \nabla^2}$, which cannot be understood with ordinary calculus. Historically the solution of such problems has been obtained by seeking a multi-component solution in coordinate space.¹⁷

17. H. Feshbach and F. Villars, Rev. Mod. Phys. 30 (1958) 24-45.

We may avoid all of these problems in the model chosen here by solving the equations in k -space and then Fourier transforming the result to examine the solution in coordinate space. We pursue this approach to avoid unnecessarily complicating the problem and obscuring relevant physics. Since the fields and couplings used in our example are all real, we can ignore the negative energy states (as the hierarchy has done automatically), and achieve a reduction of the problem. Of course, one cannot avoid the negative energy states with an electromagnetic coupling of the form ieA_μ , and one cannot avoid the multi-component formalism when dealing with fermions, even in k -space. We have chosen scalar particles specifically to avoid these complications, which could be incorporated into the theory in an obvious, systematic way, as desired.

It is important to note that it may not be possible to solve the hierarchy (23) using finite complex valued functions for ψ_{NM} . This will become clearer later as we begin to deal with the equations.

CHAPTER 3

THE CONNECTION WITH PERTURBATION THEORY

Once one has derived the hierarchy equation for a given field theory, all field theoretic problems are reduced to solving that equation. Clearly, an infinite hierarchy of differential - integral equations is a formidable challenge to solve.

Since perturbative techniques are so well known and understood, it seems appropriate to show that time dependent perturbation theory applied to the hierarchy equation (23) will indeed produce the usual diagrammatic expansion of the S matrix with the correct factor associated to each diagram. In doing this, we will encounter all of the usual problems with renormalization and convergence of the perturbation series. We need not worry about these questions in this chapter, though. It will be sufficient to derive the perturbation series in terms of bare fields and bare parameters. Then, if so desired, the theory can be reinterpreted in terms of physical quantities in the same way it always is.

The first step in deriving the perturbation expansion is to express S^A as a power series in the interaction terms using standard time dependent

perturbation theory. Supposing the Hamiltonian is of the form

$$H=H_0+\lambda V \quad (30)$$

The perturbation expansion for $S^{\beta\alpha}$ is well known.¹⁸ It is given by

$$S^{\beta\alpha}=\delta^{\beta\alpha}+\sum_{n=1}^{\infty}(-i)^n S_n^{\beta\alpha} \quad (31)$$

where

$$S_n^{\beta\alpha}=\lambda^n \sum_{\gamma_m, m=1, \dots, n-1} \int_{-\infty}^{\infty} \int_{-\infty}^{t_n} \dots \int_{-\infty}^{t_1} [V_{\beta\gamma_{n-1}} V_{\gamma_{n-1}\gamma_{n-2}} \dots V_{\gamma_1\alpha}] \times e^{iE_{\beta}t_n} e^{iE_{\gamma_{n-1}}(t_{n-1}-t_n)} \dots e^{iE_{\gamma_1}(t_1-t_2)} e^{-iE_{\alpha}t_1} dt_1 \dots dt_n \quad (32)$$

where E_{γ_j} is the eigenvalue of H_0 for the eigenstate $|\gamma_j\rangle$, and the sum over γ_m is either a sum or integral over all possible intermediate states. This expression is quite general, since it applies to any quantum mechanical system which has scattering states. To use it in the present situation, define H_0 and V according to standard practice, H_0 being the Hamiltonian for free particles,

$$\langle p_1, \dots, p_N; q_1, \dots, q_M | \Pi_{\alpha\beta} H_0 | \Psi \rangle = \left(\sum_{j=1}^N \omega_{p_j} + \sum_{j=1}^M \omega'_{q_j} \right) \psi_{\alpha\beta}(p_1, \dots, p_N; q_1, \dots, q_M) \quad (33)$$

and V the sum of the six interaction terms,

18. A. Messiah, Quantum Mechanics, (J. Wiley, New York), pp.722-726.

$$\begin{aligned}
& \langle p_1, \dots, p_N; q_1, \dots, q_M | \Pi_{NM} V | \Psi \rangle = \\
& - (2\pi)^{\sigma/2} \sqrt{(N+2)(N+1)(M+1)} \times \\
& \quad \int \int \int \psi_{N+2, M+1}(p_1, \dots, p_N, r, u; q_1, \dots, q_M, s) \delta^\sigma(r+u+s) \frac{d^3 r d^3 u d^3 s}{8 \omega_r \omega_u \omega_s} \\
& - \frac{1}{\sqrt{(2\pi)^\sigma M N (N-1)}} \sum_{i=1}^M \sum_{\substack{j=1 \\ i \neq j}}^N \delta^\sigma(q_i + p_i + p_j) \times \\
& \quad \psi_{N-2, M-1}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, \hat{q}_i, \dots, q_M) \\
& - 2(2\pi)^{-\sigma/2} \sqrt{M+1} \sum_{j=1}^N [\\
& \quad \int \int \psi_{N, M+1}(p_1, \dots, \hat{p}_j, \dots, p_N, s; q_1, \dots, q_M, r) \delta^\sigma(r+s-p_j) \frac{d^3 r d^3 s}{4 \omega_s \omega_r}] \\
& - 2(2\pi)^{-\sigma/2} M^{-1/2} \sum_{i=1}^M \sum_{j=1}^N [\\
& \quad \int \psi_{N, M-1}(p_1, \dots, \hat{p}_j, \dots, p_N, s; q_1, \dots, \hat{q}_i, \dots, q_M) \delta^\sigma(q_i + p_j + s) \frac{d^3 s}{2 \omega_s}] \\
& - \sqrt{\frac{(N+1)(N+2)}{M(2\pi)^\sigma}} \sum_{i=1}^M [\\
& \quad \int \psi_{N+2, M-1}(p_1, \dots, p_N, r, s; q_1, \dots, \hat{q}_i, \dots, q_M) \delta^\sigma(q_i - r - s) \frac{d^3 r d^3 s}{4 \omega_r \omega_s} \\
& - \sqrt{\frac{M+1}{(2\pi)^\sigma N(N-1)}} \sum_{\substack{i, j=1 \\ i \neq j}}^N [\\
& \quad \int \psi_{N-2, M+1}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, q_M, r) \delta^\sigma(p_i + p_j - r) \frac{d^3 r}{2 \omega_r}]
\end{aligned}$$

First, we show that $S_n^{\beta\alpha}$ contains terms due to all possible diagrams with n vertices which are consistent with the initial and final states $|\alpha\rangle$ and $|\beta\rangle$. After that, we will derive the form of the contribution to $S^{\beta\alpha}$ from any given diagram.

Note that V occurs n times in $S_n^{\beta\alpha}$, and V contains all six of the elementary processes which are expected from the Lagrange density (1). Essentially, the n matrix elements

$$V_{\beta\gamma_{n-1}} V_{\gamma_{n-1}\gamma_{n-2}} \dots V_{\gamma_1\alpha} \quad (35)$$

describe n interactions at times $\tau_1, \dots, \tau_n$ in which two m -particles and one μ -particle interact, causing a change from the state $|\gamma_j\rangle$ to $|\gamma_{j+1}\rangle$ at each interaction. This immediately suggests a diagrammatic interpretation of $S_n^{\theta\alpha}$.

Diagrams may be obtained by:

- A) Drawing n horizontal lines and labeling them $\tau_1, \dots, \tau_n$ in order,
- B) Drawing N_{τ_j} solid lines and M_{τ_j} dotted lines between τ_j and τ_{j+1} to represent the state $|\gamma_j\rangle$, and
- C) Including a vertex linking two solid lines and one dotted line at τ_j , providing the transition from $|\gamma_j\rangle$ to $|\gamma_{j+1}\rangle$ described by V_{τ_{j+1}, τ_j} .

Clearly, $S_n^{\theta\alpha}$ will contain all possible diagrams of order n which are consistent with the proper number and type of incoming and outgoing particles.

The time ordering of these diagrams is important, because different time orderings correspond to different terms in the sum over intermediate states $|\gamma_j\rangle$ (Fig. 1). In general, there will be $n!$ time orderings of a diagram of order n , corresponding to every possible permutation of the times $\tau_1, \dots, \tau_n$. If two diagrams differ from one another only by a time ordering, we will call them

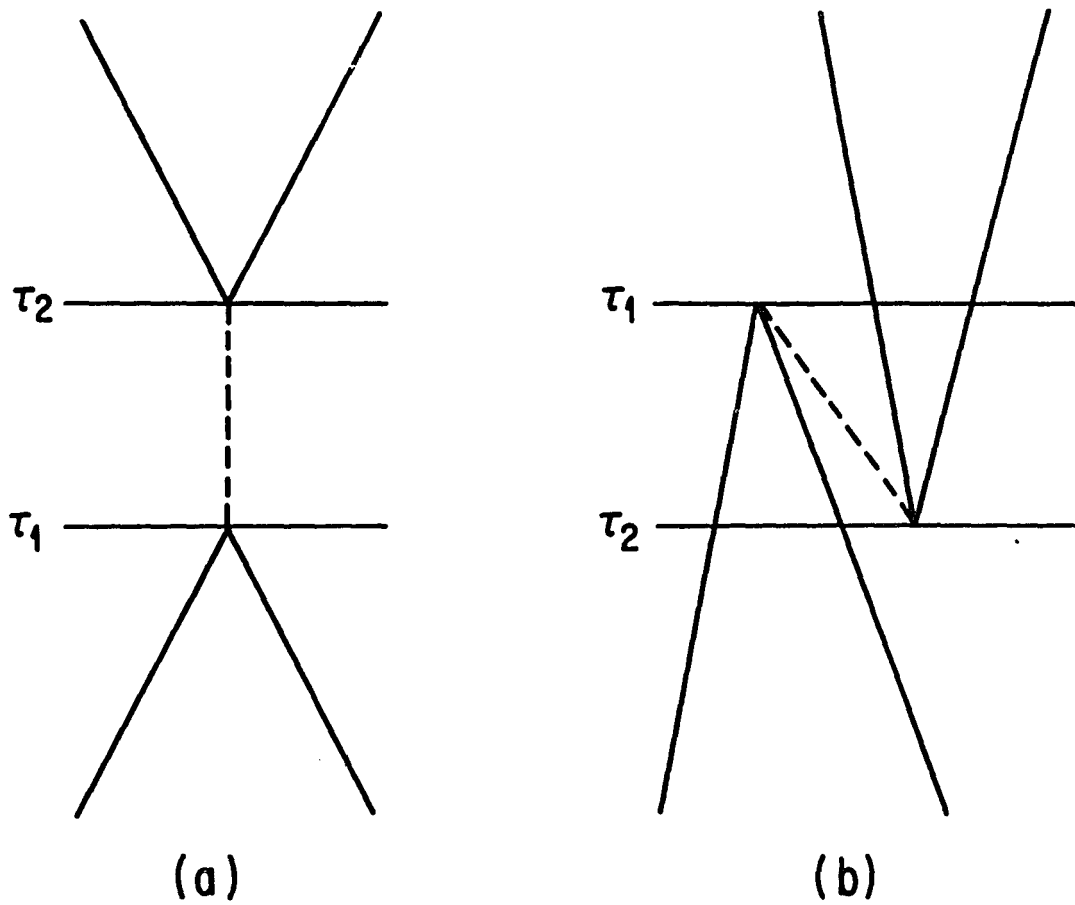


Figure 1: Different time orderings of a diagram. The intermediate states are (a) ψ_{01} and (b) ψ_{41} .

"equivalent diagrams", and we will speak of the equivalence class D of a diagram D_j as the set of all diagrams equivalent to D_j . This makes it possible to write

$$S_n^{aa} = \sum_{\substack{\text{equiv classes } D \\ \text{of order } n}} S_D^{aa} \quad (36)$$

and

$$S_D^{aa} = \sum_{D_j \text{ in } D} S_{D_j}^{aa} \quad (37)$$

thereby viewing S^{aa} as a sum of contributions due to every conceivable diagram. S_D^{aa} corresponds to the contribution to S^{aa} from a single Feynman diagram. That is, the sum of all of the time ordered diagrams in a given equivalence class results in a single Feynman diagram (Fig. 2). This result is necessary in order to achieve Lorentz covariance, because a Lorentz transformation can change the order in which interactions occur.

Now that we can view S^{aa} as a sum of diagrams, we must determine the contribution of a given diagram to S^{aa} . This is accomplished in two steps. First, the matrix element (35), due to the diagram D_j is calculated, and then the sum over time orderings and the time integrals are performed.

Consider a general diagram with (N_i, M_i) initial particles and (N_f, M_f) final particles. In the state $|\gamma_j\rangle$ there are (N_j, M_j) particles. Let us evaluate a single

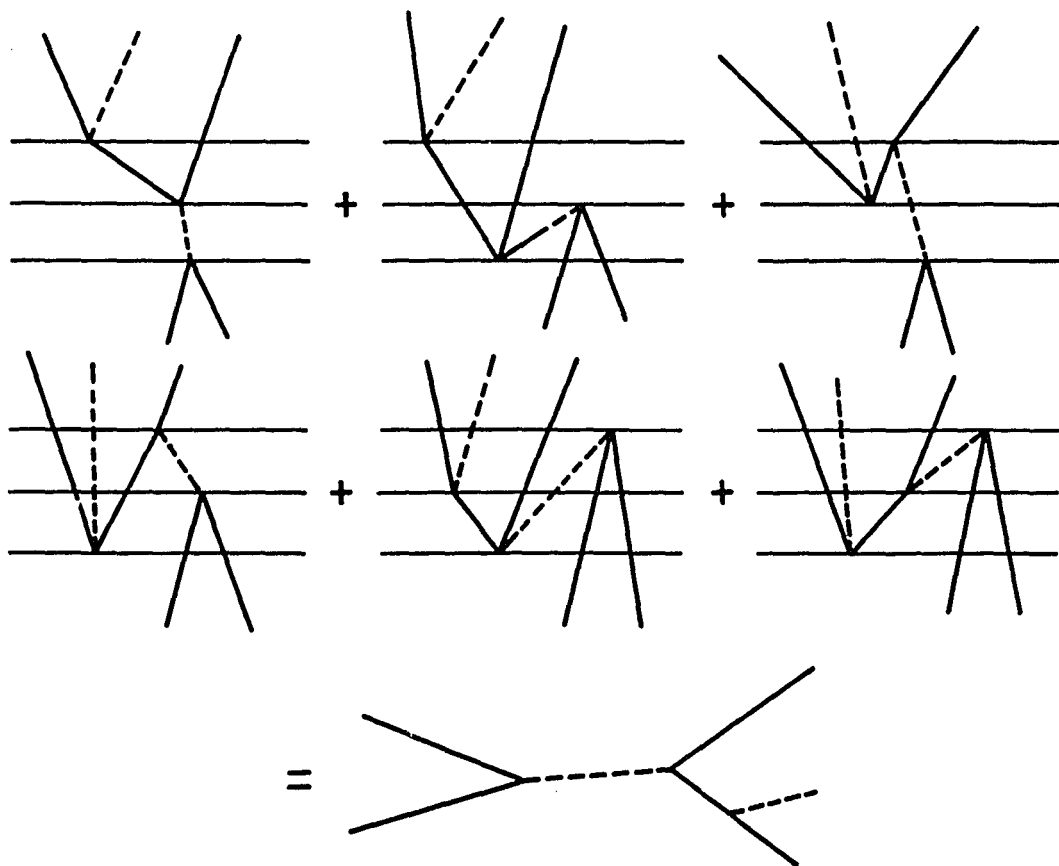


Figure 2: Six time ordered diagrams which combine to form a single Feynman Diagram.

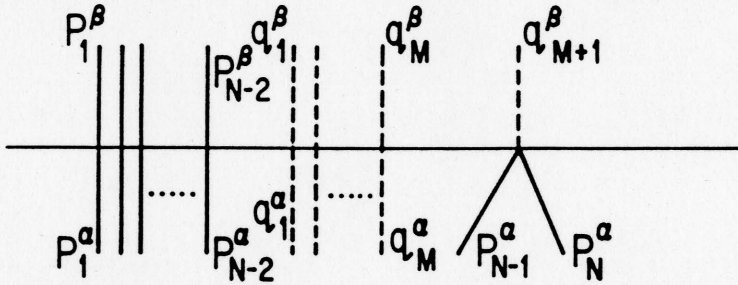
matrix element $\langle \beta | V | \alpha \rangle$ first. Then we can combine them, and determine what the contribution from a particular diagram is. A general state is given by

$$\Pi_{NM} | \alpha \rangle = \frac{1}{\sqrt{N!M!}} \prod_{j=1}^N 2\omega_{p_j} \prod_{j=1}^M 2\omega_{q_j} \sum_{\substack{\text{permutations} \\ \{j\}}} \prod_{j=1}^N \delta(p_{\{j\}} - p_j^*) \prod_{j=1}^M \delta(q_{\{j\}} - q_j^*) \quad (38)$$

if the number of particles in the state $| \alpha \rangle$ are N and M , and zero otherwise. The normalization of such a state is

$$\langle \beta | \alpha \rangle = \delta_{N_\beta N_\alpha} \delta_{M_\beta M_\alpha} \prod_{j=1}^N 2\omega_{p_j} \prod_{j=1}^M 2\omega_{q_j} \sum_{\substack{\text{permutations} \\ \{j\}}} \prod_{j=1}^N \delta(p_{\{j\}}^* - p_j^*) \prod_{j=1}^M \delta(q_{\{j\}}^* - q_j^*) \quad (39)$$

There are six different interactions to consider, each of which connects a given initial state to a unique final state, and they must all be worked out individually. Here we consider one, given by the diagram segment



$$(40)$$

The explicit form of the interaction is obtained from (34). It is given by

$$\langle p_1, \dots, p_{N-2}; q_1, \dots, q_{M+1} | \Pi_{N-2, M+1} V | \alpha \rangle = - (2\pi)^{-\sigma/2} \sqrt{\frac{N(N-1)}{M+1}} \sum_{i=1}^{M+1} \int \psi_{NM}^*(p_1, \dots, p_N; q_1, \dots, \hat{q}_1, \dots, q_M) \delta'(q_1 - p_N - p_M) \frac{dp_N dp_{N-1}}{4\omega_{p_N} \omega_{p_{N-1}}} \quad (41)$$

So,

$$\langle \beta | V | \alpha \rangle = \quad (42)$$

$$\begin{aligned} & \int \langle \beta | p_1, \dots, p_{N-2}; q_1, \dots, q_{M+1} \rangle \langle p_1, \dots, p_{N-2}; q_1, \dots, q_{M+1} | V | \alpha \rangle \prod_{j=1}^{N-2} \frac{d^3 p_j}{2\omega_{p_j}} \prod_{j=1}^{M+1} \frac{d^3 q_j}{2\omega_{q_j}} \\ &= \frac{-1}{(2\pi)^{\sigma/2} (N-2)! (M+1)!} \int \prod_{j=1}^{N-2} 2\omega_{p_j} \prod_{j=1}^{M+1} 2\omega_{q_j} \left[\sum_{\substack{\text{perms} \\ \tau \in (N-2)!(M+1)!}} \prod_{j=1}^{N-2} \delta^\sigma(p_{\tau(j)} - p_j^\beta) \times \right. \\ & \quad \left. \prod_{j=1}^{M+1} \delta^\sigma(q_{\tau'(j)} - q_j^\beta) \right] \sum_{l=1}^{M+1} \frac{1}{2\omega_{q_l}} \left[\sum_{\substack{\text{perms} \\ \eta \in N! M!}} \prod_{j=1}^N \delta^\sigma(p_{\eta(j)} - p_j^\alpha) \prod_{j=1}^M \delta^\sigma(q_{\eta'(j)} - q_j^\alpha) \times \right. \\ & \quad \left. \delta^\sigma(q_1 - p_N - p_{N-1}) \right] \prod_{j=1}^N d^3 p_j \prod_{j=1}^M d^3 q_j \end{aligned}$$

One wants only the terms in the sums over permutations which contribute to the process described by the diagram (40). Other terms will correspond to different diagrams in which different legs interact. Contributions to (40) are contributions for which

$$q_1 = q_{M+1} \quad p_{N-1} = p_{N-1} \quad p_N = p_N \quad (43)$$

or, in other words,

$$\zeta'(1) = M+1 \quad \eta(N-1) = N-1 \quad \eta(N) = N \quad (44)$$

or, alternatively,

$$\zeta'(1) = M+1 \quad \eta(N-1) = N \quad \eta(N) = N-1 \quad (45)$$

and additionally,

$$\zeta(j) = \eta(j) \quad j=1, \dots, N-2 \quad (46)$$

$$\zeta'(j) = \eta'(j) \quad j=1, \dots, M$$

There are

$$2(\text{Number of poss. values of } l)(N-2)! M! = 2N!(M+1)! \quad (47)$$

such terms in the sum, each having the same value.

Retaining only these terms, the result is

$$\langle \beta | V | \alpha \rangle_{D_j} = \frac{-2\lambda}{(2\pi)^{e/2}} \delta^e(q_{M+1}^\beta - p_N^\alpha - p_{N-1}^\alpha) \prod_{j=1}^{N-2} 2\omega_{p_j^\alpha} \delta^e(p_j^\alpha - p_j^\beta) \prod_{j=1}^M 2\omega_{q_j^\beta} \delta^e(q_j^\alpha - q_j^\beta) \quad (48)$$

Likewise, evaluating the other five interaction terms, one finds that the prescription for writing down the contribution of any interaction term to $\langle \beta | V_{D_j} | \alpha \rangle$ is to include a factor

$$-\frac{2\lambda}{(2\pi)^{e/2}} \delta^e\left(\sum_{\text{vertex}} k_{\text{out}}\right) \quad (49)$$

for the particles that interact at the vertex, and a factor

$$2\omega_k \delta^e(k^\alpha - k^\beta) \quad (50)$$

for all unaffected particles. The matrix element

$\langle \beta | V_{D_j}^n | \alpha \rangle$ for the diagram D_j may be obtained by putting in all n factors together and performing the sum over intermediate states,

$$\langle \beta | V_{D_j}^n | \alpha \rangle = \frac{1}{\eta} \int \prod_{\text{all vert's } j} \left\{ \frac{-2\lambda}{(2\pi)^{e/2}} \delta^e\left(\sum_{\text{vertex } j} k_{\text{out}}\right) \prod_{\substack{\text{particles not} \\ \text{interacting at } j}} 2\omega_{k_i^{(j)}} \delta^e(k_i^{\gamma_{j-1}} - k_i^{\gamma_j}) \frac{d^e k_i^{\gamma_j}}{2\omega_{k_i^{(j)}}} \right\}$$

where η is a symmetry factor which keeps states with identical particles serving the same function from being counted twice as is the case in Fig. 3. It is found by including a factor $n!$ for each set of internal lines which serve the same purpose. The δ -functions remove all integrations over particles which don't interact at a particular vertex, just keeping momenta the same before

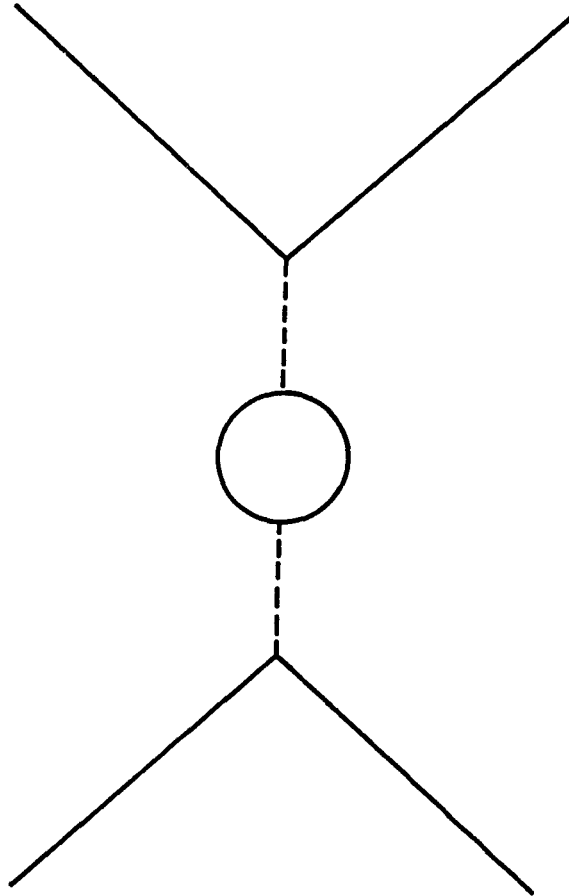


Figure 3: A diagram in which a double counting factor is required. The factor in this case is $1/2!$.

and after the interaction. The result is

$$\langle \beta | V_{D_j}^n | a \rangle = \int \prod_{\text{vertices } j} \left\{ \frac{-2\lambda}{(2\pi)^{e/2}} \delta^e \left(\sum_{i,j} k_{ij}^{\text{out}} \right) \right\} \prod_{\text{disconnected lines}} \left\{ 2\omega_k \delta^e(k_{\text{init}} - k_{\text{final}}) \right\} \prod_{\text{internal lines}} \frac{d^3 k_i}{2\omega_{k_i}^{(i)}} \quad (52)$$

For fixed internal momenta, one simply drops the integral sign and denotes the matrix element as $\langle \beta | V_{D_j}^n | a \rangle_F$,

$$\langle \beta | V_{D_j}^n | a \rangle = \int \langle \beta | V_{D_j}^n | a \rangle_F \prod_{\substack{\text{internal} \\ \text{momenta } i}} \frac{d^3 k_i}{2\omega_{k_i}^{(i)}} \quad (53)$$

Now one may write

$$S_D^{\beta a} = \sum_{\substack{\text{time} \\ \text{orderings } l}} \int \left\{ \prod_{\substack{\text{internal} \\ \text{momenta } i}} \frac{d^3 k_i}{2\omega_{k_i}^{(i)}} \right\} \langle \beta | V_{D_j}^n | a \rangle_F \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\tau_2} e^{iE_{\beta} \tau_n} \dots e^{iE_a \tau_1} d\tau_1 \dots d\tau_n \quad (54)$$

Clearly the factor $\langle \beta | V_{D_j}^n | a \rangle_F$, is independent of the time orderings, so one may write

$$S_D^{\beta a} = \int \langle \beta | V_{D_j}^n | a \rangle_F I \prod_{\substack{\text{internal} \\ \text{momenta } i}} \frac{d^3 k_i}{2\omega_{k_i}^{(i)}} \quad (55)$$

where

$$I = \sum_{\substack{\text{time} \\ \text{orderings } l}} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\tau_2} e^{iE_{\beta} \tau_n} \dots e^{iE_1^1 (\tau_1 - \tau_2)} e^{iE_a \tau_1} d\tau_1 \dots d\tau_n \quad (56)$$

To evaluate $S_D^{\beta a}$, one must evaluate I. To do this, one must first sum over the time orderings. To take the sum, note that

$$e^{iE_{\gamma} (\tau_j - \tau_{j+1})} = \prod_{\substack{\text{all particles } i \\ \text{in } \gamma_j}} e^{i\omega_{p_i}^{(i)} (\tau_j - \tau_{j+1})} \quad (57)$$

so I contains a factor

$$e^{i\omega_{p_1}^{(1)} (\tau_j - \tau_{j+1})} \quad (58)$$

for each particle and each time interval in which that particle exists. Now, if the j'th particle is created at

$$a(j), \quad 1 \leq a(j) \leq n \quad (59)$$

and annihilated at

$$b(j), \quad 1 \leq a(j) < b(j) \leq n \quad (60)$$

the resulting factor in I will be

$$e^{i\omega_{p_j}(r_{a(j)} - r_{b(j)})} \quad (61)$$

Likewise, if the particle is external, there will be a

factor

$$e^{i\omega_{k_j}^{(j)} r_{a(j)}} \quad (62)$$

if it is created at $a(j)$ and a factor

$$e^{-i\omega_{k_j}^{(j)} r_{b(j)}} \quad (63)$$

if it is annihilated at $b(j)$. With this notation, one may write

$$I = \sum_{\substack{\text{time} \\ \text{orderings}}} \int_0^\infty \dots \int_0^{r_2} \prod_{\substack{j: \text{final} \\ \text{particles}}} e^{i\omega_{k_j}^{(j)} r_{a(j)}} \prod_{\substack{j: \text{initial} \\ \text{particles}}} e^{-i\omega_{k_j}^{(j)} r_{b(j)}} \times \quad (64) \\ \prod_{\substack{j: \text{internal} \\ \text{particles}}} e^{i\omega_{p_j}^{(j)} (r_{a(j)} - r_{b(j)})} d\tau_1 \dots d\tau_n$$

Now, two time orderings differ only by a permutation

$$\zeta: (\tau_1, \dots, \tau_n) \longrightarrow (\tau_{\zeta(1)}, \dots, \tau_{\zeta(n)}) \quad (65)$$

of the times. The sum and integral combination in (64)

amount to the integration of some function of $\tau_1, \dots, \tau_n$

over all values of $\tau_1, \dots, \tau_n$. This function can be found

from (61), since a particle must be created before it can

be annihilated,

$$\tau_{b(j)} - \tau_{a(j)} > 0 \quad (66)$$

no matter which permutation is examined. Thus,

$$e^{i\omega_{p_j}^{(j)}(r_{a(j)}-r_{b(j)})} = e^{-i\omega_{p_j}^{(j)}|r_{a(j)}-r_{b(j)}|} \quad (67)$$

and so

$$I = \quad (68)$$

$$\begin{aligned} & \sum_i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{\text{final}} e^{i\omega_{k_j}^{(j)} r_{f(a(j))}} \prod_{\text{initial}} e^{-i\omega_{q_j}^{(j)} r_{f(b(j))}} \prod_{\text{internal}} e^{-i\omega_{p_j}^{(j)} |r_{f(a(j))}-r_{f(b(j))}|} d\tau_1 \dots d\tau_n \\ &= \sum_i \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{\text{final}} e^{i\omega_{k_j}^{(j)} r_{a(j)}} \prod_{\text{initial}} e^{-i\omega_{q_j}^{(j)} r_{b(j)}} \prod_{\text{internal}} e^{-i\omega_{p_j}^{(j)} |r_{a(j)}-r_{b(j)}|} d\tau_1 \dots d\tau_n \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \prod_{\text{final}} e^{i\omega_{k_j}^{(j)} r_{a(j)}} \prod_{\text{initial}} e^{-i\omega_{q_j}^{(j)} r_{b(j)}} \prod_{\text{internal}} e^{-i\omega_{p_j}^{(j)} |r_{a(j)}-r_{b(j)}|} d\tau_1 \dots d\tau_n \end{aligned}$$

Now, one may evaluate this integral in a straightforward way to obtain an expression in terms of Feynman propagators. Firstly, define some numbers associated to the topology of different diagrams:

n = number of vertices

N_e = number of external m -particles

M_e = number of external μ -particles

N_I = number of internal m -particles

M_I = number of internal μ -particles

ν = total number of internal lines

l = number of independent loops

Since the rules for drawing diagrams are the same as spinor QED, the following relations hold:

$$N_I = n - \frac{1}{2} N_e \quad (69)$$

$$M_I = \frac{1}{2} (n - M_e)$$

$$l = N_I + M_I - (n - 1) = \frac{1}{2} (n - N_e - M_e) + 1$$

$$\nu = \frac{3}{2} n - \frac{1}{2} (N_e + M_e)$$

To evaluate (68), change variables from $\tau_1, \dots, \tau_n$ to

$$T = \tau_1 \quad (70)$$

$$t_j = \tau_{\beta(j)} - \tau_{\alpha(j)} \quad j=1, \dots, \nu$$

One must write the factor

$$F = \prod_{\text{final}} e^{i\omega_{k_j}^{(j)} \tau_{\alpha(j)}} \prod_{\text{initial}} e^{-i\omega_{q_j}^{(j)} \tau_{\beta(j)}} \quad (71)$$

in terms of new variables. First, write

$$F = \exp[i(\sum_{\text{final}} k_{0j} - \sum_{\text{initial}} q_{0j})\tau_1] \exp[i\sum_{\text{final}} k_{0j}(\tau_{\alpha(j)} - \tau_1) - i\sum_{\text{initial}} q_{0j}(\tau_{\beta(j)} - \tau_1)]$$

where $q_{0i} = \omega_{q_i}^{(i)}$, etc. Setting

$$E_f = \sum_{\text{final}} k_{0j} \quad E_i = \sum_{\text{initial}} q_{0j} \quad (73)$$

one seeks to write

$$F = e^{i(E_f - E_i)\tau_1} \exp[i\sum_{\text{internal}} p_{0j}(\tau_{\beta(j)} - \tau_{\alpha(j)})] \quad (74)$$

Equating (71) and (74), one obtains n different equations

for the p_{0j} in terms of the k_{0j} and q_{0j} by equating the coefficients of τ_j , $j=1, \dots, n$. There is, however, one relation among these equations since the sum of all the coefficients is zero. The equations are a conservation law for the k_{0j} , q_{0j} , and p_{0j} at each vertex except one, the vertex at τ_1 .

$$\sum_{\substack{\text{internal lines } j \\ \text{ending at vertex } v}} p_{0j} - \sum_{\substack{\text{internal lines } j \\ \text{starting at } v}} p_{0j} = \sum_{\substack{\text{final lines } j \\ \text{starting at } v}} k_{0j} - \sum_{\substack{\text{initial lines } j \\ \text{ending at } v}} q_{0j} \quad (75)$$

In general there will be 1 degrees of freedom in choosing the p_{0j} . we will assume that they have been chosen in such a way that (74) holds. Then one may write

$$I = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} e^{iE_f \tau_1} \prod_{j, \text{ internal}} \exp[ip_{0j}(\tau_{\beta(j)} - \tau_{\alpha(j)} - i\omega_{p_j}^{(j)}|\tau_{\alpha(j)} - \tau_{\beta(j)}|)] d\tau_1 \dots d\tau_n \quad (76)$$

To change variables requires $(\nu+1)-n=1$ δ -functions, since there are $\nu+1$ new variables and only n old variables. It is easy to see what these δ -functions should be, since for any loop in a diagram, the sum of the times should be zero, as illustrated in Fig. 4. Thus, the needed δ -functions may be written as

$$\prod_{\text{loops}} \delta\left(\sum_{j=1} \sigma_{1j} t_j\right) \quad (77)$$

where $\sigma_{1j} = \pm 1$. Then I may be expressed as

$$I = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} e^{i(E_f - E_i)T} \prod_{\substack{\text{internal} \\ \text{lines}}} e^{ip_{0j} t_j - \omega_{p_j}^{(j)} |t_j|} \prod_{\text{loops}} \delta\left(\sum_{j=1} \sigma_{1j} t_j\right) dT dt_1 \dots dt_n \quad (78)$$

$$= 2\pi \delta(E_f - E_i) \int_{-\infty}^{\infty} \prod_{\text{loops}} \delta\left(\sum_{j=1} \sigma_{1j} t_j\right) \prod_{j=1} e^{-i\omega_{p_j}^{(j)} |t_j|} \prod_{j=1} e^{ip_{0j} t_j} dt_1 \dots dt_n.$$

This is a Fourier transform of a product of two functions.

It may be evaluated by writing it as a convolution

$$(fg)^{\wedge}(k) = (\hat{f} * \hat{g})(k) = \frac{1}{2\pi} \int \hat{f}(k-p) \hat{g}(p) dp \quad (79)$$

where

$$f = \prod_{\text{loops}} \delta\left(\sum_{j=1} \sigma_{1j} t_j\right) \quad (80)$$

and

$$g = \prod_{j=1} \exp(-i\omega_{p_j}^{(j)} |t_j|) \quad (81)$$

The convolutions of f and g are easily evaluated. Then, the result is

$$I = (2\pi)^{n-\nu} \delta\left(\sum_{\text{final}} k_{0j} - \sum_{\text{initial}} q_{0j}\right) \int_{-\infty}^{\infty} \prod_{\substack{j=2 \\ \text{vertices}}}^n \delta\left(\sum_{\text{incoming}} (p_{0i} - p_i')\right) \prod_{j=1} \frac{2i\omega_{p_j}^{(j)}}{p_j^2 - \omega_{p_j}^{(j)2} + i\epsilon_j} dp_1' \dots dp_n' \quad (82)$$

The energy conservation δ -function outside the integral can then be changed into a δ -function for energy conservation at

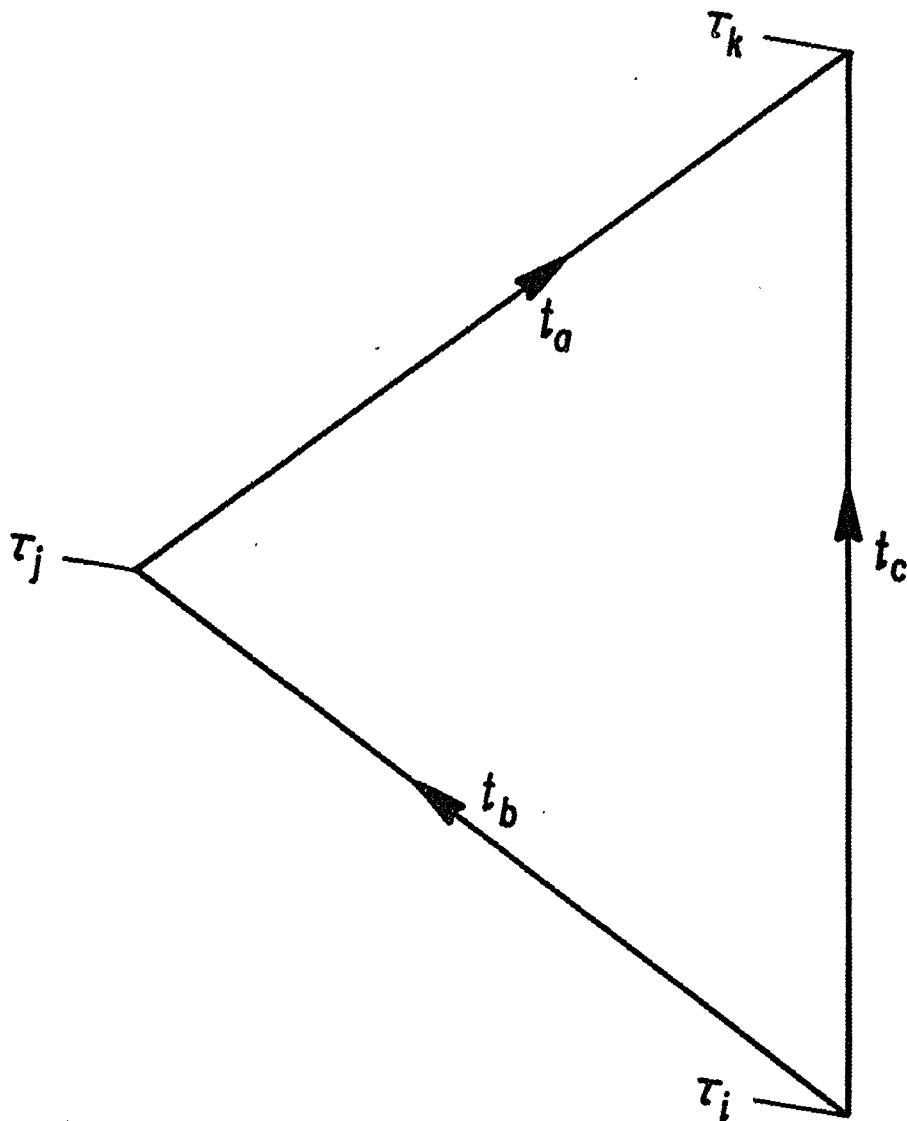


Figure 4: A loop for which a delta function is needed in Eq. (77). Since $t_a + t_b = t_c$, a $\delta(t_a + t_b + t_c)$ is required.

vertex one. Therefore

$$I = (2\pi)^{n-\nu} \int_{-\infty}^{\infty} \prod_{\text{vertices}} \delta\left(\sum_{\text{incoming}} p_{0i}\right) \prod_{\text{internal}} \frac{2i\omega_{p_j}^{(j)}}{p_{0j}^2 - \omega_{p_j}^{(j)2} + i\epsilon_j} dp_{01} \dots dp_{0\nu} \quad (83)$$

The only terms which contribute to this integral are terms in which the necessary conditions (75) are fulfilled. The final result for the contribution to the S-matrix due to one diagram is thus

$$S_D^{\beta\alpha} = \frac{1}{\eta} (2i\lambda)^n (2\pi)^{\sigma(N_\bullet + M_\bullet)/2} \int \prod_{\text{vertices}} (2\pi)^{\sigma+1} \delta^{\sigma+1}\left(\sum_{\text{incoming}} p_j\right) \prod_{\text{internal lines}} \frac{i}{p_j^2 - m_j^2 + i\epsilon_j} \frac{d^{\sigma+1}p_j}{(2\pi)^{\sigma+1}} \quad (84)$$

where we have switched to a covariant $\sigma+1$ dimensional space time notation. This gives exactly the Feynman rules we desired to derive:

1. Draw all diagrams with the correct number of initial and final states, and vertices with two m -particle lines and one μ -particle line.
2. To obtain the contribution to S from one diagram, include

- a. A factor of

$$2i\lambda(2\pi)^{\sigma+1} \delta^{\sigma+1}\left(\sum_{\text{incoming}} p_j\right)$$

for each vertex.

- b. A factor of

$$\frac{i}{p_j^2 - m_j^2 + i\epsilon_j} \frac{d^{\sigma+1}p_j}{(2\pi)^{\sigma+1}}$$

for each internal line.

and integrate over all dp 's.

3. Finally, include a factor

$$\frac{1}{\eta} (2\pi)^{\sigma(N_{\bullet} + M_{\bullet})/2}$$

which is due to the chosen normalization of the states, and the aforementioned symmetry factor.

The full S matrix may thus be expressed as a sum over all Feynman diagrams, with each diagram making the usual contribution as derived in the standard perturbative approach to field theory. This completes the proof that the hierarchy equation (23) is equivalent to the usual understanding of field theory in the appropriate small λ limit.

The difficulties inherent to this perturbation expansion are well known. They include the whole question of convergence and the troubles associated with bound states. The failure of the expansion occurs in equation (32), and is endemic to perturbation theory in general, so we will not discuss it further here.

It is also important to note that even though the result we have derived here applies only to one specific field theory, it can easily be generalized to much more complicated systems. In particular, the integral I is dependent only on the structure of the diagram it

represents, and not on V . Thus, the propagators for internal particles will always be the same (for scalar bosons), and energy will be conserved at each vertex. Additionally, any local interaction will have a δ -function for the momenta at each vertex, so energy - momentum will be conserved at each vertex. The only changes will be the structure of the diagrams and the constant factor at each vertex.

CHAPTER 4

RENORMALIZATION CONSTANTS FOR THE HIERARCHY

Before attempting to make any calculations with the hierarchy (23), we must develop a method of regularizing and renormalizing it. All of the usual infinities associated with quantized fields will be present here.

It seems natural to introduce a lattice renormalization scheme here, in view of the numerical work we intend to do, and in view of our already obvious lack of concern for manifest Lorentz covariance. The lattice cuts off all infrared and ultraviolet divergences and reduces all integrals in (23) to finite sums.

We do not concern ourselves with the details of choosing a lattice here, because it is only a device to define singular integrals, counterterms, and wavefunctions systematically. Such lattices have been used frequently in the context of Lattice Gauge theories, and studied within that context. Obviously, defining a continuum limit for wavefunctions, etc., is a difficult mathematical task, the depth of which can only be properly appreciated after doing some calculations. As such, we defer a fuller discussion until Chapter 7. For now, we will just calculate some of the renormalization constants for the regularized

hierarchy, and then use them in Chapter 5 to subtract off infinities and derive some meaningful results.

The connection between the hierarchy equation and perturbation theory provides some clues about renormalizing the hierarchy. Essentially, all of the counterterms needed in the perturbation theory will be needed for the hierarchy, or else the perturbation series derived from the hierarchy will not be finite.

In particular, for the model field theory we have been dealing with, write the bare Lagrangian as

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \psi_0 \partial^\mu \psi_0 - m_0^2 \psi_0^2 + \partial_\mu \varphi_0 \partial^\mu \varphi_0 - \mu_0^2 \varphi_0^2) + \lambda_0 \varphi_0 \psi_0^2 \quad (88)$$

which is singular in the continuum. Defining

$$\psi = Z_1^{-1/2} \psi_0 \quad \varphi = Z_2^{-1/2} \varphi_0 \quad (89)$$

the Lagrangian may be rewritten as

$$\begin{aligned} \mathcal{L} = & \frac{1}{2} (\partial_\mu \psi \partial^\mu \psi - m^2 \psi^2) + \frac{1}{2} (\partial_\mu \varphi \partial^\mu \varphi - \mu^2 \varphi^2) + \lambda \psi^2 \varphi \\ & + \frac{1}{2} (Z_1 - 1) (\partial_\mu \psi \partial^\mu \psi - m^2 \psi^2) - \frac{1}{2} (m_0^2 - m^2) Z_1 \psi^2 \\ & + \frac{1}{2} (Z_2 - 1) (\partial_\mu \varphi \partial^\mu \varphi - \mu^2 \varphi^2) - \frac{1}{2} (\mu_0^2 - \mu^2) Z_2 \varphi^2 \\ & + (Z_1 Z_2^{1/2} \lambda_0 - \lambda) \psi^2 \varphi \end{aligned} \quad (90)$$

where the terms on the second, third, and fourth lines vanish in the classical limit.

On the lattice, all of the renormalization parameters, Z_1 , Z_2 , m_0 , μ_0 , and λ_0 will be finite functions of m , μ , λ , and Δk . To renormalize the field

theory, one must determine these functions and plug them into the hierarchy. To do this, it is necessary to analyze the hierarchy equation resulting from the Lagrangian (90). This equation, which can be derived by a procedure identical to that of Chapter 2, is given by

$$\begin{aligned}
 i \frac{\partial \psi_{NM}}{\partial t} = & \gamma \psi_{NM} + (Z_1 \sum_{i=1}^N \omega_{p_i} + Z_2 \sum_{j=1}^M \omega'_{q_j}) \psi_{NM}(p_1, \dots, p_N; q_1, \dots, q_M) \\
 & + \sum_{i=1}^N Z_1 (\mu_0^2 - m^2) \frac{1}{2\omega_{p_i}} \psi_{NM}(p_1, \dots, p_N; q_1, \dots, q_M) \\
 & + \frac{1}{2} Z_1 (\mu_0^2 - m^2) \sqrt{(N+1)(N+2)} \int \psi_{N+2, M}(p_1, \dots, p_N, r, s; q_1, \dots, q_M) \delta^\sigma(r+s) \frac{d^3 r d^3 s}{4\omega_r \omega_s} \\
 & + \frac{1}{2} Z_1 (\mu_0^2 - m^2) (N(N-1))^{-1/2} \sum_{i,j=1}^N \delta^\sigma(p_i + p_j) \times \\
 & \quad \psi_{N-2, M}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, q_M) \\
 & + \sum_{j=1}^M Z_2 (\mu_0^2 - \mu^2) \frac{1}{2\omega_{q_j}} \psi_{NM}(p_1, \dots, p_N; q_1, \dots, q_M) \\
 & + \frac{1}{2} Z_2 (\mu_0^2 - \mu^2) \sqrt{(M+1)(M+2)} \int \psi_{N, M+2}(p_1, \dots, p_N; q_1, \dots, q_M, r, s) \delta^\sigma(r+s) \frac{d^3 r d^3 s}{4\omega_r \omega_s} \\
 & + \frac{1}{2} Z_2 (\mu_0^2 - \mu^2) (M(M-1))^{-1/2} \sum_{i,j=1}^M \delta^\sigma(q_i + q_j) \times \\
 & \quad \psi_{N, M-2}(p_1, \dots, p_N; q_1, \dots, \hat{q}_i, \dots, \hat{q}_j, \dots, q_M) \\
 & - Z_1 Z_2^{-1/2} \lambda_0 ((2\pi)^\sigma M N (N-1))^{-1/2} \sum_{i=1}^N \sum_{j=1}^M \delta^\sigma(p_i + p_j + q_1) \times \\
 & \quad \psi_{N-2, M-1}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, \hat{q}_1, \dots, q_M) \\
 & - Z_1 Z_2^{-1/2} \lambda_0 (2\pi)^{-\sigma/2} \sqrt{(M+1)(N+1)(N+2)} \iiint (\delta^\sigma(r+u+s) \times \\
 & \quad \psi_{N+2, M+1}(p_1, \dots, p_N, r, u; q_1, \dots, q_M, s)) \frac{d^3 r d^3 u d^3 s}{8\omega_r \omega_u \omega_s} \\
 & - 2 Z_1 Z_2^{1/2} \lambda_0 (2\pi)^{-\sigma/2} \sqrt{M+1} \sum_{j=1}^M \int \int (\delta^\sigma(r+s-p_j) \times \\
 & \quad \psi_{N, M+1}(p_1, \dots, \hat{p}_j, \dots, p_N, s; q_1, \dots, q_M, r)) \frac{d^3 s d^3 r}{4\omega_s \omega_r}
 \end{aligned} \tag{91}$$

$$\begin{aligned}
& -2Z_1 Z_2^{1/2} \lambda_0 (2\pi)^{-\sigma/2} M^{-1/2} \sum_{i=1}^M \sum_{j=1}^N \int \{ \delta^\sigma(p_j + q_i - s) \times \\
& \quad \psi_{N,M-1}(p_1, \dots, \hat{p}_j, \dots, p_N, s; q_1, \dots, \hat{q}_i, \dots, q_M) \} \frac{d^s s}{2\omega_s} \\
& -Z_1 Z_2^{1/2} \lambda_0 \sqrt{\frac{(N+1)(N+2)}{M(2\pi)^\sigma}} \sum_{i=1}^M \iint \delta^\sigma(q_i - r - s) \times \\
& \quad \psi_{N+2,M-1}(p_1, \dots, p_N, r, s; q_1, \dots, \hat{q}_i, \dots, q_M) \frac{d^s s d^r r}{4\omega_r \omega_s} \\
& -Z_1 Z_2^{1/2} \lambda_0 \sqrt{\frac{M+1}{(2\pi)^\sigma N(N-1)}} \sum_{\substack{i,j=1 \\ i \neq j}}^N \int \delta^\sigma(r - p_i - p_j) \times \\
& \quad \psi_{N-2,M+1}(p_1, \dots, \hat{p}_i, \dots, \hat{p}_j, \dots, p_N; q_1, \dots, q_M, r) \} \frac{d^r r}{2\omega_r}
\end{aligned}$$

where an additional counterterm γ_{NM} has been included to explicitly modify the vacuum energy.

The counterterms are determined by setting the vacuum energy to zero,

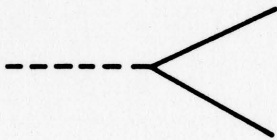
$$\epsilon_v = 0 \quad (92)$$

and the single particle energies equal to free particle energies

$$\epsilon_p = \omega_p \quad (93)$$

$$\epsilon'_p = \omega'_p$$

and, finally, looking at a transition of the form



$$(94)$$

and fixing the transition rate.

In general, it is not possible to determine these counterterms analytically, so some sort of approximation

method is necessary. This method will depend on the type of problem one is attempting to solve. At first we will be concerned with the study of bound states in weakly coupled systems and their contribution to the dynamics of those systems. In weak coupling problems, one can (presumably) expand the renormalization parameters in powers of the coupling constant and determine the coefficients of the expansions order by order. Thus, we may write

$$\gamma = \gamma_1 \lambda + \gamma_2 \lambda^2 + \dots \quad (95)$$

$$Z_1 = 1 + Z_{11} \lambda + Z_{12} \lambda^2 + \dots$$

$$Z_2 = 1 + Z_{21} \lambda + Z_{22} \lambda^2 + \dots$$

$$m_0 = m(1 + m_1 \lambda + m_2 \lambda^2 + \dots)$$

$$\mu_0 = \mu(1 + \mu_1 \lambda + \mu_2 \lambda^2 + \dots)$$

$$\lambda_0 = \lambda(1 + \Lambda_1 \lambda + \Lambda_2 \lambda^2 + \dots)$$

where none of the coefficients of λ^n are functions of λ .

Note that

$$\gamma_1 = Z_{11} = Z_{21} = m_1 = \mu_1 = \Lambda_1 = 0 \quad (96)$$

because all of the diagrams which the counterterms are designed to cancel are at least of order λ^2 . To determine the coefficients to order λ^2 and higher, one must write the hierarchy equations out and impose the conditions (92), etc. We will determine γ_2 , Z_{12} , Z_{22} , m_2 and μ_2 ,

which will be sufficient to do some non-trivial calculations in the next section.

First, consider the vacuum energy to order λ^2 . When $\lambda=0$, the vacuum state is given by

$$\psi_{00}=1 \quad \psi_{NM}=0 \text{ if } N \neq 0 \text{ or } M \neq 0 \quad (97)$$

ψ_{00} couples only to ψ_{21} , so one would expect ψ_{21} to be of order λ and ψ_{00} of order unity for small λ , and every other component of $|\Psi\rangle$ to be at least of order λ^2 . Thus, the hierarchy for the vacuum to order λ^2 is of the form

$$i\frac{\partial\psi}{\partial t}{}^{00} = \epsilon\psi_{00} = \gamma_2\lambda^2\psi_{00} - \frac{\sqrt{2}\lambda}{(2\pi)^{3/2}} \int \psi_{21}(r,s;p) \delta'(r+s+p) \frac{d'r d's d'p}{8\omega_r \omega_s \omega_p} \quad (98)$$

$$i\frac{\partial\psi}{\partial t}{}^{21} = \epsilon\psi_{21}(r,s;p) = (\omega_r + \omega_s + \omega_p) \psi_{21} - \frac{\sqrt{2}\lambda}{(2\pi)^{3/2}} \psi_{00} \delta'(r+s+p) \quad (99)$$

Setting $\epsilon_v=0$ will determine γ_2 . To see this, solve (99)

for ψ_{21} ,

$$\psi_{21}(r,s;p) = \frac{\sqrt{2}\lambda}{(2\pi)^{3/2}} \frac{\psi_{00} \delta'(r+s+p)}{\omega_r + \omega_s + \omega_p} \quad (100)$$

and plug it into (98), giving

$$0 = \gamma_2\lambda^2\psi_{00} - \frac{2\lambda^2\psi_{00}}{(2\pi)^3} \delta'(0) \int \frac{\delta'(r+s+p) d'r d's d'p}{(\omega_r + \omega_s + \omega_p) 8\omega_r \omega_s \omega_p} \quad (101)$$

which gives

$$\gamma_2 = \frac{2}{(2\pi)^3} \delta'(0) \int \frac{\delta'(r+s+p) d'r d's d'p}{(\omega_r + \omega_s + \omega_p) 8\omega_r \omega_s \omega_p} \quad (102)$$

which, of course, is finite only on a lattice.

Next, we seek to determine m_2 and Z_{12} . To do so, look at the equations for a single m-particle state to order λ^2 . In the $\lambda=0$ limit, the state will be of the form

$$\psi_{10}(p) = 2\omega_p \delta'(p-p_0) \quad (103)$$

$$\psi_{NM}=0 \quad \text{if } N \neq 1 \text{ or } M \neq 0$$

Thus, to order λ^2 , the hierarchy will be

$$\epsilon_p \psi_{10}(p) = (\gamma_2^2 + \omega_p + Z_{12} \lambda^2 \omega_p + \frac{m^2 m \lambda^2}{\omega_p}) \psi_{10}(p) - \frac{2\lambda}{(2\pi)^{3/2}} \int \psi_{11}(r; s) \delta'(r+s-p) \frac{d'r d's}{4\omega_r \omega'_s} \quad (104)$$

$$- \frac{\sqrt{6}\lambda}{(2\pi)^{3/2}} \int \psi_{31}(p, r, k; s) \delta'(r+k+s) \frac{d'r d'k d's}{8\omega_r \omega_k \omega'_s}$$

$$\epsilon_p \psi_{11}(r; s) = (\omega_r + \omega'_s) \psi_{11}(r; s) - \frac{2\lambda}{(2\pi)^{3/2}} \int \psi_{10}(k) \delta'(r+s-k) \frac{d'k}{2\omega_k} \quad (105)$$

$$\epsilon_p \psi_{31}(p, r, k; s) = (\omega_p + \omega_r + \omega_k + \omega'_s) \psi_{31}(p, r, k; s) - \frac{2\lambda}{(2\pi)^{3/2} \sqrt{6}} \{ \psi_{10}(p) \delta'(r+k+s) + \psi_{10}(r) \delta'(p+k+s) + \psi_{10}(k) \delta'(p+r+s) \} \quad (106)$$

Now, set $\epsilon_p = \omega_p$ so the physical particle will have the correct mass and energy. Solving (105) and (106) for ψ_{11} and ψ_{31} gives

$$\psi_{11}(r; s) = \frac{-2\lambda}{(2\pi)^{3/2} (\omega_p - \omega_r - \omega'_s)} \int \psi_{10}(k) \delta'(r+s-k) \frac{d'k}{2\omega_k} \quad (107)$$

$$\psi_{31}(p, r, k; s) = \frac{2\lambda}{(2\pi)^{3/2} \sqrt{6} (\omega_p + \omega_k + \omega'_s)} \{ \psi_{10}(p) \delta'(r+k+s) + \psi_{10}(r) \delta'(p+k+s) + \psi_{10}(k) \delta'(p+r+s) \} \quad (108)$$

Plugging these into (104) gives, after some rearranging,

$$(\epsilon_p - \omega_p) \psi_{10}(p) = (\gamma_2 + Z_{12} \omega_p + \frac{m^2 m}{\omega_p}) \lambda^2 \psi_{10}(p) \quad (109)$$

$$- \frac{2\lambda^2}{(2\pi)^3} \psi_{10}(p) \delta'(0) \int \frac{\delta'(r+k+s) d'r d'k d's}{(\omega_r + \omega_k + \omega'_s) 8\omega_r \omega_k \omega'_s} + \frac{4\lambda^2}{(2\pi)^3} \frac{\psi_{10}(p)}{2\omega_p} \int (\omega_p^2 - (\omega_{p-k} + \omega'_k)^2)^{-1} \frac{2(\omega'_k + \omega_{p-k})}{4\omega'_k \omega_{p-k}} d'k$$

γ_2 cancels the second term, leaving

$$Z_{12} \omega_p + \frac{m^2 m}{\omega_p} = \frac{-4}{(2\pi)^3 2\omega_p} \int \frac{2(\omega'_k + \omega_{p-k})}{\omega_p^2 - (\omega_{p-k} + \omega'_k)^2} \frac{d'k}{4\omega'_k \omega_{p-k}} \quad (110)$$

The integral

$$I(p) = \int \frac{2(\omega'_k + \omega_{p-k})}{\omega_p^2 - (\omega_{p-k} + \omega'_k)^2} \frac{d^3k}{4\omega'_k \omega_{p-k}} \quad (111)$$

may be rewritten as

$$I(p) = \int \frac{2p_0}{\omega_p^2 - p_0^2} \delta^{(4)}(p-r-k) \delta(k^2 - \mu^2) \delta(r^2 - m^2) \theta(r_0) \theta(k_0) dp_0 d^{(4)}k d^{(4)}r \quad (112)$$

It is a messy proposition to separate Z_{12} and m_2 out in this integral, just as it is in ordinary perturbation theory, but fortunately we will not need to. These terms will occur together whenever we encounter them, so they can be subtracted together.

Next, the physical μ -particle state is described to order λ^2 by the equations

$$\epsilon'_p \psi_{01}(p) = (\gamma \lambda^2 + \omega'_p + Z_{12} \lambda^2 \omega'_p + \frac{\mu_2 \mu^2 \lambda^2}{\omega'_p}) \psi_{01}(p) \quad (116)$$

$$\begin{aligned} & - \frac{\sqrt{2}\lambda}{(2\pi)^{3/2}} \int \psi_{20}(r, s) \delta'(r+s+p) \frac{d^3r d^3s}{4\omega_r \omega_s} \\ & - \frac{2\lambda}{(2\pi)^{3/2}} \int \psi_{22}(r, s; p, k) \delta'(r+s+k) \frac{d^3r d^3s d^3k}{8\omega_r \omega_s \omega'_k} \\ \epsilon'_p \psi_{20}(r, s) &= (\omega_r + \omega_s) \psi_{20}(r, s) - \frac{\sqrt{2}\lambda}{(2\pi)^{3/2}} \int \psi_{01}(k) \delta'(k-r-s) \frac{d^3k}{2\omega'_k} \end{aligned} \quad (117)$$

$$\begin{aligned} \epsilon'_p \psi_{22}(r, s; p, k) &= (\omega_r + \omega_s + \omega'_p + \omega'_k) \psi_{22}(r, s; p, k) \\ & - \frac{\lambda}{(2\pi)^{3/2}} (\psi_{01}(p) \delta'(r+s+k) + \psi_{01}(k) \delta'(r+s+p)) \end{aligned} \quad (118)$$

Setting $\epsilon'_p = \omega'_p$ and solving (117) and (118) for ψ_{20} and

ψ_{22} gives

$$\psi_{20}(r, s) = \frac{-\sqrt{2}\lambda}{(2\pi)^{3/2}(\omega'_p - \omega_r - \omega_s)} \int \psi_{01}(k) \delta'(k-r-s) \frac{d^3k}{2\omega'_k} \quad (119)$$

$$\begin{aligned} \psi_{22}(r, s; p, k) &= \frac{\lambda}{(2\pi)^{3/2}(\omega'_p + \omega_r + \omega_s)} (\psi_{01}(p) \delta'(r+s+k) \\ & + \psi_{01}(k) \delta'(r+s+p)) \end{aligned} \quad (120)$$

Plugging these into (116) gives

$$\begin{aligned}
\epsilon'_p \psi_{o1}(p) = & (\omega'_p + \gamma_2 \lambda^2 + Z_{22} \lambda^2 \omega'_p + \frac{\mu_2 \mu^2 \lambda^2}{\omega_p}) \psi_{o1}(p) \\
& - \frac{2\lambda^2}{(2\pi)^3} \delta'(0) \psi_{o1}(p) \int \frac{\delta'(r+s+k) d'r d's d'k}{(\omega'_r + \omega'_s + \omega'_k) 8\omega_r \omega_s \omega'_k} \\
& + \frac{4\lambda^2}{(2\pi)^3} \frac{\psi_{o1}(p)}{2\omega'_p} \int \frac{\omega'_s + \omega_{p-s}}{\omega'^2_p - (\omega'_s + \omega_{p-s})^2} \frac{d's}{4\omega_s \omega_{p-s}}
\end{aligned} \tag{121}$$

Again, γ_2 cancels the second term leaving

$$Z_{22} \omega'_p + \frac{\mu_2 \mu^2}{\omega_p} = \frac{-2}{(2\pi)^3 \omega'_p} \int \frac{(\omega'_s + \omega_{p-s}) d's}{(\omega'^2_p - (\omega'_s + \omega_{p-s})^2) 4\omega_p \omega_{p-s}} \tag{122}$$

This integral is almost identical in form to (110). Again, it will only occur in this form, so we do not need to separate it.

CHAPTER 5

BOUND STATES AND RESONANCES

Renormalization constants in hand, we can go on to look at some physical systems. In doing so we make one important assumption that we will weigh only after looking at the results we can obtain under its umbrella. That is, we assume that a truncated, discretized hierarchy is a relevant approximation of the real world. This is not an obvious truth, because the equations describe bare particles, not physical particles. As we shall see, though, the reason it works is essentially the same as the reason one may calculate low order Feynman diagrams and get good results.

Here we consider bound states and resonances of a pair of m -particles. Such states are suspected to exist on the basis of the similarity of this system to Q.E.D. Their derivation from a truncated hierarchy will provide a good illustration of how to use the hierarchy to solve problems which might not be otherwise accessible.

The standard method for handling bound states is the Bethe - Saltpeter equation, which takes the form

$$(\square_1 + m_1^2)(\square_2 + m_2^2)\Psi(x_1, t_1; x_2, t_2) - I(x_1, t_1; x_2, t_2)\Psi(x_1, t_1; x_2, t_2) = 0 \quad (125)$$

for a pair of spinless particles.¹⁹

While this approach has been very important to our understanding of bound states, one cannot go much beyond a simple bound state with it. Abnormal solutions of the Bethe-Salpeter equation have been suggested to correspond to different scattering channels²⁰, but the Bethe - Saltpeter formalism provides no simple apparatus for dealing with particle creation and annihilation processes, such as the annihilation of positronium, or the annihilation of two m -particles into a pair of μ -particles in the model we are considering here.

The power of the hierarchy equations (23) to deal with multi - particle dynamics, including bound states and resonances, goes beyond the Bethe - Saltpeter formalism. This is true simply because the hierarchy is a complete set of equations for a quantized field, which can potentially be approximated to deal with any situation, at least in theory.

To order λ^2 , the hierarchy for an m -particle

19. C. Zemach, "Interpretation of Relativistic Wave Equations", Latin American School of Physics, (M. Behperad, E Ferrerira, Eds.), pp. 389-425, Gordon and Breach, New York, 1966.

20. Op. Cit., Zemach, pp. 422-424.

pair will take the form

$$i\frac{\partial\psi_{20}}{\partial t}=E\psi_{20}(p,q)=(\gamma\lambda^2+\omega_p+\omega_q+\frac{m_1m^2\lambda^2}{\omega_p}+\frac{m_2m^2\lambda^2}{\omega_q})\psi_{20}(p,q) \quad (126)$$

$$\begin{aligned} & -\frac{\sqrt{12}\lambda}{(2\pi)^{3/2}}\iiint\psi_{41}(p,q,r,u;s)\delta'(r+u+s)\frac{d'r d'u d's}{8\omega_r\omega_u\omega_s'} \\ & -\frac{2\lambda}{(2\pi)^{3/2}}\iint[\psi_{21}(p,s;r)\delta'(r+s-q)+\psi_{21}(q,s;r)\delta'(r+s-p)]\frac{d's d'r}{4\omega_s\omega_r'} \\ & -\frac{\sqrt{2}\lambda}{(2\pi)^{3/2}}\int\psi_{01}(r)\delta'(r-p-q)\frac{d'r}{2\omega_r'} \end{aligned}$$

$$i\frac{\partial\psi_{01}}{\partial t}=E\psi_{01}(r)=\omega_r'\psi_{01}(r)-\frac{\sqrt{2}\lambda}{(2\pi)^{3/2}}\iint\psi_{20}(u,v)\delta'(u+v-r)\frac{d'u d'v}{4\omega_u\omega_v} \quad (127)$$

$$i\frac{\partial\psi_{21}}{\partial t}=E\psi_{21}(p,q;r)=(\omega_p+\omega_q+\omega_r')\psi_{21}(p,q;r) \quad (128)$$

$$\begin{aligned} & -\frac{4\lambda}{(2\pi)^{3/2}}\int\psi_{02}(r,s)\delta'(s-p-q)\frac{d's}{2\omega_s'} \\ & -\frac{2\lambda}{(2\pi)^{3/2}}\int[\psi_{20}(p,s)\delta'(q+r-s) \\ & +\psi_{20}(q,s)\delta'(p+r-s)]\frac{d's}{2\omega_s'} \end{aligned}$$

$$i\frac{\partial\psi_{41}}{\partial t}=E\psi_{41}(p,q,r,u;s)=(\omega_p+\omega_q+\omega_r+\omega_u+\omega_s')\psi_{41}(p,q,r,u;s) \quad (129)$$

$$\begin{aligned} & -\frac{\lambda}{(2\pi)^{3/2}\sqrt{2}}\{\psi_{20}(p,q)\delta'(r+u+s)+\psi_{20}(p,r)\delta'(q+u+s) \\ & +\psi_{20}(p,u)\delta'(r+q+s)+\psi_{20}(r,q)\delta'(p+u+s) \\ & +\psi_{20}(u,q)\delta'(p+r+s)+\psi_{20}(r,u)\delta'(p+q+s)\} \end{aligned}$$

$$i\frac{\partial\psi_{02}}{\partial t}=E\psi_{02}(r,s)=(\omega_r'+\omega_s')\psi_{02}(r,s) \quad (130)$$

$$\begin{aligned} & -\frac{\lambda}{(2\pi)^{3/2}}\int\{\psi_{21}(p,q;r)\delta'(p+q-s) \\ & +\psi_{21}(p,q,s)\delta'(p+r-s)\}\frac{d'p d'q}{4\omega_p\omega_q} \end{aligned}$$

where terms containing ψ_{02} have been included to account for the possible decay of the pair of m -particles into a pair of μ -particles. ψ_{02} will contribute only a λ^4 term to the

interaction, which is neglected here. Solving (127), (128) and (129) for ψ_{01} , ψ_{21} , and ψ_{41} , and substituting into (126) will give an equation for ψ_{20} . However, since the energy E is unknown, one must write

$$E = \epsilon_0 + \epsilon_2 \lambda^2 + \epsilon_4 \lambda^4 + \dots \quad (131)$$

where clearly

$$\epsilon_0 = \omega_p + \omega_q \quad (132)$$

On substitution, one obtains, to order λ^2 ,

$$E\psi_{20}(p, q) = (\omega_q + \omega_p)\psi_{20}(p, q) + \left(\gamma_2 + \frac{m_1 m^2}{\omega_p} + \frac{m_2 m^2}{\omega_q} \right) \lambda^2 \psi_{20}(p, q) \quad (133)$$

$$\begin{aligned} & + \frac{2\lambda^2}{(2\pi)^3} \iint \frac{\psi_{20}(u, v) \delta'(u+v-p-q)}{\omega_p + \omega_q - \omega'_{p+q}} \frac{d'u d'v}{8\omega_u \omega_v \omega'_{u+v}} \\ & + \frac{2\lambda^2}{(2\pi)^3} \iiint \left(\frac{\psi_{20}(p, u) \delta'(s+r-q) \delta'(r+s-u)}{\omega_q - \omega_s - \omega'_r} \right. \\ & \quad + \frac{\psi_{20}(s, u) \delta'(p+r-u) \delta'(r+s-q)}{\omega_q - \omega_s - \omega'_r} \\ & \quad + \frac{\psi_{20}(q, u) \delta'(s+r-u) \delta'(r+s-p)}{\omega_p - \omega_s - \omega'_r} \\ & \quad \left. + \frac{\psi_{20}(s, u) \delta'(q+r-u) \delta'(r+s-p)}{\omega_p - \omega_s - \omega'_r} \right) \frac{d'r d'u d's}{8\omega_s \omega_u \omega'_r} \\ & + \frac{2\lambda^2}{(2\pi)^3} \iiint \frac{-1}{\omega_r + \omega_k + \omega'_s} \{ \psi_{20}(p, q) \delta'(r+u+s) \\ & \quad + \psi_{20}(p, r) \delta'(q+u+s) + \psi_{20}(p, u) \delta'(r+q+s) \\ & \quad + \psi_{20}(r, q) \delta'(p+u+s) + \psi_{20}(u, q) \delta'(r+p+s) \\ & \quad + \psi_{20}(r, u) \delta'(p+q+s) \} \delta'(r+u+s) \frac{d'r d'u d's}{8\omega_r \omega_u \omega'_s} \\ & + \frac{4\lambda^2}{(2\pi)^3} \iiint \frac{\psi_{02}(r, u) \delta'(u-p-s) \delta'(r+s-q)}{\omega_q - \omega_s - \omega'_r} \end{aligned}$$

$$+\frac{\psi_{20}(r,u)\delta'(u-q-s)\delta'(r+s-p)}{\omega_p-\omega_s-\omega'_r}\bigg]\frac{d'sd'u d'r}{8\omega_s\omega'_u\omega'_r}$$

Rewriting,

$$E\psi_{20}(p,q)=(\omega_p+\omega_q)\psi_{20}(p,q) \quad (134)$$

$$\begin{aligned} & +\gamma_2\lambda^2\psi_{20}(p,q)-\frac{2\lambda^2}{(2\pi)^2}\delta'(0)\psi_{20}(p,q)\iint\frac{d'r d's}{(\omega_r+\omega_s+\omega'_{r+s})8\omega_r\omega_s\omega'_{r+s}} \\ & +\frac{m_2m^2\lambda^2}{\omega_p}\psi_{20}(p,q) \\ & -\frac{2\lambda^2}{(2\pi)^2\omega_p}\psi_{20}(p,q)\int\left[\frac{1}{2\omega'_{p-s}(\omega_p-\omega_s-\omega'_{p-s})}-\frac{1}{2\omega'_{p+s}(\omega_p+\omega_s+\omega'_{p+s})}\right]\frac{d's}{2\omega_s} \\ & +\frac{m_2m^2\lambda^2}{\omega_q}\psi_{20}(p,q) \\ & -\frac{2\lambda^2}{(2\pi)^2\omega_q}\psi_{20}(p,q)\int\left[\frac{1}{2\omega'_{q-s}(\omega_q-\omega_s-\omega'_{q-s})}-\frac{1}{2\omega'_{q+s}(\omega_q+\omega_s+\omega'_{q+s})}\right]\frac{d's}{2\omega_s} \\ & +\frac{2\lambda^2}{(2\pi)^2}\iint\left[\frac{1}{\omega_p+\omega_q-\omega'_{p+q}}-\frac{1}{\omega_r+\omega_s+\omega'_{r+s}}\right]\psi_{20}(r,s)\delta'(p+q-r-s)\frac{d'r d's}{8\omega_r\omega_s\omega'_{r+s}} \\ & +\frac{4\lambda^2}{(2\pi)^2}\iint\left[\frac{1}{2\omega'_{q-s}(\omega_q-\omega_s-\omega'_{q-s})}+\frac{1}{2\omega'_{p-s}(\omega_p-\omega_s-\omega'_{p-s})}\right]\times \\ & \quad \psi_{20}(r,s)\delta'(p+q-r-s)\frac{d'r d's}{4\omega_r\omega_s} \\ & +\frac{4\lambda^2}{(2\pi)^2}\iint\left[\frac{1}{2\omega_{q-r}(\omega_q-\omega_{q-r}-\omega'_r)}+\frac{1}{2\omega_{p-r}(\omega_p-\omega'_r-\omega_{p-r})}\right]\times \\ & \quad \psi_{02}(r,s)\delta'(p+q-r-s)\frac{d'r d's}{4\omega'_r\omega'_s} \end{aligned}$$

Substituting γ_2 and m_2 into (134) cancels all vacuum and self energy terms, leaving

$$E\psi_{20}(p,q) = (\omega_p + \omega_q)\psi_{20}(p,q) \quad (135)$$

$$\begin{aligned} & + \frac{2\lambda^2}{(2\pi)^2} \iint \left[\frac{1}{\omega_p + \omega_q - \omega_{p+q}} - \frac{1}{\omega_r + \omega_s + \omega_{r+s}} \right] \psi_{20}(r,s) \delta'(p+q-r-s) \frac{d'r d's}{8\omega_r \omega_s \omega_{r+s}} \\ & + \frac{4\lambda^2}{(2\pi)^2} \iint \left[\frac{1}{2\omega_{q-r}(\omega_q - \omega_s - \omega_{q-s})} + \frac{1}{2\omega_{p-s}(\omega_p - \omega_s - \omega_{p-s})} \right] \times \\ & \quad \psi_{20}(r,s) \delta'(p+q-r-s) \frac{d'r d's}{4\omega_r \omega_s} \\ & + \frac{4\lambda^2}{(2\pi)^2} \iint \left[\frac{1}{2\omega_{q-r}(\omega_q - \omega_r - \omega_{q-r})} + \frac{1}{2\omega_{p-r}(\omega_p - \omega_r - \omega_{p-r})} \right] \times \\ & \quad \psi_{02}(r,s) \delta'(p+q-r-s) \frac{d'r d's}{4\omega_r \omega_s} \end{aligned}$$

In a similar manner, one may derive an equation for ψ_{02} by solving (128) and substituting it into (130). The result is

$$\begin{aligned} i\frac{\partial \psi_{02}}{\partial t} = E\psi_{02}(p,q) & = (\omega'_p + \omega'_q)\psi_{02}(p,q) \quad (136) \\ & + \frac{4\lambda^2}{(2\pi)^2} \iint \left[\frac{1}{2\omega_{q-r}(\omega_p + \omega_q - \omega_r - \omega_{q-r} - \omega'_p)} + \frac{1}{\omega_{p-r}(\omega_p + \omega_q - \omega_r - \omega_{p-r} - \omega'_q)} \right] \times \\ & \quad \psi_{20}(r,s) \delta'(p+q-r-s) \frac{d'r d's}{4\omega_r \omega_s} \end{aligned}$$

to order λ^2 . Equations (135) and (136) form a coupled pair of equations in momentum space which are accurate as long as λ is small enough and ψ_{02} remains of order λ^2 . These equations are too complex to solve outright, although numerical approximation techniques may prove useful in solving them. Here we will consider only the non-relativistic limit where

$$\lambda^2 \frac{p^2}{m} \ll \frac{p^2}{m} \ll m \quad (137)$$

but where, at least to start with, we do not assume μ is much greater than the characteristic momentum of the system. If we drop the ψ_{02} term in eq. (135) because it is of fourth order, eq. (135) then becomes

$$\begin{aligned} E\psi_{20}(p,q) = & \frac{p^2+q^2}{2m} \psi_{20}(p,q) \\ & + \frac{2\lambda^2}{(2\pi)^4 4m^2(4m^2-\mu^2)} \int \psi_{20}(r,s) \delta'(r+s-p-q) d'r d's \\ & - \frac{4\lambda^2}{(2\pi)^4 8m^2} \int \left[\frac{1}{\mu^2+(q-s)^2} + \frac{1}{\mu^2+(p-s)^2} \right] \psi_{20}(r,s) \delta'(r+s-p-q) d's d'r \end{aligned} \quad (138)$$

where $2m$ was absorbed into E . Next, changing to the center of mass variables,

$$\begin{aligned} P &= p+q & Q &= p-q \\ R &= r+s & S &= r-s \end{aligned} \quad (139)$$

gives

$$\begin{aligned} E\psi_{20}(P,Q) = & \left(\frac{P^2}{4M} + \frac{Q^2}{M} \right) \psi_{20}(P,Q) \\ & + \frac{\lambda^2}{(2\pi)^4 2m^2(4m^2-\mu^2)} \int \psi_{20}(P,S) d'S \\ & - \frac{2\lambda^2}{(2\pi)^4 2m^2} \int \left[\frac{1}{\mu^2+(S+Q)^2} \right] \psi_{20}(P,S) d'S \end{aligned} \quad (140)$$

Now one may separate variables by setting

$$\psi_{20}(P,Q) = \delta'(P-P_0) \psi(Q) \quad (141)$$

$$E = \frac{P_0^2}{4m} + \epsilon \quad (142)$$

where ψ obeys

$$\epsilon\psi(Q) = \frac{Q^2}{m}\psi(Q) + \frac{\lambda^2}{(2\pi)^3 2m^2(4m^2 - \mu^2)} \int \psi(S) d^3S - \frac{\lambda^2}{(2\pi)^3 m^2} \int \frac{\psi(S)}{\mu^2 + (Q+S)^2} d^3S$$

Fourier transforming gives a standard Schrodinger equation

$$\epsilon\psi(r) = -\frac{1}{m} \nabla^2 \psi(r) + \frac{\lambda^2}{2m^2(4m^2 - \mu^2)} \delta^3(r) \psi(r) + V(r) \psi(r) \quad (144)$$

where

$$V(r) = \frac{-\lambda^2}{(2\pi)^3 m^2} \int \frac{e^{ik \cdot r}}{\mu^2 + k^2} d^3k \quad (145)$$

The potential V depends on the dimension σ . For $\sigma=1$,

$$V(r) = \frac{-\lambda^2}{2m^2 \mu} e^{-\mu|r|} \quad (146)$$

and for $\sigma=3$,

$$V(r) = \frac{-\lambda^2}{4\pi m^2 |r|} e^{-\mu|r|} \quad (147)$$

which is simply a Yukawa potential.

Clearly this equation will have bound states, provided μ is not too close to $2m$. When $\mu=2m$, the pair annihilation channel becomes infinite, so there is no bound state.

In the non - relativistic limit where one further assumes μ is much greater than the characteristic momentum of the system, the Yukawa potential can be approximated as a δ -function,

$$V(r) \approx \frac{-\lambda^2}{(2\pi)^3 m^2 \mu^2} \int e^{ik \cdot r} d^3k = \frac{-\lambda^2}{m^2 \mu^2} \delta^3(r) \quad (148)$$

whereby equation (144) becomes

$$\epsilon\psi(r) = -\frac{1}{m} \nabla^2 \psi(r) + \left(\frac{1}{2m^2(4m^2 - \mu^2)} - \frac{1}{m^2 \mu^2} \right) \lambda^2 \delta^3(r) \psi(r) \quad (149)$$

which is just a standard δ -function potential, the solutions of which are well known. In one dimension,

there is one bound state when the coefficient of $\delta(r)$ is negative, for which ψ takes the form

$$\psi(r) = Ae^{-a|r|} \quad a = \sqrt{-mE_b} \quad (150)$$

and the binding energy is

$$E_b = -\frac{m\lambda^4}{4} \left(\frac{1}{2m^2(4m^2 - \mu^2)} - \frac{1}{m^2\mu^2} \right) \quad (151)$$

The bound state wavefunction from equation (144) with $m=1$, $\mu=1.25$ and $\lambda=0.3$ is depicted in Fig. 5, while a comparison between the states of (144) and (149) is depicted in Fig.

6. The computer program used to derive these results and the data it produced is detailed in Appendix A.

In neglecting the coupling to ψ_{02} , in eq (135), we have ignored the possibility of the decay $2m \rightarrow 2\mu$, which one expects when $\mu < m$. Let us include it in the non-relativistic limit in which all characteristic momenta are much less than μ . Then equations (135) and (136) become

$$E\psi_{20}(p, q) = \left(2m + \frac{p^2 + q^2}{2m} \right) \psi_{20}(p, q) \quad (152)$$

$$+ \left(\frac{2\lambda^2}{(2\pi)^4 m^2 (4m^2 - \mu^2)} - \frac{4\lambda^2}{(2\pi)^4 m^2 \mu^2} \right) \int \psi_{20}(r, s) \delta'(p + q - r - s) d^3r d^3s$$

$$- \frac{\lambda^2}{(2\pi)^4 m \mu} \int \psi_{02}(r, s) \delta'(r + s - p - q) d^3r d^3s$$

and

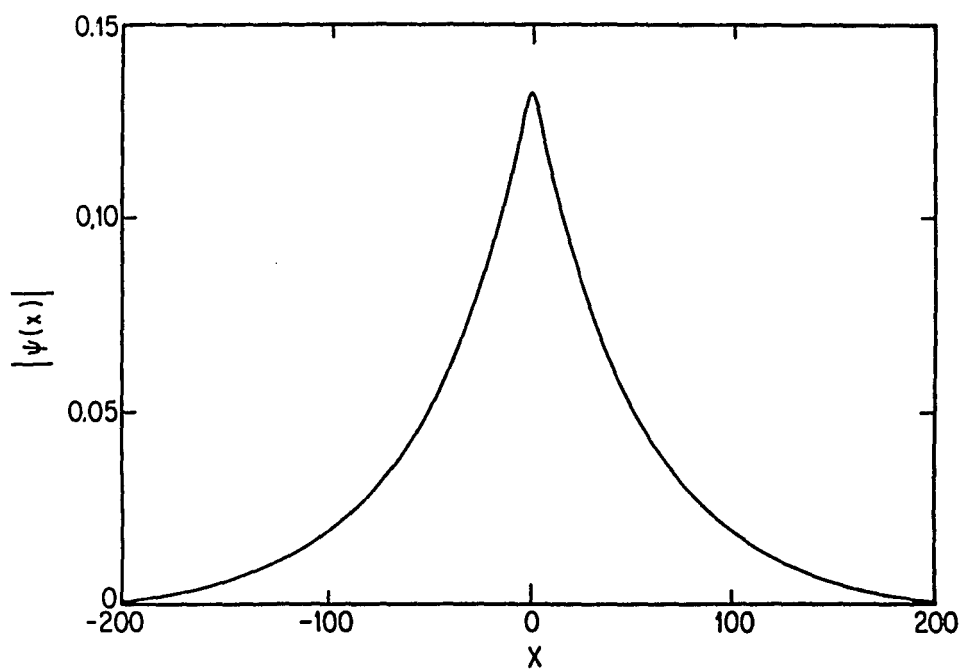


Figure 5: A typical bound state wavefunction.

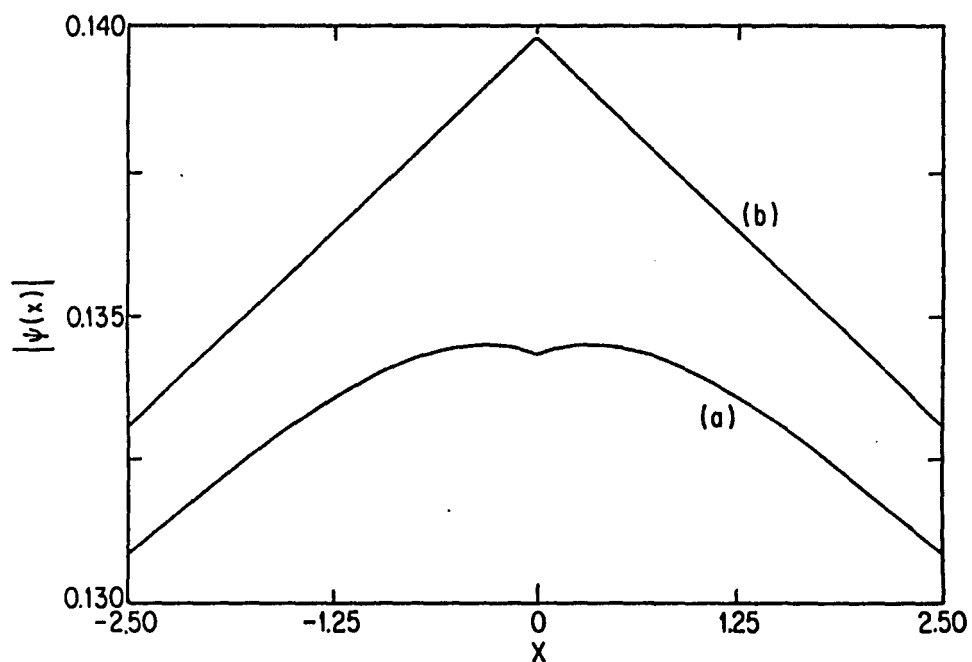


Figure 6: Comparison of bound state wavefunctions from solving (a) Eq. (144) and (b) Eq. (149)

$$E\psi_{02}(p, q) = (2\mu + \frac{p^2 + q^2}{2\mu})\psi_{02}(p, q) \quad (153)$$

$$-\frac{\lambda^2}{(2\pi)^3 m^3 \mu} \int \psi_{20}(r, s) \delta'(r+s-p-q) d'r d's$$

Transforming to center of mass coordinates and separating variables by writing

$$\psi_{20}(P, Q) = \delta'(P - P_0) \psi(Q) \quad (154)$$

$$\psi_{02}(P, Q) = \delta'(P - P_0) \varphi(Q) \quad (155)$$

$$E = 2m + \frac{P_0^2}{4m} + \epsilon \quad (156)$$

gives

$$\epsilon \psi(Q) = \frac{Q^2}{m} \psi(Q) + \frac{2\lambda^2}{(2\pi)^3 4m^2(4m^2 - \mu^2)} \int \psi(S) d'S \quad (157)$$

$$-\frac{\lambda^2}{(2\pi)^3 m^2 \mu^2} \int \psi(S) d'S - \frac{\lambda^2}{(2\pi)^3 m^3 \mu} \int \varphi(S) d'S$$

$$(\epsilon + 2m + \frac{P_0^2}{4m} - 2\mu - \frac{P_0^2}{4\mu}) \varphi(Q) = \frac{Q^2}{\mu} \varphi(Q) - \frac{\lambda^2}{(2\pi)^3 m^3 \mu} \int \psi(S) d'S \quad (158)$$

These Fourier transform into

$$\epsilon \psi(r) = -\frac{1}{m} \nabla^2 \psi(r) + [\frac{1}{2m^2(4m^2 - \mu^2)} - \frac{1}{m^2 \mu^2}] \lambda^2 \delta'(r) \psi(r) - \frac{\lambda^2}{m\mu} \delta'(r) \varphi(r) \quad (159)$$

$$\epsilon \varphi(r) = [2\mu + \frac{P_0^2}{4\mu} - 2m - \frac{P_0^2}{4m}] - \frac{1}{\mu} \nabla^2 \varphi(r) - \frac{\lambda^2}{m^3 \mu} \delta'(r) \psi(r) \quad (160)$$

These equations can be solved exactly. In one dimension, write

$$\psi(r) = A e^{-a|r|} \quad (161)$$

$$\varphi(r) = B e^{-b|r|} \quad (162)$$

For $r=0$, one must have

$$\epsilon = -\frac{a^2}{m} \quad (163)$$

$$b = \sqrt{\mu \left(2\mu + \frac{P_a^2}{4\mu} - 2m - \frac{P_a^2}{4m} - \epsilon \right)} \quad (164)$$

and for $r=0$ one obtains the equations

$$0 = \left(\frac{2a}{m} - a \right) A - \beta B \quad (165)$$

$$0 = \frac{2b}{\mu} B - \gamma A \quad (166)$$

where

$$\alpha = \lambda^2 \left(\frac{1}{m^2 \mu^2} - \frac{1}{2m^2(4m^2 - \mu^2)} \right) \quad (167)$$

$$\beta = \frac{\lambda^2}{m\mu^2} \quad \gamma = \frac{\lambda}{m^2 \mu}$$

Combining (163) to (167) with

$$\frac{A^2}{a^2} + \frac{B^2}{b^2} = 1 \quad (168)$$

one may solve for A, B, a and b to obtain the solution.

One obtains a correction of order $\epsilon \lambda^2$ to the value obtained using equation (149).

Of prime importance is the fact that b is real only if

$$2\mu + \frac{P_a^2}{4\mu} > 2m + \frac{P_a^2}{4m} + \epsilon \quad (169)$$

That is, if the bound state has less energy than a pair of free μ -particles. If condition (169) does not hold, b is purely imaginary. Then there must be a flux of μ -particles to and from infinity, whereby one must interpret the m-particle pair as a resonance. Thus, equation (169) provides the stability condition for the bound state.

The time dependent form of equations (159) and (160) provide a simple model of the decay process $2m \rightarrow 2\mu$

when condition (169) is not fulfilled. This system was modeled numerically to show a typical decay, using the parameters $m=1$, $\mu=0.75$, and $\lambda=0.3$. The results are depicted in Fig. 7, which show the decay of ψ into φ , and the subsequent dispersal of the μ -particle pair. The program used to do this calculation is listed in Appendix B. Note that the figure does not correspond to the data output by the program. The program is correct.

This analysis should illustrate the power of the hierarchy techniques discussed herein for deriving approximate equations of motion for complex systems. The possible variety is practically unlimited, depending only on the approximations one wishes to make and the trouble one wishes to go to to solve such equations. All of this is predicated, of course, upon the validity of using a truncated hierarchy.

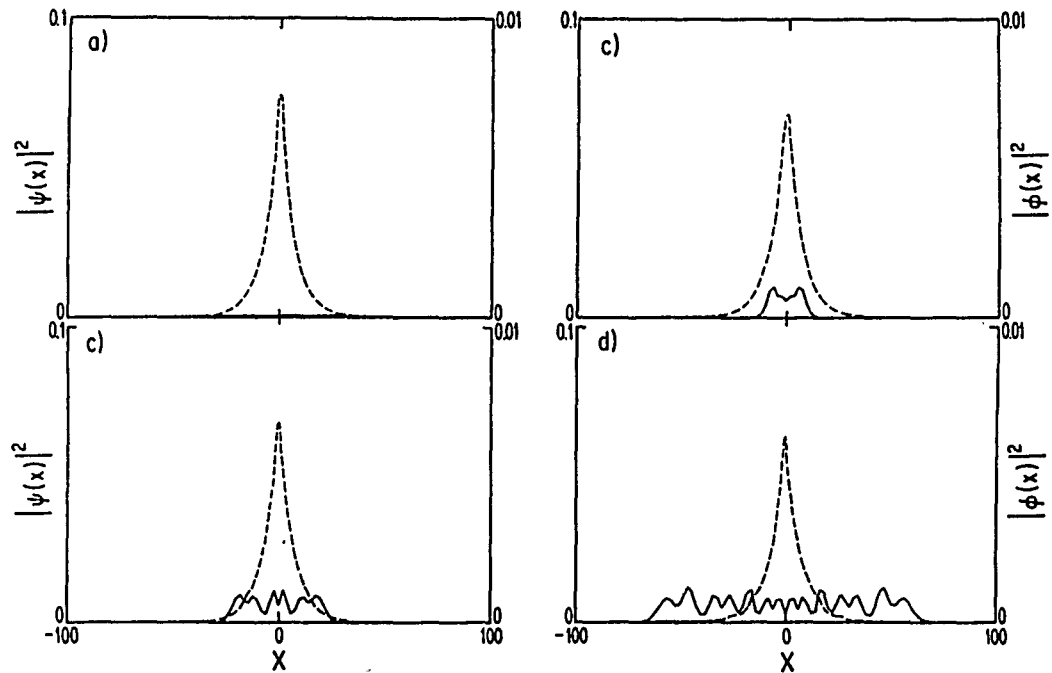


Figure 7: The decay of a bound state when it is unstable.

ψ is represented by a dotted line and ϕ by a solid line.

(a) $t=0$, (b) $t=10$, (c) $t=20$, (d) $t=50$

CHAPTER 6

NUMERICAL SOLUTIONS

A considerable amount of numerical work in quantum field theory has been done in the last decade within the context of lattice gauge theories²¹, and lattice Hamiltonian field theories²². The usual problem with this type of numerical analysis is that the amount of data which must be stored in computer memory, and the number of calculations required in a computation quickly becomes unmanagable because the wavefunction is a complex functional of the field configuration²³. Typically, for a system of M particles on p coordinate points, one needs

$$W_1 \approx M^p \quad (170)$$

words of memory, since one must have at least M field values for each point in space, and the wavefunctional $\Psi(\phi(x))$ must assign an amplitude to each possible combination. Although there are a number of methods

 21 - Claudio Rebbi, Editor, "Lattice Gauge Theories and Monte Carlo Simulations" pp. 23-37, Interscience, New York, 1983.

22 - T. Barnes, G. Daniell, Phys. Rev. D, 28 (1983) 2045-2058.

23 - R.C. Brower, Discrete QCD, in "Gauge Theories in High Energy Physics, Les Houches, Session 37, August 3, 1981 to September 11, 1981" (M. Gailard, R. Stora, Eds.) p. 585, North Holland, New York, 1983.

available to reduce these requirements to some extent²⁴, computation remains a formidable obstacle to treating complex problems, especially when it comes to particle dynamics.

The truncation and discretization of a hierarchy equation of the type we have been discussing presents the possibility of avoiding this explosive growth of required computing capacity. (Which is why it was attempted in the first place.) Since a truncation of the hierarchy at M particles involves a wavefunction with M variables, it allows one to represent a system of M particles on P space points with

$$W_2 \approx P^M \quad (171)$$

words of memory. In many cases, one is interested in the behavior of a relatively small number of interacting particles, so M is small and P is large. In such situations, $W_1 \ll W_2$. For example, to model 3 particles on a grid of 32 space points, $W_1 \approx 2 \times 10^{15}$ and $W_2 \approx 3 \times 10^4$. While the functional approach utilizes a much larger space of states, most of these states are not necessary to provide an accurate approximation in many instances.

24 - M. Kovarik, J. Darewych, R. Koniuk, Phys. Rev. D, 33 (1986) 3654-3657.

Accordingly, some attempts were made to do some numerical calculations with the hierarchy (23). All of this work was done in 1+1 dimensions, mostly because of computer limitations.

A number of techniques had to be developed in order to make the hierarchy amenable to numerical approximation, some of which are worth noting:

Firstly, the hierarchy must be truncated symmetrically, or the Hamiltonian of the truncated system will not be Hermitian. To truncate symmetrically, one must simply make sure that if an interaction

$$\psi_{\alpha\beta} \longrightarrow \psi_{\gamma\delta} \quad (172)$$

is included in the system, then the reverse interaction,

$$\psi_{\gamma\delta} \longrightarrow \psi_{\alpha\beta} \quad (173)$$

is also included.

Secondly, useful values of the grid size Δk , the number of grid points N , and the time step Δt had to be found. One must be sure that all of the momenta relevant to a problem are taken into account, or channels for likely reactions will be artificially closed. Additionally, Δk must be chosen small enough so that the numerical approximations will not blow up. This is further complicated by the uncertainty principle,

because the grid must be large enough to accomodate particles in both k-space and coordinate space. In a unitless system with $\hbar=c=1$ and masses close to unity, I found a value of $\Delta k=2\pi/16$ and $N=16$ the minimum for doing some interesting problems.

Next, one must be careful how initial states are set up in any computer simulated experiments. This is non-trivial because one must initialize a state as a linear combination of physical particle states, and not bare particle states. If one uses bare particles, they are not stable.

Next, the system had to be renormalized numerically, so that physical particle masses were what we expected them to be. This was accomplished here using a feedback loop which guessed at initial values of the bare masses m_0 and μ_0 , calculated the energy eigenvalues for the single particle states and the vacuum, calculated the physical masses m and μ , and went back to guess (using Newton's method) at new m_0 and μ_0 , continuing this process until m and μ were within a prescribed tolerance.

After some of these details were worked out, a number of simple tests were carried out to make sure that the system was working as expected. These included tests

of energy and norm conservation, tests of the stability of physical particles, and tests of the propagation of wave packets. All of these tests seemed to produce the expected results once the bugs in the programming and in my thinking were worked out. Then, several more serious tests of the numerical methods were done, two of which I detail here.

Firstly, the physical states of the system were compared with the predictions of the perturbation theory developed in Chapter 4. For a given number of grid points and a given grid size Δk , the vacuum correction term γ_2 and the mass corrections m_2 and μ_2 could be compared with the numerical results from a hierarchy truncated to include ψ_{00} , ψ_{21} , ψ_{01} , ψ_{20} , ψ_{11} , ψ_{31} , and ψ_{22} . The program which accomplished this is listed and commented on in Appendix C.

For $m=\mu=1$, one obtains, from (102), (110) and

(122),

$$\gamma_2 = \frac{1}{8\pi\Delta k} \sum_{i,j} \frac{(\Delta k)^2}{(\omega_{k_i} + \omega_{k_j} + \omega_{k_i+k_j})\omega_{k_i}\omega_{k_j}\omega_{k_i+k_j}} \quad (174)$$

$$m_2 = \frac{1}{\pi} \sum_j \frac{\Delta k}{(1-4\omega_{k_j}^2)\omega_{k_j}}$$

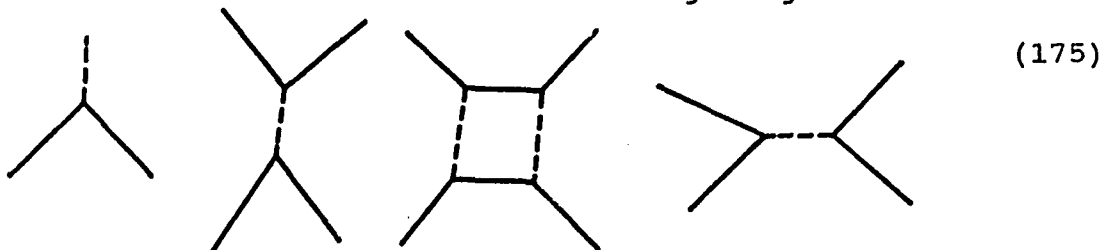
$$\mu_2 = \frac{1}{2} m_2$$

The numerical results approximated these perturbative

constants very accurately (i.e. to order λ^4) for small values of λ , as illustrated in Table I for $\Delta k = 2\pi/16$ on 16 grid points, and detailed more thoroughly in Appendix C.

The second and more exciting test of the numerical work was a scattering experiment in which two m-particles collided and the byproducts were examined. This type of a calculation is a good illustration of how the hierarchy can be used to do problems inaccessible by traditional techniques, being the first numerical attempt at a quantized field dynamics problem. The hierarchy used in this calculation was truncated to include ψ_{00} , ψ_{01} , ψ_{10} , ψ_{20} , ψ_{11} , ψ_{02} , ψ_{30} , ψ_{21} , and ψ_{12} . Appendix D provides a listing of the program, MESN2A, used to do these calculations.

In the scattering of two m-particles, relevant processes would include the following diagrams:



the first usually being disallowed by energy-momentum conservation. We are particularly interested in particle creation events here, as the hierarchy should be able to

Table I : Comparison of Perturbation Theory with Numerical Calculations. Vacuum energy, and mass corrections.

LAMDA	Numerical Calculation			Perturbation Calculation		
	E VAC	M	MU	E VAC	M	MU
0.10	-.1771E-02	.10018E+01	.10009E+01	-.1779E-02	.10018E+01	.10009E+01
0.20	-.7002E-02	.10071E+01	.10035E+01	-.7117E-02	.10074E+01	.10036E+01
0.30	-.1545E-01	.10155E+01	.10077E+01	-.1601E-01	.10166E+01	.10083E+01
0.40	-.2681E-01	.10265E+01	.10132E+01	-.2847E-01	.10296E+01	.10148E+01
0.50	-.4070E-01	.10395E+01	.10197E+01	-.4448E-01	.10462E+01	.10231E+01
0.60	-.5676E-01	.10539E+01	.10268E+01	-.6405E-01	.10665E+01	.10333E+01
0.70	-.7468E-01	.10693E+01	.10345E+01	-.8718E-01	.10906E+01	.10453E+01
0.80	-.9417E-01	.10853E+01	.10425E+01	-.1139E+00	.11183E+01	.10592E+01
0.90	-.1149E+00	.11017E+01	.10507E+01	-.1441E+00	.11496E+01	.10748E+01
1.00	-.1369E+00	.11184E+01	.10590E+01	-.1779E+00	.11848E+01	.10924E+01
1.10	-.1598E+00	.11350E+01	.10674E+01	-.2153E+00	.12236E+01	.11118E+01
1.20	-.1836E+00	.11516E+01	.10757E+01	-.2562E+00	.12661E+01	.11331E+01
1.30	-.2080E+00	.11681E+01	.10840E+01	-.3007E+00	.13123E+01	.11562E+01
1.40	-.2331E+00	.11843E+01	.10922E+01	-.3487E+00	.13622E+01	.11811E+01
1.50	-.2587E+00	.12004E+01	.11003E+01	-.4004E+00	.14158E+01	.12079E+01

handle them easily, and they are a fairly clean test for it.

Figure 8 shows the progress of the m-particle wave packets as a function of time, and figure 9 shows the probabilities of finding one and two μ -particles as a function of time with $m=\mu=1$ and $\lambda=0.1$. Figure 9 exhibits the qualitative behavior one would expect, the single particle exchange being much larger than the two particle exchange, but going to zero when the m-particles are not in close proximity.

The results were first checked for convergence as the time increment Δt used to advance the system in time using

$$|\Psi(t+\Delta t)\rangle = |\Psi(t)\rangle - i\Delta t H |\Psi(t)\rangle \quad (175)$$

was allowed to go to zero. Table II lists the maximum probability for finding one μ -particle, P_{01} , and the final probabilities for finding one and two μ -particles, P_{01} and P_{02} , as a function of Δt . The convergence as $\Delta t \rightarrow 0$ is clear.

The scattering problem was worked on lattices of $N=16, 24$ and 32 points (See Table III). The solutions appeared to converge rapidly, with the results on 24 and 32 points differing by less than 0.5% . The 16 point

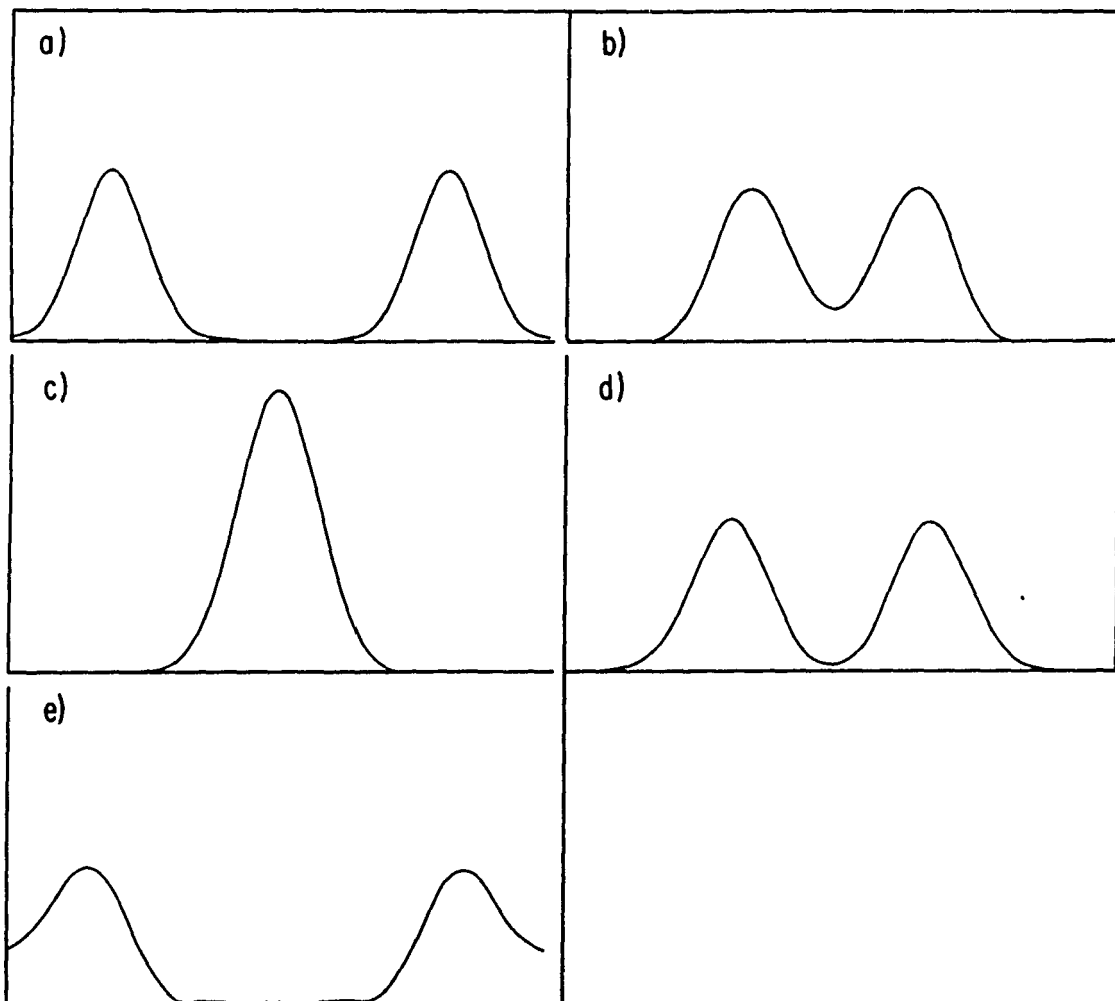


Figure 8: Particle configuration in the scattering simulation as a function of time. (a) $t=0$, (b) $t=3$, (c) $t=6$, (d) $t=9$, (e) $t=12$.

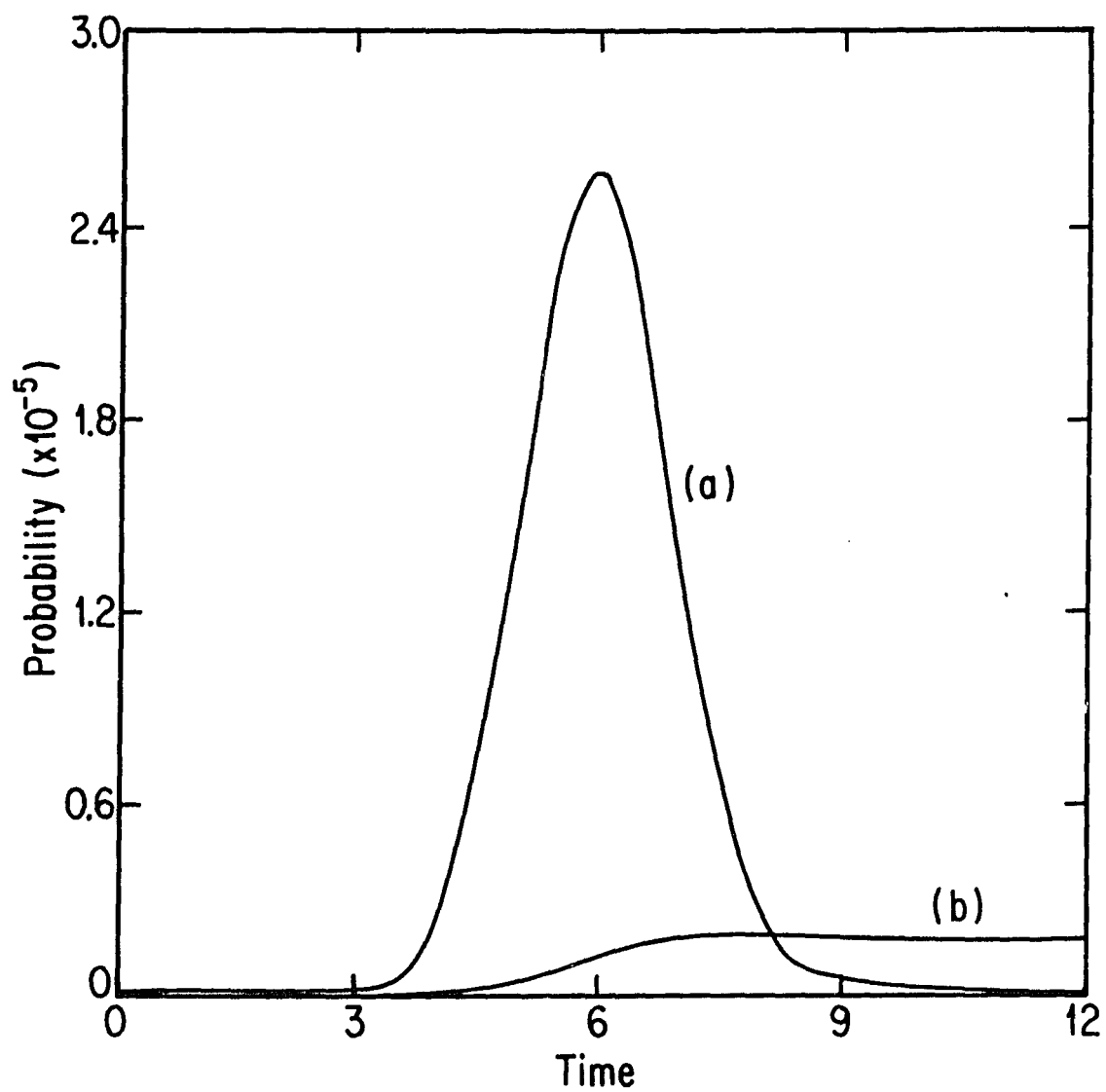


Figure 9: Probability of finding one and two created particles as a function of time. (a) One μ -particle, (b) Two μ -particles.

Table II : Convergence of scattering results as a function of time increment for $\lambda=0.1$, $N=16$ and $\Delta k=2\pi/16$.

$$\Delta E = (E(12)-E(0))/E(0).$$

Δt	ΔE		P01(T=6)	P01(T=11)	P02(T=11)
.02	22	%	.17667E-04	.15707E-06	.11242E-05
.01	10	%	.22766E-04	.11500E-06	.14858E-05
.005	4.8	%	.25516E-04	.13853E-06	.17888E-05
.0025	2.3	%	.26924E-04	.17226E-06	.19782E-05
.00125	1.2	%	.27634E-04	.19647E-06	.20829E-05
.000625	0.57	%	.27990E-04	.21069E-06	.21378E-05
--> 0	0	%	.2835E-04	.23E-06	.219E-05

Table III : Convergence of scattering results as a function of grid size, for $\lambda=0.1$, $\Delta k=2\pi/N$.

N	ΔE	ΔT	P01(T=6)	P01(T=11)	P02(T=11)
16	2.3%	.0025	.269240E-04	.172259E-06	.197816E-05
24	2.4%	.0025	.280152E-04	.124604E-06	.201411E-05
32	2.4%	.0025	.280205E-04	.124600E-06	.202329E-05
16	1.1%	.00125	.276340E-04	.196470E-06	.208285E-05
24	1.1%	.00125	.286832E-04	.148757E-06	.213710E-05
32	1.2%	.00125	.286887E-04	.148762E-06	.214548E-05
16	-->0	-->0	.2834E-04	.2207E-06	.2188E-05
24	-->0	-->0	.2935E-04	.1729E-06	.2260E-05
23	-->0	-->0	.2936E-04	.1729E-06	.2268E-05

results were only accurate to a few percent, probably because a small portion of the high momentum components were cut off. Modeling the system on more than 32 points was impossible with this program due to computational limitations, while uncertainty principle limits would not allow the problem to be modeled with less than 16 points.

A perturbative calculation of the final probability of finding two μ -particles in this scattering experiment gives, with $m=\mu=1$ and $\lambda=0.1$, (Appendix E)

$$P_{02}^{\text{pert}} = 2.309 \times 10^{-6} + O(\lambda^6) \quad (176)$$

as compared with

$$P_{02}^{\text{Numerical}} = 2.268 \times 10^{-6} \quad (177)$$

for 32 lattice points in the $\Delta t \rightarrow 0$ limit (See Table III), a difference of only 1.7% $\approx \lambda$. These results look fairly good. Of course, it is difficult to test them at larger values of λ as the perturbation theory loses accuracy and we have nothing to compare the results to.

CHAPTER 7

CONVERGENCE OF THE HIERARCHY

So far, we have seen that standard perturbation theory can be derived from the hierarchy equations and that accurate numerical work can be done with a truncated, discretized hierarchy. We have yet to address the convergence of this numerical approach. This is a very complex question which cannot be answered analytically without the invention of some new form of mathematical limit-taking process. While such mathematical work is beyond the scope of this paper, we can examine and understand the problems involved.

To do this, consider a model field theory which can be solved exactly,

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \varphi \partial^\mu \varphi - \varphi^2 - \mu^2 \varphi^2) \quad (178)$$

which describes a free scalar field of mass $(1+\mu^2)^{1/2}$.

If the mass term $\frac{1}{2}\mu^2 \varphi^2$ is treated as a perturbation, the perturbation theory will converge only for $|\mu| < 1$ because it involves the expansion of roots of the form $(1+\mu^2)^{1/2}$. Using the techniques of Chapter 2, one can derive a hierarchy for this model. The hierarchy is given by

$$\begin{aligned}
i\frac{\partial \psi_N}{\partial t}(q_1, \dots, q_N) = & \sum_{j=1}^N \left(1 + \frac{\mu^2}{2\omega_{q_j}^2}\right) \omega_{q_j} \psi_N(q_1, \dots, q_N) \\
& + \frac{\mu^2}{2} (N+2)^{1/2} (N+1)^{1/2} \int \psi_{N+2}(q_1, \dots, q_N, k, p) \delta(k+p) \frac{dk dp}{4\omega_k \omega_p} \\
& + \frac{\mu^2}{2} (N-1)^{1/2} N^{1/2} \sum_{\substack{i,j=1 \\ i \neq j}}^N \psi_{N-2}(q_1, \dots, \hat{q}_i, \dots, \hat{q}_j, \dots, q_N) \delta(q_i + q_j)
\end{aligned} \tag{179}$$

It can be truncated and analyzed numerically, and compared with exact results and with perturbation theory. Here we truncate the theory at one, three, and five particles and solved the eigenvalue problem for the physical vacuum and for the single particle, $k=0$ state to calculate the mass of a physical particle. We can examine this mass prediction in a variety of ways to better understand how the truncation of the hierarchy works.

The five particle truncated hierarchy is given by

$$i\frac{\partial \psi_0}{\partial t} = \frac{\mu^2}{\sqrt{2}} \int \psi_2(k, p) \delta(k+p) \frac{dk dp}{4\omega_k \omega_p} \tag{180a}$$

$$i\frac{\partial \psi_2}{\partial t} = \left[\omega_k + \frac{\mu^2}{2\omega_k} + \omega_p + \frac{\mu^2}{2\omega_p} \right] \psi_2(k, p) + \frac{\mu^2}{\sqrt{2}} \delta(k+p) \psi_0 \tag{180b}$$

$$+ \sqrt{3} \mu^2 \int \psi_4(k, p, q, s) \delta(q+s) \frac{dq ds}{4\omega_q \omega_s}$$

$$i\frac{\partial \psi_4}{\partial t} = \left[\omega_k + \frac{\mu^2}{2\omega_k} + \omega_p + \frac{\mu^2}{2\omega_p} + \omega_q + \frac{\mu^2}{2\omega_q} + \omega_s + \frac{\mu^2}{2\omega_s} \right] \psi_4(k, p, q, s) \tag{180c}$$

$$+ \frac{\mu^2}{2\sqrt{3}} \left[\delta(k+p) \psi_2(q, s) + \delta(k+q) \psi_2(p, s) + \delta(k+s) \psi_2(p, q) \right.$$

$$\left. + \delta(p+q) \psi_2(k, s) + \delta(p+s) \psi_2(k, q) + \delta(q+s) \psi_2(k, p) \right]$$

$$i\frac{\partial\psi_1}{\partial t} = [\omega_k + \frac{\mu^2}{2\omega_k}] \psi_1(k) + \sqrt{\frac{3}{2}} \mu^2 \int \psi_3(k, r, s) \delta(r+s) \frac{dr ds}{4\omega_r \omega_s} \quad (181a)$$

$$i\frac{\partial\psi_2}{\partial t} = [\omega_k + \frac{\mu^2}{2\omega_k} + \omega_r + \frac{\mu^2}{2\omega_r} + \omega_s + \frac{\mu^2}{2\omega_s}] \psi_2(k, r, s) \quad (181b)$$

$$+ \frac{\mu^2}{\sqrt{6}} [\delta(k+r) \psi_1(s) + \delta(k+s) \psi_1(r) + \delta(r+s) \psi_1(k)]$$

$$+ \sqrt{5} \mu^2 \int \psi_3(k, r, s, p, q) \delta(p+q) \frac{dp dq}{4\omega_p \omega_q}$$

$$i\frac{\partial\psi_3}{\partial t} = [\omega_k + \frac{\mu^2}{2\omega_k} + \omega_r + \frac{\mu^2}{2\omega_r} + \omega_p + \frac{\mu^2}{2\omega_p} + \omega_q + \frac{\mu^2}{2\omega_q} + \omega_s + \frac{\mu^2}{2\omega_s}] \psi_3(k, r, s, p, q) \quad (181c)$$

$$+ \frac{\mu^2}{2\sqrt{5}} [\delta(k+r) \psi_3(s, p, q) + \delta(k+s) \psi_3(r, p, q) + \delta(k+p) \psi_3(r, s, q)$$

$$+ \delta(k+q) \psi_3(r, s, p) + \delta(k+s) \psi_3(k, p, q) + \delta(r+p) \psi_3(k, s, q)$$

$$+ \delta(r+q) \psi_3(k, s, p) + \delta(s+p) \psi_3(k, r, q) + \delta(s+q) \psi_3(k, r, p)$$

$$+ \delta(p+q) \psi_3(k, r, s)]$$

The vacuum energy is derived from eqs. (180) by making a number of substitutions as in Chapter 4. By doing this, we derive the equations

$$E_v = \frac{\mu^2}{4\sqrt{2}} \sum_k f(k) \frac{\Delta k}{\omega_k^2} \quad (182a)$$

$$f(k) = \frac{\mu^2}{\sqrt{2\Delta k}(E_v - v_k)} + \frac{\mu^4}{8(E_v - v_k)} \left(\frac{2f(k)}{\omega_k^2(E_v - 4v_k)} + \sum_q \frac{f(q) - f(k)}{\omega_q^2(E_v - 2v_q - 2v_k)} \right) \quad (182b)$$

for the five particle truncation, where $v_k = (1 + \mu^2/2\omega_k^2)\omega_k$.

For the three particle truncation, eq. (182b) reduces to

$$f(k) = \frac{\mu^2}{\sqrt{2\Delta k}(E_v - v_k)} \quad (183)$$

Equations (182ab) or (182a) and (183) are non-linear

simultaneous equations for E_v and $f(k)$, $k=k_{-j}, \dots, k_j$. Likewise, to find the single particle energy at zero momentum, write

$$\psi_j(k, r, s) = \psi_1(0) \{ \delta_{k,-r} \delta_{s,0} g(k) + \delta_{k,-s} \delta_{r,0} g(k) + \delta_{r,-s} \delta_{k,0} g(r) \} \quad (184)$$

and perform some substitutions in (181a,b,c) to get

$$E_0 = v_0 + \frac{\sqrt{6}\mu^2}{8} (2g(0)\Delta k + \sum_k g(k) \frac{\Delta k}{\omega_k}) \quad (185a)$$

$$g(k) = \frac{\mu^2}{\sqrt{6}\Delta k(E_0 - v_0 - 2v_k)} + \frac{\mu^4}{8(E_0 - v_0 - 2v_k)} \left(\frac{4g(k)}{\omega_k^2(E_0 - v_0 - 4v_k)} + \frac{2(g(0) - g(k))}{\omega_k^2(E_0 - 3v_0 - 2v_k)} + \sum_p \frac{g(k) + g(p)}{\omega_p^2(E_0 - v_0 - 2v_k - 2v_p)} \right) \quad (185b)$$

for the five particle truncation. Eq. (185b) reduces to

$$g(k) = \frac{\mu^2}{\sqrt{6}\Delta k(E_0 - v_0 - 2v_k)} \quad (186)$$

for the three particle truncation. Again the result is a set of simultaneous equations for E_v and $g(k)$, $k=k_{-j}, \dots, k_j$.

The programs used to solve these simultaneous equations are detailed in Appendix F. The results of this truncation in the range $0 < \mu < 3$ are plotted in Fig. 10. Clearly the numerical approximations are accurate and well behaved in this range, and they appear to converge to the theoretical result quickly as the truncation point is increased, even at large values of μ . For example, at $\mu=1.4$, the perturbation term in (178) is twice as large

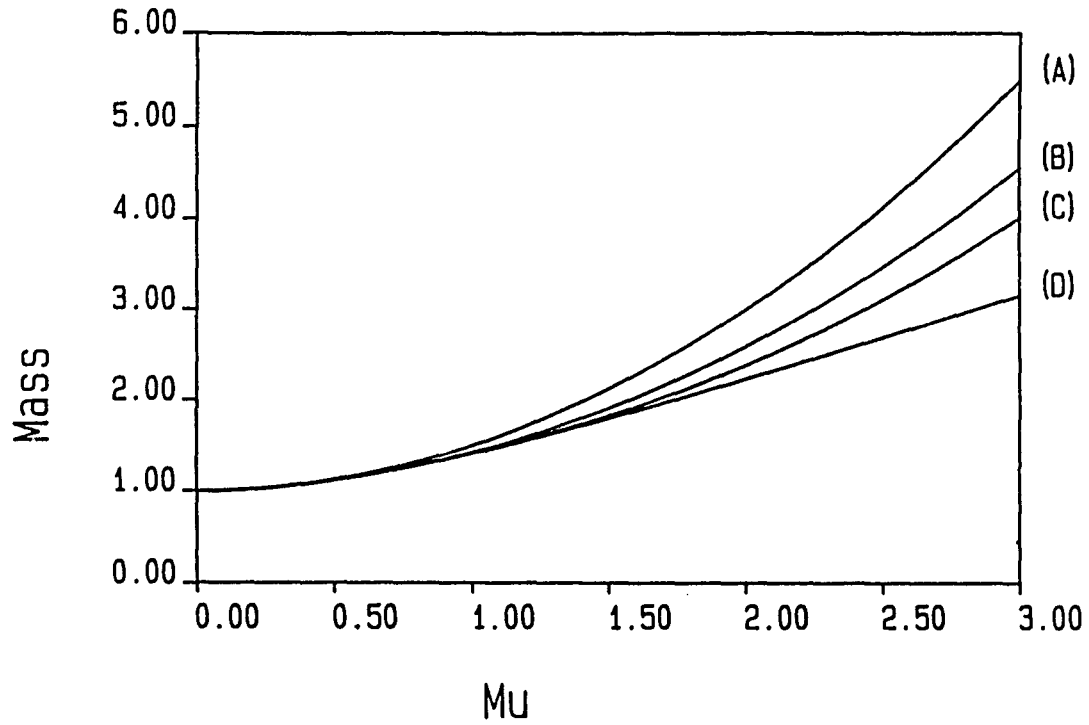


Figure 10: Physical particle mass prediction of truncated hierarchy. Physical mass versus μ for 16 points and $\Delta k = 2\pi/16$. (A) 1-particle truncation, (B) 3-particle truncation, (C) 5-particle truncation, (D) Theoretical mass

as the original mass term, yet the five particle truncation predicts the physical mass to within one percent. For comparison, the perturbation series expansion for the physical mass

$$(1+\mu^2)^{1/2} = 1 + \frac{\mu^2}{2} - \frac{\mu^4}{8} + \frac{\mu^6}{16} - \frac{5\mu^8}{128} + \frac{25\mu^{10}}{125} - \dots \quad (187)$$

is plotted in Fig. 11. The non-convergence for $\mu > 1$ is clear in the figure.

At the same time, one can create problems by looking at how well the mass is approximated as the number of space points is increased for a fixed truncation. Fig. 12 illustrates the mass predictions for a three particle truncation with 4, 16 and 64 points, and they appear to get worse as the number of points is increased. This should not be too surprising, since both the vacuum energy E_v and the single particle energy E_0 in this truncation become infinite in the continuum limit. To see this, let us look at the vacuum energy, derived from (180). This is given by

$$E_v = \frac{\mu^4}{8} \sum_k \left((E_v - 2\omega_k - \frac{\mu^2}{\omega_k}) \omega_k^2 \right)^{-1} \quad (188)$$

Since there is no Δk in this expression, it diverges as the sum becomes infinite in a finite range of k .

Clearly a more complex limiting process than either increasing the number of space points or

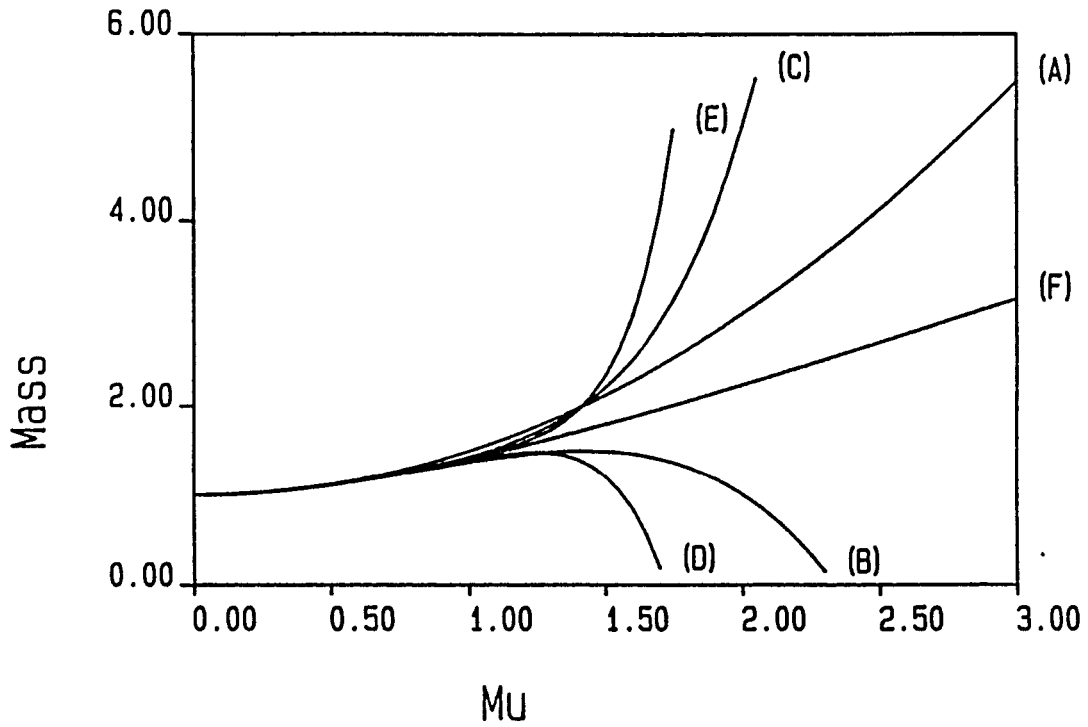


Figure 11: Physical particle mass prediction of the perturbation expansion versus μ . (A) 2 terms, (B) 3 terms, (C) 4 terms, (D) 5 terms, (E) 6 terms, (F) Theoretical mass.

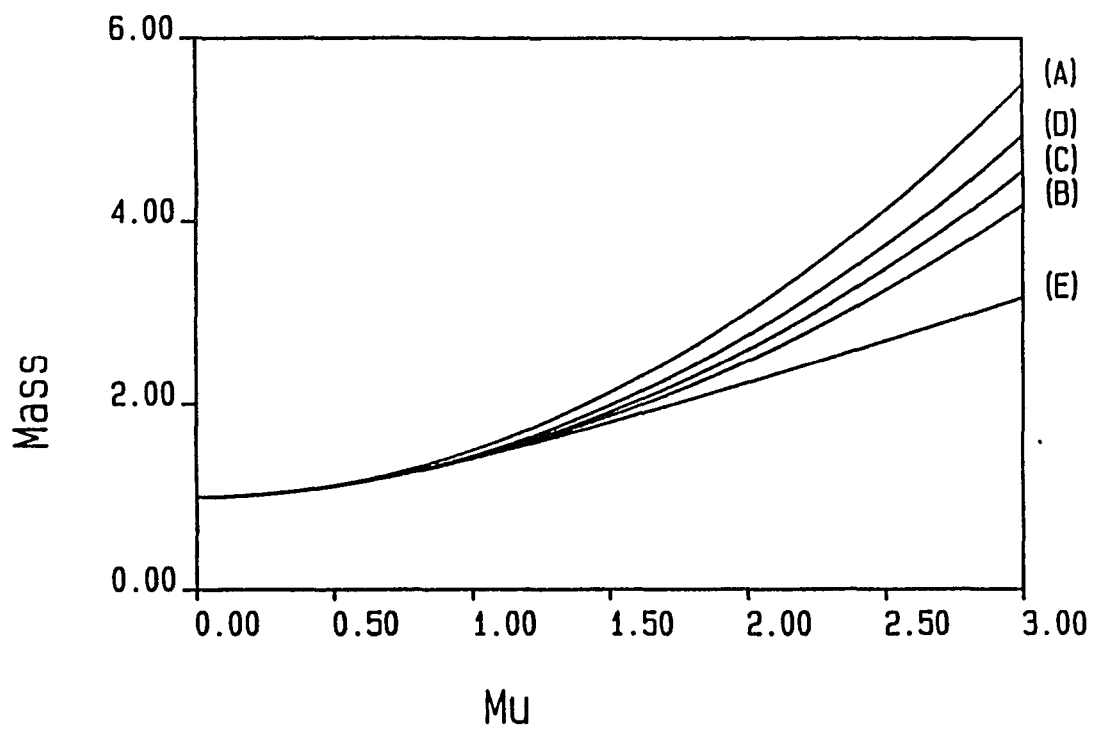


Figure 12: Physical particle mass predictions for varying numbers of grid points P in k -space, for the 2-particle truncation. In each case, $\Delta k = 2\pi/P$. (A) 1-particle truncation mass, (B) $P=4$, (C) $P=16$, (D) $P=64$, (E) Theoretical mass.

increasing the truncation point is necessary if one is to systematically approximate the continuous, infinite hierarchy. The problem becomes even more complex when dealing with a theory which is not super-renormalizable like this mass-perturbation theory. Then, the bare input parameters have no physical meaning, and must be adjusted to reflect the proper physical values, and, as we have seen, this depends on both the truncation point and the number of space points used for the approximation.

These difficulties are not some peculiarity of the hierarchy equations, but a problem which shows up in one form or another in any attempt to solve a quantized field theory. It is a consequence of renormalization. To see this, consider this model theory when attempting to model a single particle in N space points with a truncation point of K particles. In addition to the single particle, one must consider vacuum bubbles,

$$\begin{array}{c} | \\ * \end{array} + \begin{array}{c} \bigcirc \\ \times \end{array} \begin{array}{c} | \\ * \end{array} + \begin{array}{c} \bigcirc \\ \times \end{array} \begin{array}{c} | \\ * \end{array} \begin{array}{c} \bigcirc \\ \times \end{array} \quad (189)$$

By truncating at K , we limit the number of vacuum bubbles which can be accurately modeled. For example, if $K=3$, then the last diagram in (189) cannot be accounted for.

If its amplitude is small enough, though, one doesn't need it, and the $K=3$ cutoff will be accurate. The discretization of the hierarchy effectively limits the amplitude of these vacuum bubbles by closing most of the channels through which they can be produced. Increasing N opens more and more of those channels, causing the amplitude of vacuum bubbles to increase, and making the approximations lose accuracy. Similar problems occur in more complex theories too. Every infinite mass correction and vertex correction is regularized by discretization and will cause the same problems with the accuracy of the truncation.

The obvious solution is to take a limit in which both the truncation point K and the grid size N go to infinity and the bare masses, coupling constants, etc., are adjusted all in such a way that K grows fast enough to accommodate the increasing numbers of particles in the system. Clearly this is a difficult mathematical problem. It involves a limiting process more complex than what is encountered in ordinary calculus, and to the best of my knowledge, it has not been studied, although it does not seem totally insurmountable in this form.

The problem we are facing here is directly analogous to ordinary perturbation theory. When doing perturbation theory, one generally calculates a few regularized low order diagrams, subtracts off the infinities, and produces an accurate prediction. One can calculate a low order diagram and get a good number for the same reason one can discretize and truncate the hierarchy and get a good number. The mathematical problems involved with subtracting infinity from infinity to get a finite number are troubling to a pure mathematician, but we have learned to deal with them. The lack of a rigorous mathematical convergence proof for the hierarchy equations at this time should be no more of a hinderance to us.

It also seems that this approach would work very nicely with field theories used in solid state physics, since the discretization process seems to have some physical relevance, so convergence questions would lose their importance.

The End

APPENDIX A

The Turbo Pascal program GETPSI determines the ground state of a two m-particle system, which is a bound state. Then it calculates the wavefunction for that state and reports it in the range of $x=-250$ to $x=250$, reporting it in detail for $x=-2.5$ to $x=2.5$, where the wavefunction is expected to differ significantly from a decaying exponential.

The program is based on equation (144) in the text. To calculate the energy, it uses the ground state energy of the delta function approximation (149), and then does eight Newton's method approximations to find the correct number, as follows.

The wavefunction is calculated by assuming it is a dying exponential of the form

$$\psi(x) = A \exp(-\sqrt{-mE}|x|) \quad (A.1)$$

sufficiently far away from the potential well ($x=20$ in this case). Then, the wavefunction is calculated in from that point to $x=0$ using the formula

$$\psi(x_{n-1}) = (2 - m\Delta x^2 (E + \frac{\lambda^2}{2m^2\mu} e^{-\mu\Delta x})) \psi(x_n) - \psi(x_{n-1}) \quad (A.2)$$

Finally, the slope at $x=0$ is calculated and compared with the predicted slope due to the delta function at $x=0$.

When the slope is equal to the predicted value, the

correct wavefunction and energy have been found.

The program listing, and a summary of the data it produced is presented here.

```

PROGRAM Getpsi;                                (Find wavefunction for ground state of 2 m-particle system)
CONST
  No           =2048;                          (Number of points in grid)
  N1           =2049;                          (No+1)
VAR
  m,mu,l      :Real;                          (Various constants)
  dx,A,G,So,Slope :Real;
  F,Fl,E,El,D  :Real;
  Nrm          :Real;                          (Norm of wavefunction)
  P            :Array[0..N1] of Real;         (Wavefunction array)
  J,N,Y        :Integer;                      (Index variables)
  Nrgs         :Text;                         (File of energies)
  Detail       :Text;                         (Detailed scan of values of Psi)
  Coarse       :Text;                         (Coarse scan of values of Psi)

PROCEDURE SetPsi(EE:Real);                    (Set up the array P with the wavefunction)
BEGIN
  P[No+1]:=Exp(-Sqrt(-m*EE)*(No+1)*dx);        (Assume end value is exponential)
  P[No]:=Exp(-Sqrt(-m*EE)*No*dx);
  FOR N:=No DOWNT0 1 DO P[N-1]:=(2-m*dx*dx*(EE+A*Exp(-mu*n*dx)))*P[N]-P[N+1];
END;

FUNCTION Diff(EE:Real) : Real;                (Difference between desired and computed slope at zero.)
BEGIN
  SetPsi(EE);                                (Compute wavefunction)
  Diff:=(P[1]-P[0])/(P[0]*dx)-So;             (Compute slope difference)
END;

PROCEDURE Normalize;                          (Normalize the wavefunction)
BEGIN
  Nrm:=Sqr(P[0])*dx;
  FOR J:=1 TO No DO Nrm:=Nrm+2*Sqr(P[J])*dx;    (Calculated part)
  Nrm:=Nrm*Exp(-2*Sqrt(-m*E)*20.5)/Sqrt(-m*E); (Exponential part)
  Nrm:=1/Sqrt(Nrm);                            (Now Nrm is the coefficient in front of exp)
  For J:=0 TO No DO P[J]:=Nrm*P[J];           (Normalize calculated part)
END;

FUNCTION Psi(X:Real) : Real;                  (Give the value of the wavefunction at x)
BEGIN
  IF Abs(X)<No*dx THEN BEGIN                  (In range of the numerical aproximation)
    Y:=Trunc(Abs(X)/dx); D:=Abs(X)-Y*dx;
    Psi:=(P[Y]*(dx-D)+P[Y+1]*D)/dx;
  END
  ELSE Psi:=Nrm*Exp(-Sqrt(-m*E)*Abs(X));      (Out of range of the numerical aproximation)
END;

BEGIN                                         (Main Routine)
  m:=1; mu:=1.25; l:=0.3;                   (Initialize variables)
  dx:=0.01;
  A:=l*l/(2*m*m*mu);
  G:=l*l/(2*m*m*(4*m*m-mu*mu))-l*l/(m*m*mu*mu);
  So:=l*l/(4*m*m*(4*m*m-mu*mu));

  Assign(Nrgs,'NRG.PSI');                    (Open file of energy aproximations)
  Rewrite(Nrgs);                             (and determine energy of the bound state)
  El:=-m*G*G/4; E:=1.1*El;                  (First guess - delta fctn soln)
  Fl:=Diff(El); F:=Diff(E);
  Write('Estimating energy. ');
  WriteLn(Nrgs,'Energy = ',E,' gives F = ',F); (Save result in output file)

```

```

FOR J:=1 TO 8 DO BEGIN
  Write(J, ' ');
  Slope:=(F-Fl)/(E-El);
  El:=E; Fl:=F; E:=E-F/Slope; F:=Diff(E);
  WriteLn(Nrgs, 'Energy = ', E, ' gives F = ', F);
  END;
WriteLn;
Close(Nrgs);
SetPsi(E);
Normalize;

Assign(Detail, 'DETAIL.PSI');
Rewrite(Detail);
FOR J:=-250 TO 250 DO
  WriteLn(Detail, J*0.01, Psi(J*0.01), J*0.01, 0.1398849*exp(-1.95678E-02*abs(j*0.01)));
Close(Detail);

Assign(Coarse, 'COARSE.PSI');
Rewrite(Coarse);
FOR J:=-250 TO 250 DO WriteLn(Coarse, J*1.0, Psi(J*1.0));
Close(Coarse);
END.

```

{Apply Newton's approximation 8 times}
 {Energy computed, close energy file}
 {Compute wavefunction with proper E}
 {And normalize it}
 {Detail of wavefunction inside x=2.5}
 {Report in detail, compare with exponential}
 {Coarse examination of psi inside x=250}
 {Record it in Coarse file}
 {All done}

Bound state energy predicted by GETPSI program:
 Energy = -3.55711E-04 gives F = 6.130E-11

Wavefunction produced by GETPSI program:

X	Psi(X)
0	1.3438E-01
0.02	1.3440E-01
0.1	1.3448E-01
0.2	1.3454E-01
0.4	1.3456E-01
0.5	1.3452E-01
0.6	1.3446E-01
0.7	1.3437E-01
0.8	1.3427E-01
0.9	1.3414E-01
1.0	1.3400E-01
1.1	1.3385E-01
1.2	1.3369E-01
1.3	1.3352E-01
1.4	1.3333E-01
1.5	1.3314E-01
1.6	1.3295E-01
1.7	1.3274E-01
1.8	1.3253E-01
1.9	1.3232E-01
2.0	1.3210E-01
2.1	1.3188E-01
2.2	1.3166E-01
2.3	1.3143E-01
2.4	1.3121E-01
3	1.2981E-01
4	1.2743E-01
5	1.2506E-01
10	1.1381E-01
20	9.4252E-02
30	7.8052E-02
40	6.4636E-02
50	5.3526E-02
60	4.4325E-02
70	3.6706E-02
80	3.0397E-02
90	2.5172E-02
100	2.0845E-02
110	1.7262E-02
120	1.4295E-02
130	1.1838E-02
140	9.8035E-03
150	8.1184E-03
160	6.7230E-03
170	5.5674E-03
180	4.6104E-03
190	3.8180E-03
200	3.1617E-03

APPENDIX B

BOUND STATE DECAY CALCULATIONS

The Turbo Pascal program DECAY numerically examines the resonance state of a bound pair of two m -particles decaying into a pair of μ -particles. The program is based on equations (159) and (160) in the text. It sets up an initial state in which only ψ is non-zero. Then it propagates the system in time using the standard Hamiltonian formula

$$|\psi(t+\Delta t)\rangle = e^{-i\Delta t H} |\psi(t)\rangle$$

where Ψ is a vector combination of ψ and φ . After every ten time steps, it reports the squared norms of ψ and φ , and after every one hundred time steps, it produces a detailed description of ψ and φ as functions of x .

The data produced by this program is displayed graphically in the text, in Figure 7. It is not reproduced in tabular form here because it is excessively long. To reproduce the data, it is simpler to re-run the program, even if it does take a half an hour to run.

```

program resonance;                                     {display resonance decay}

var
  psi          :array[1..2,0..101] of real;    {M particle wavefunction}
  phe          :array[1..2,0..101] of real;    {Mu particle wavefunction}
  t,g,dt,dx    :real;                          {Miscellaneous constants}
  m,mu,l       :real;
  No,i,j,k     :integer;                       {Index variables}
  pfilem,s     :string[20];                   {File names}
  timefile     :text;                          {File for norms as function of t}
  psifile      :text;                          {File for wavefunction data}

procedure norm;                                       {Normalize wavefunction}
var
  nrm1,nrm2,nrm :real;
  n              :integer;
begin
  nrm1:=(sqr(psi[1,0])+sqr(psi[2,0]))*dx;
  nrm2:=(sqr(phe[1,0])+sqr(phe[2,0]))*dx;
  for n:=1 to No do
    begin
      nrm1:=nrm1+2*(sqr(psi[1,n])+sqr(psi[2,n]))*dx;
      nrm2:=nrm2+2*(sqr(phe[1,n])+sqr(phe[2,n]))*dx;
    end;
  nrm:=1/sqrt(nrm1+nrm2);
  for n:=0 to No do
    begin
      psi[1,n]:=nrm*psi[1,n]; psi[2,n]:=nrm*psi[2,n];
      phe[1,n]:=nrm*phe[1,n]; phe[2,n]:=nrm*phe[2,n];
    end;
end;

procedure readnorm;                                   {Print time and square magnitude of each wavefunction}
var
  nrm1,nrm2,nrm :real;
  n              :integer;
begin
  nrm1:=(sqr(psi[1,0])+sqr(psi[2,0]))*dx;
  nrm2:=(sqr(phe[1,0])+sqr(phe[2,0]))*dx;
  for n:=1 to No do
    begin
      nrm1:=nrm1+2*(sqr(psi[1,n])+sqr(psi[2,n]))*dx;
      nrm2:=nrm2+2*(sqr(phe[1,n])+sqr(phe[2,n]))*dx;
    end;
  writeln(timefile,t,nrm1,nrm2);
end;

procedure push;                                       {Advance system in time by increment dt}
var
  t1,t2,tl1,tl2 :real;
  s1,s2,sl1,sl2 :real;
  n              :integer;
begin
  t:=t+dt;
  t1:=psi[1,0]; t2:=psi[2,0]; s1:=phe[1,0]; s2:=phe[2,0];
  psi[1,0]:=t1+dt*(-2*(psi[2,1]-t2)/(m*dx*dx)-g*t2/dx-l*l*s2/(m*mu*mu*mu*dx));
  psi[2,0]:=t2-dt*(-2*(psi[1,1]-t1)/(m*dx*dx)-g*t1/dx-l*l*s1/(m*mu*mu*mu*dx));
  phe[1,0]:=s1+dt*((2*mu-2*m)*s2-2*(phe[2,1]-s2)/(mu*dx*dx)-l*l*t2/(m*m*mu*mu*dx));
  phe[2,0]:=s2-dt*((2*mu-2*m)*s1-2*(phe[1,1]-s1)/(mu*dx*dx)-l*l*t1/(m*m*mu*mu*dx));

```

```

for n:=1 to No do
begin
  t1:=t1; t2:=t2; s1:=s1; s2:=s2;
  t1:=psi[1,n]; t2:=psi[2,n]; s1:=phe[1,n]; s2:=phe[2,n];
  psi[1,n]:=t1+dt*(2*t2-t1-psi[2,n+1])/(m*dx*dx);
  psi[2,n]:=t2-dt*(2*t1-t2-psi[1,n+1])/(m*dx*dx);
  phe[1,n]:=s1+dt*((2*mu-2*m)*s2+(2*s2-s1-phe[2,n+1])/(mu*dx*dx));
  phe[2,n]:=s2-dt*((2*mu-2*m)*s2+(2*s2-s1-phe[2,n+1])/(mu*dx*dx));
end;
norm;
end;

procedure setpsi;      {Set up the wavefunction, psi using delta function solution, phe zeroed}
var
  a      :real;
  n      :integer;
begin
  a:=m*g/2;
  for n:=0 to No do
  begin
    psi[1,n]:=sqrt(a)*exp(-a*n*dx); psi[2,n]:=0;
    phe[1,n]:=0; phe[2,n]:=0;
  end;
  psi[1,No+1]:=0; psi[2,No+1]:=0;
  phe[1,No+1]:=0; phe[2,No+1]:=0;
  norm;
end;

begin
  m:=1; mu:=0.75; l:=0.3; No:=100; dx:=2.0; dt:=0.1; t:=0;
  g:=l*(1/(m*mu)-1/(2*m*(4*m-mu)));
  assign(timefile,'NORM.DAT'); rewrite(timefile);
  setpsi;
  pfilenm:='PSI0.DAT';
  assign(psifile,pfilenm); rewrite(psifile);
  for j:=0 to No do writeln(psifile,psi[1,j],psi[2,j],phe[1,j],phe[2,j]);
  close(psifile);
  for i:=1 to 20 do
  begin
    for j:=1 to 10 do
    begin
      for k:=1 to 10 do
      begin
        write('Pushing ',100*(i-1)+10*(j-1)+k,#13);
        push;
      end;
      readnorm;
    end;
    str(i,s);
    pfilenm:='PSI'+s+'.DAT';
    assign(psifile,pfilenm); rewrite(psifile);
    for j:=0 to No do writeln(psifile,psi[1,j],psi[2,j],phe[1,j],phe[2,j]);
    close(psifile);
  end;
  close(timefile);
end.

```

{main control routine}
 {Initialize variables}
 {Open file for norms printout}
 {Set up wavefunction at t=0}
 {Print norm of psi and phe 200 times}
 {Report wavefunctions 200 times}

APPENDIX C

RENORMALIZATION CALCULATIONS

The program MESN1 listed here calculates the ground state energy and the masses of the bare m -particle and the μ -particle for a given values of the physical masses. This is accomplished for a given number of grid points and a given grid size Δk . All of the work here is essentially done by the subroutine RENORM, which is the Newton's approximation feedback calculator discussed in Chapter 6.

Data produced by this program is then presented in tabular form after the listing. Grid sizes which were examined ranged from 8 points to 512 points. The program took quite a long time to execute at 512 points, so the calculation was shortened somewhat for that run. This paper may be freely reproduced by anyone without permission, regardless of what it says in the front.

This particular listing was run on the SCS-40 at the University of Arizona.


```

program mesn1

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real x1,k1,x2,k2,ph,sig,x
complex f1(32),f2(32),f3(32)
integer r,p,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

call link("unit6=tty//")

call init
dt=.005
mur=1.0

write(6,1000)
write(6,1001)
1000 format(44hlamda      e vac      m      mu)
1001 format(44h-----)

do 10 r=1,30
  lr=float(r)/20.0
  call renorm
10 continue

call exit

end

c* *****

subroutine init

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
integer r,p,q,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c* initialize all standard variables

spc=2
pi=acos(-1.0)
dx=1.0
dk=2.0*pi/(16.0*dx)
mr=1.0
t=0.0

return
end

```

```

c* *****
subroutine renorm

c* this routine finds the solution of a set of eqns of the form
c* ar(ao)=r, where ao and b are vectors.

real u(32),ao(3),an(3),ar(3),arn(3),d(3,3),r(3),sq2
real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
integer n,p,count,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c* count is a loop index to see how many times the loop has
c* been used. initialize to zero
count=0

c* assign the physical values we want to end up with
r(1)=mr
r(2)=mur
r(3)=lr

c* guess at what the unrenormalized values should be
c* here we just guess that they are the same as the physical
do 10 n=1,3
ao(n)=r(n)
10 continue

c* get renormalized values for initial guess
call nrgs(ao(1),ao(2),ao(3),ar)

c* loop begins here
c* shift ao by a small amount in each direction to calculate derivatives s
30 do 50 n=1,3
an(n)=ao(n)+(ao(n)/ar(n))*(r(n)-ar(n))
50 continue

do 55 n=1,3
do 52 p=1,3
d(n,p)=0.
52 continue
55 continue

call nrgs(an(1),ao(2),ao(3),arn)
do 60 n=1,3
if(arn(n).ne.ar(n)) d(n,1)=(arn(n)-ar(n))/(an(1)-ao(1))
60 continue
if(arn(1).eq.ar(1).and.an(1).eq.ao(1)) d(1,1)=1.

call nrgs(ao(1),an(2),ao(3),arn)
do 70 n=1,3
if(arn(n).ne.ar(n)) d(n,2)=(arn(n)-ar(n))/(an(2)-ao(2))
70 continue
if(arn(2).eq.ar(2).and.an(2).eq.ao(2)) d(2,2)=1.

call nrgs(ao(1),ao(2),an(3),arn)
do 80 n=1,3
if(arn(n).ne.ar(n)) d(n,3)=(arn(n)-ar(n))/(an(3)-ao(3))
80 continue
if(arn(3).eq.ar(3).and.an(3).eq.ao(3)) d(3,3)=1.

c* invert the tensor d(i,j)
call matinv(d)

```

```

c*      use d inv to make a new guess at ao
      do 100 p=1,3
      do 90 n=1,3
      ao(p)=ao(p)+d(n,p)*(r(n)-ar(n))
90      continue
100     continue

c*      calculate a new value of a
      call nrgs(ao(1),ao(2),ao(3),ar)

c*      see if a is small enough:
c*          if so, we're done
c*          if not, loop again, unless stuck
      sq2=0.0
      do 120 n=1,3
      sq2=sq2+(ar(n)-r(n))**2
120     continue
      count=count+1
      if(count.eq.10) go to 130
      if(sq2.lt.1.0e-15) go to 130
      go to 30

c*      if done, set bare values for output, read actual
c*      renormalized values out again
130     call nrgs(ao(1),ao(2),ao(3),ar)
      mr=ar(1)
      mur=ar(2)
      lr=ar(3)
      m=ao(1)
      mu=ao(2)
      l=ao(3)
      call nrgs(m,mu,l,ar)

      call vnrg(sq2)
      write(6,1000) l,sq2,m,mu

1000    format(e9.3,e16.9,e16.9,e16.9)

      return
      end

c*      *****

      subroutine matinv(s)

      real s(3,3),si(3,3),det
      integer i,j

c*      first calculate determinant
      det=s(1,1)*(s(2,2)*s(3,3)-s(2,3)*s(3,2))
      det=det-s(2,1)*(s(1,2)*s(3,3)-s(1,3)*s(3,2))
      det=det+s(3,1)*(s(1,2)*s(2,3)-s(1,3)*s(2,2))

c*      if det=0 matrix is singular - can't invert
      if(det.le.1e-20) write(6,1000)
1000    format(15hsingular matrix)

```

```

c*      then the inverse matrix elements
      si(1,1)=(s(2,2)*s(3,3)-s(2,3)*s(3,2))/det
      si(1,2)=-(s(2,1)*s(3,3)-s(2,3)*s(3,1))/det
      si(1,3)=(s(2,1)*s(3,2)-s(2,2)*s(3,1))/det
      si(2,1)=-(s(1,2)*s(3,3)-s(1,3)*s(3,2))/det
      si(2,2)=(s(1,1)*s(3,3)-s(1,3)*s(3,2))/det
      si(2,3)=-(s(1,1)*s(3,2)-s(1,2)*s(3,1))/det
      si(3,1)=(s(1,2)*s(2,3)-s(1,3)*s(2,2))/det
      si(3,2)=-(s(1,1)*s(2,3)-s(1,3)*s(2,1))/det
      si(3,3)=(s(1,1)*s(2,2)-s(1,2)*s(2,1))/det

c*      and then substitute si into s
      do 20 i=1,3
      do 10 j=1,3
10      s(i,j)=si(i,j)
      continue
20      continue

      return
      end

c*      *****

      subroutine nrgs(me,mue,le,a)

      real me,mue,le,a(3),e
      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
      integer spc

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

      m=me
      mu=mue
      l=le

      call vnrg(e)
      call mnrg(17,a(1))
      a(1)=a(1)-e
      call munrg(17,a(2))
      a(2)=a(2)-e
      call lamda(a(3))

      return
      end

c*      *****

      subroutine lamda(lt)

      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
      real lt
      integer spc

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c*      just a dummy for now
      lt=l

      return
      end

```

```

c* *****

subroutine vnrg(s)

c* finds the physical vacuum energy

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real f,e,lf,le,s
integer k,n,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

c* calculate omega's for the given values of m and mu
do 10 n=1,32
s=dk*float(n-17)
w(n)=sqrt(m**2+s**2)
wu(n)=sqrt(mu**2+s**2)
10 continue

e=-0.0001
call fg
le=e
lf=f
e=-1.0
call fg

20 s=(f-lf)/(e-le)
le=e
lf=f
e=le-lf/s
call fg
if(sqrt((e-le)**2).le.1.0e-11) go to 30
if(sqrt(f*f).gt.1.0e-40) go to 20

30 s=e

return
end

c* *****

subroutine fg

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real a,e,f
integer spc,r,p

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

c* calculate f for a given e
a=l**2*dk/(8.*pi)
f=0.0
do 40 r=1,32
do 30 p=1,32
if(r+p-17.lt.1.or.r+p-17.gt.32) go to 30
f=f+a/((e-w(r)-w(p)-wu(r+p-17))*w(r)*w(p)*wu(r+p-17))
30 continue
40 continue
f=f-e

return
end

```

```

c* *****
subroutine mnrg(k,s)

c* finds the physical m-particle energy level

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real f,e,lf,le,s
integer k,n,r,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

c* set up the constants w,wu
do 10 r=1,32
w(r)=sqrt(m**2+(float(r-17)*dk)**2)
wu(r)=sqrt(mu**2+(float(r-17)*dk)**2)
10 continue

e=1.0001
call fm(k)
le=e
lf=f
e=1.0
call fm(k)

20 s=(f-lf)/(e-le)
le=e
lf=f
e=le-lf/s
call fm(k)
if(sqrt((e-le)**2).le.1.0e-11) go to 30
if(sqrt(f*f).gt.1.0e-40) go to 20

30 s=e

return
end

c* *****
subroutine fm(p)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real f,e,a
integer spc,r,s,p

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

f=0.0
a=l*dk/(8*pi)
do 30 r=1,32
if(p+r-17.gt.32.or.p+r-17.lt.1) go to 10
f=f+2.*a/((e-2*w(p)-w(r)-wu(p+r-17))*w(p)*w(r)*wu(p+r-17))
10 do 20 s=1,32
if(r+s-17.eq.p) f=f+2.*a/((e-w(r)-wu(s))*w(p)*w(r)*wu(s))
if(s+r-17.gt.32.or.s+r-17.lt.1) go to 20
f=f+a/((e-w(p)-w(r)-w(s)-wu(r+s-17))*w(r)*w(s)*wu(r+s-17))
20 continue
30 continue
f=f+w(p)-e
return
end

```

```

c* *****

subroutine munrg(k,s)

c* finds the physical mu-particle energy level

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real f,e,lf,le,s,u(32)
integer k,n,r,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

c* set up the constants w,wu
do 10 r=1,32
w(r)=sqrt(m**2+(float(r-17)*dk)**2)
wu(r)=sqrt(mu**2+(float(r-17)*dk)**2)
10 continue

e=0.01
call fmu(k)
le=e
lf=f
e=1.0
call fmu(k)

20 s=(f-lf)/(e-le)
le=e
lf=f
e=le-lf/s
call fmu(k)
if(sqrt((e-le)**2).le.1.0e-11) go to 30
if(sqrt(f*f).gt.1.0e-40) go to 20

30 s=e

return
end

c* *****

subroutine fmu(p)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(32),wu(32),t
real a,e,f
integer spc,r,s,p

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

f=0.0
a=l*l*dk/(8*pi)

do 20 r=1,32
do 10 s=1,32
if(r+s-17.eq.p) f=f+a/((e-w(r)-w(s))*w(r)*w(s)*wu(p))
if(r+s-17.eq.p) f=f+a/((e-2*wu(p)-w(r)-w(s))*w(r)*w(s)*wu(p))
if(r+s-17.lt.1.or.r+s-17.gt.32) go to 10
f=f+a/((e-wu(p)-w(r)-w(s)-wu(r+s-17))*w(r)*w(s)*wu(r+s-17))
10 continue
20 continue
f=f-e+wu(p)
return
end

```

Table C-1 : 8 GRID POINTS, DK=2PI/16, M =MU =1

LAMDA	E VAC	M	MU
-----	-----	-----	-----
.500E-01	-.2764003E-03	.1000404E+01	.1000201E+01
.100E+00	-.1102079E-02	.1001608E+01	.1000799E+01
.150E+00	-.2466700E-02	.1003591E+01	.1001784E+01
.200E+00	-.4353764E-02	.1006316E+01	.1003137E+01
.250E+00	-.6741587E-02	.1009739E+01	.1004835E+01
.300E+00	-.9604475E-02	.1013805E+01	.1006850E+01
.350E+00	-.1291396E-01	.1018456E+01	.1009154E+01
.400E+00	-.1663997E-01	.1023632E+01	.1011714E+01
.450E+00	-.2075185E-01	.1029272E+01	.1014501E+01
.500E+00	-.2521922E-01	.1035319E+01	.1017486E+01
.550E+00	-.3001261E-01	.1041716E+01	.1020641E+01
.600E+00	-.3510391E-01	.1048412E+01	.1023940E+01
.650E+00	-.4046665E-01	.1055363E+01	.1027360E+01
.700E+00	-.4607620E-01	.1062524E+01	.1030881E+01
.750E+00	-.5190975E-01	.1069860E+01	.1034484E+01
.800E+00	-.5794639E-01	.1077337E+01	.1038154E+01
.850E+00	-.6416697E-01	.1084927E+01	.1041876E+01
.900E+00	-.7055404E-01	.1092605E+01	.1045638E+01
.950E+00	-.7709175E-01	.1100348E+01	.1049430E+01
.100E+01	-.8376567E-01	.1108137E+01	.1053243E+01
.105E+01	-.9056273E-01	.1115958E+01	.1057068E+01
.110E+01	-.9747108E-01	.1123795E+01	.1060899E+01
.115E+01	-.1044800E+00	.1131636E+01	.1064731E+01
.120E+01	-.1115796E+00	.1139472E+01	.1068559E+01
.125E+01	-.1187611E+00	.1147294E+01	.1072378E+01
.130E+01	-.1260165E+00	.1155094E+01	.1076185E+01
.135E+01	-.1333385E+00	.1162866E+01	.1079978E+01
.140E+01	-.1407203E+00	.1170605E+01	.1083753E+01
.145E+01	-.1481559E+00	.1178306E+01	.1087509E+01
.150E+01	-.1556398E+00	.1185966E+01	.1091244E+01

Table C-2 : 16 GRID POINTS, $DK=2\pi/16$, $M = MU = 1$

LAMDA	E VAC	M	MU

.050E+00	-.4443253E-03	.1000459E+01	.1000229E+01
.100E+00	-.1771807E-02	.1001828E+01	.1000913E+01
.150E+00	-.3966347E-02	.1004077E+01	.1002036E+01
.200E+00	-.7002342E-02	.1007162E+01	.1003576E+01
.250E+00	-.1084630E-01	.1011024E+01	.1005504E+01
.300E+00	-.1545877E-01	.1015597E+01	.1007785E+01
.350E+00	-.2079630E-01	.1020808E+01	.1010384E+01
.400E+00	-.2681329E-01	.1026584E+01	.1013264E+01
.450E+00	-.3346355E-01	.1032851E+01	.1016388E+01
.500E+00	-.4070153E-01	.1039541E+01	.1019723E+01
.550E+00	-.4848324E-01	.1046589E+01	.1023236E+01
.600E+00	-.5676686E-01	.1053938E+01	.1026899E+01
.650E+00	-.6551312E-01	.1061535E+01	.1030687E+01
.700E+00	-.7468546E-01	.1069332E+01	.1034576E+01
.750E+00	-.8425007E-01	.1077290E+01	.1038546E+01
.800E+00	-.9417585E-01	.1085373E+01	.1042581E+01
.850E+00	-.1044343E+00	.1093550E+01	.1046666E+01
.900E+00	-.1149992E+00	.1101795E+01	.1050787E+01
.950E+00	-.1258469E+00	.1110086E+01	.1054934E+01
.100E+01	-.1369554E+00	.1118403E+01	.1059098E+01
.105E+01	-.1483048E+00	.1126730E+01	.1063270E+01
.110E+01	-.1598770E+00	.1135053E+01	.1067444E+01
.115E+01	-.1716552E+00	.1143361E+01	.1071615E+01
.120E+01	-.1836243E+00	.1151643E+01	.1075777E+01
.125E+01	-.1957703E+00	.1159892E+01	.1079927E+01
.130E+01	-.2080803E+00	.1168101E+01	.1084060E+01
.135E+01	-.2205425E+00	.1176264E+01	.1088176E+01
.140E+01	-.2331461E+00	.1184377E+01	.1092270E+01
.145E+01	-.2458811E+00	.1192435E+01	.1096342E+01
.150E+01	-.2587382E+00	.1200436E+01	.1100389E+01

Table C-3 : 32 GRID POINTS, $DK=2\pi/16$, $M = \mu = 1$

LAMDA	E VAC	M	MU

.050E+00	-.5387949E-03	.1000475E+01	.1000237E+01
.100E+00	-.2148950E-02	.1001890E+01	.1000945E+01
.150E+00	-.4812221E-02	.1004213E+01	.1002106E+01
.200E+00	-.8499615E-02	.1007396E+01	.1003697E+01
.250E+00	-.1317325E-01	.1011378E+01	.1005686E+01
.300E+00	-.1878857E-01	.1016086E+01	.1008039E+01
.350E+00	-.2529656E-01	.1021444E+01	.1010715E+01
.400E+00	-.3264584E-01	.1027372E+01	.1013677E+01
.450E+00	-.4078441E-01	.1033796E+01	.1016887E+01
.500E+00	-.4966102E-01	.1040642E+01	.1020308E+01
.550E+00	-.5922617E-01	.1047843E+01	.1023908E+01
.600E+00	-.6943277E-01	.1055339E+01	.1027658E+01
.650E+00	-.8023655E-01	.1063075E+01	.1031530E+01
.700E+00	-.9159624E-01	.1071004E+01	.1035502E+01
.750E+00	-.1034736E+00	.1079084E+01	.1039554E+01
.800E+00	-.1158332E+00	.1087280E+01	.1043667E+01
.850E+00	-.1286428E+00	.1095559E+01	.1047827E+01
.900E+00	-.1418723E+00	.1103897E+01	.1052022E+01
.950E+00	-.1554943E+00	.1112269E+01	.1056240E+01
.100E+01	-.1694838E+00	.1120657E+01	.1060471E+01
.105E+01	-.1838177E+00	.1129045E+01	.1064709E+01
.110E+01	-.1984745E+00	.1137419E+01	.1068947E+01
.115E+01	-.2134349E+00	.1145768E+01	.1073178E+01
.120E+01	-.2286809E+00	.1154083E+01	.1077399E+01
.125E+01	-.2441957E+00	.1162355E+01	.1081605E+01
.130E+01	-.2599640E+00	.1170579E+01	.1085794E+01
.135E+01	-.2759718E+00	.1178748E+01	.1089962E+01
.140E+01	-.2922056E+00	.1186859E+01	.1094107E+01
.145E+01	-.3086534E+00	.1194907E+01	.1098227E+01
.150E+01	-.3253038E+00	.1202891E+01	.1102321E+01

Table C-4 : 64 GRID POINTS, DK=2PI/16, M =MU =1

LAMDA	E VAC	M	MU
.050E+00	-.5757192E-03	.1000479E+01	.1000239E+01
.100E+00	-.2296449E-02	.1001905E+01	.1000952E+01
.150E+00	-.5143368E-02	.1004247E+01	.1002123E+01
.200E+00	-.9086573E-02	.1007454E+01	.1003726E+01
.250E+00	-.1408701E-01	.1011465E+01	.1005731E+01
.300E+00	-.2009875E-01	.1016205E+01	.1008100E+01
.350E+00	-.2707131E-01	.1021597E+01	.1010794E+01
.400E+00	-.3495177E-01	.1027560E+01	.1013775E+01
.450E+00	-.4368660E-01	.1034017E+01	.1017003E+01
.500E+00	-.5322304E-01	.1040895E+01	.1020442E+01
.550E+00	-.6351012E-01	.1048127E+01	.1024060E+01
.600E+00	-.7449938E-01	.1055650E+01	.1027827E+01
.650E+00	-.8614520E-01	.1063410E+01	.1031715E+01
.700E+00	-.9840503E-01	.1071359E+01	.1035701E+01
.750E+00	-.1112394E+00	.1079456E+01	.1039766E+01
.800E+00	-.1246119E+00	.1087664E+01	.1043891E+01
.850E+00	-.1384889E+00	.1095952E+01	.1048062E+01
.900E+00	-.1528395E+00	.1104293E+01	.1052266E+01
.950E+00	-.1676353E+00	.1112665E+01	.1056492E+01
.100E+01	-.1828502E+00	.1121050E+01	.1060731E+01
.105E+01	-.1984602E+00	.1129430E+01	.1064975E+01
.110E+01	-.2144431E+00	.1137793E+01	.1069217E+01
.115E+01	-.2307784E+00	.1146127E+01	.1073452E+01
.120E+01	-.2474475E+00	.1154423E+01	.1077675E+01
.125E+01	-.2644328E+00	.1162673E+01	.1081883E+01
.130E+01	-.2817182E+00	.1170871E+01	.1086071E+01
.135E+01	-.2992888E+00	.1179012E+01	.1090239E+01
.140E+01	-.3171307E+00	.1187091E+01	.1094383E+01
.145E+01	-.3352308E+00	.1195105E+01	.1098501E+01
.150E+01	-.3535772E+00	.1203052E+01	.1102592E+01

Table C-5 : 128 GRID POINTS, DK=2PI/16, M =MU =1

LAMDA	E VAC	M	MU
.050E+00	-.5879134E-03	.1000480E+01	.1000240E+01
.100E+00	-.2345175E-02	.1001909E+01	.1000954E+01
.150E+00	-.5252817E-02	.1004255E+01	.1002127E+01
.200E+00	-.9280703E-02	.1007469E+01	.1003734E+01
.250E+00	-.1438948E-01	.1011486E+01	.1005742E+01
.300E+00	-.2053289E-01	.1016234E+01	.1008114E+01
.350E+00	-.2766006E-01	.1021633E+01	.1010813E+01
.400E+00	-.3571769E-01	.1027604E+01	.1013797E+01
.450E+00	-.4465188E-01	.1034068E+01	.1017029E+01
.500E+00	-.5440948E-01	.1040953E+01	.1020473E+01
.550E+00	-.6493919E-01	.1048190E+01	.1024094E+01
.600E+00	-.7619217E-01	.1055717E+01	.1027863E+01
.650E+00	-.8812250E-01	.1063481E+01	.1031754E+01
.700E+00	-.1006873E+00	.1071432E+01	.1035742E+01
.750E+00	-.1138469E+00	.1079529E+01	.1039809E+01
.800E+00	-.1275644E+00	.1087736E+01	.1043935E+01
.850E+00	-.1418061E+00	.1096022E+01	.1048107E+01
.900E+00	-.1565407E+00	.1104360E+01	.1052311E+01
.950E+00	-.1717397E+00	.1112728E+01	.1056537E+01
.100E+01	-.1873767E+00	.1121106E+01	.1060775E+01
.105E+01	-.2034273E+00	.1129479E+01	.1065017E+01
.110E+01	-.2198693E+00	.1137833E+01	.1069258E+01
.115E+01	-.2366821E+00	.1146158E+01	.1073491E+01
.120E+01	-.2538464E+00	.1154443E+01	.1077711E+01
.125E+01	-.2713449E+00	.1162681E+01	.1081916E+01
.130E+01	-.2891610E+00	.1170866E+01	.1086102E+01
.135E+01	-.3072797E+00	.1178992E+01	.1090266E+01
.140E+01	-.3256869E+00	.1187057E+01	.1094406E+01
.145E+01	-.3443693E+00	.1195055E+01	.1098520E+01
.150E+01	-.3633148E+00	.1202984E+01	.1102607E+01

Table C-6 : 256 GRID POINTS, DK=2PI/16, M =MU =1

LAMDA	E VAC	M	MU

.050E+00	-.5916418E-03	.1000480E+01	.1000240E+01
.100E+00	-.2360076E-02	.1001910E+01	.1000955E+01
.150E+00	-.5286296E-02	.1004257E+01	.1002128E+01
.200E+00	-.9340106E-02	.1007472E+01	.1003735E+01
.250E+00	-.1448208E-01	.1011491E+01	.1005744E+01
.300E+00	-.2066586E-01	.1016240E+01	.1008118E+01
.350E+00	-.2784049E-01	.1021641E+01	.1010817E+01
.400E+00	-.3595258E-01	.1027614E+01	.1013803E+01
.450E+00	-.4494812E-01	.1034080E+01	.1017035E+01
.500E+00	-.5477387E-01	.1040966E+01	.1020479E+01
.550E+00	-.6537843E-01	.1048203E+01	.1024101E+01
.600E+00	-.7671290E-01	.1055732E+01	.1027871E+01
.650E+00	-.8873126E-01	.1063495E+01	.1031762E+01
.700E+00	-.1013906E+00	.1071446E+01	.1035751E+01
.750E+00	-.1146510E+00	.1079543E+01	.1039817E+01
.800E+00	-.1284758E+00	.1087749E+01	.1043943E+01
.850E+00	-.1428309E+00	.1096033E+01	.1048115E+01
.900E+00	-.1576853E+00	.1104369E+01	.1052318E+01
.950E+00	-.1730101E+00	.1112735E+01	.1056543E+01
.100E+01	-.1887789E+00	.1121110E+01	.1060781E+01
.105E+01	-.2049674E+00	.1129480E+01	.1065022E+01
.110E+01	-.2215533E+00	.1137831E+01	.1069262E+01
.115E+01	-.2385158E+00	.1146151E+01	.1073493E+01
.120E+01	-.2558358E+00	.1154432E+01	.1077713E+01
.125E+01	-.2734956E+00	.1162666E+01	.1081916E+01
.130E+01	-.2914789E+00	.1170846E+01	.1086101E+01
.135E+01	-.3097705E+00	.1178967E+01	.1090263E+01
.140E+01	-.3283561E+00	.1187026E+01	.1094401E+01
.145E+01	-.3472227E+00	.1195018E+01	.1098513E+01
.150E+01	-.3663579E+00	.1202942E+01	.1102598E+01

Table C-7 : 512 GRID POINTS, DK=2PI/16, M =MU =1

LAMDA	E VAC	M	MU
.10E+00	-.236445112E-02	.100190987E+01	.100095489E+01
.20E+00	-.935755583E-02	.100747281E+01	.100373576E+01
.30E+00	-.207049440E-01	.101624203E+01	.100811866E+01
.40E+00	-.360216742E-01	.102761618E+01	.101380380E+01
.50E+00	-.548811415E-01	.104096865E+01	.102048098E+01
.60E+00	-.768663415E-01	.105573462E+01	.102787278E+01
.70E+00	-.101598002E+00	.107144911E+01	.103575243E+01
.80E+00	-.128744854E+00	.108775089E+01	.104394480E+01
.90E+00	-.158023528E+00	.110436977E+01	.105231932E+01
.10E+01	-.189193732E+00	.112110878E+01	.106078084E+01
.11E+01	-.222051984E+00	.113782709E+01	.106926101E+01
.12E+01	-.256425484E+00	.115442565E+01	.107771136E+01
.13E+01	-.292166723E+00	.117083608E+01	.108609785E+01
.14E+01	-.329148978E+00	.118701232E+01	.109439695E+01
.15E+01	-.367262645E+00	.120292440E+01	.110259272E+01

APPENDIX D

THREE PARTICLE CUTOFF CALCULATIONS

The program MESN2A listed here is a generic set of routines designed to allow a wide variety of situations to be modeled with the various routines embodied in it. To use them, one must call RENORM to renormalize the input constants, and then set up F1, F2 and F3 with the desired wavepacket shapes. A call to SETUP then sets up the wavepackets as superpositions of physical particle states. PUSH then advances the system in time by an increment DT, NORM normalizes the wavefunction and KNORM and XNORM print the wavefunction out in K-space and coordinate space, respectively.

This program was run on the Cyber-175 at the University of Arizona with 16 grid points and on the SCS-40 at the University of Arizona with 24 and 32 grid points.

```

program mesn2

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
real x1,k1,x2,k2,ph,sig,x,t1,t2
complex f1(16),f2(16),f3(16)
integer r,p,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

t1=second()
call link("unit6=tty//")
call init
dt=.005
mur=1.0
lr=0.1
call renorm

ph=16.0*dx
x1=9.0*dx+5.0
k1=-5.0*dk*(16.0/16.0)
x2=9.0*dx-5.0
k2=5.0*dk*(16.0/16.0)
sig=2.0
do 10 r=1,16
x=float(r)*dx
f1(r)=cexp(cmplx(-(x-x1)/sig)**2,k1*x))
f1(r)=f1(r)+cexp(cmplx(-(x-x1-ph)/sig)**2,k1*x))
f1(r)=f1(r)+cexp(cmplx(-(x-x1+ph)/sig)**2,k1*x))
f2(r)=cexp(cmplx(-(x-x2)/sig)**2,k2*x))
f2(r)=f2(r)+cexp(cmplx(-(x-x2-ph)/sig)**2,k2*x))
10 f2(r)=f2(r)+cexp(cmplx(-(x-x2+ph)/sig)**2,k2*x))
continue

call fft(1,f1)
call fft(1,f2)

call setup(1,1,0,f1,f2,f3)

call knorm(1)

do 30 r=1,12
call ftpsi
call xnorm(1)
call ftpsi
call push(1)
call norm
do 20 p=1,199
call push(0)
20 call norm
30 continue
continue

call push(1)
call knorm(1)
call ftpsi
call xnorm(1)

t1=second()-t1
100 write(6,100) t1
format(16hExecution time: ,e14.7,9h seconds./)

call exit
end

```



```

c* *****
subroutine setvac

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
real u(16),a,e
complex aa,bb
complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
complex h30(16,16,16),h21(16,16,16),h12(16,16,16)
integer r,p,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12

c* first get the vacuum energy e
call vnrg(e,u)

c* now set up the wavefunction - use h for the wavefctn for now
call nulh
h00=cplx(1.0,0.0)
do 30 r=1,16
  if(18-r.gt.16) go to 30
  h20(r,18-r)=h00*u(r)
30 continue

aa=cplx(-l*dk/((e-mu)*4.*sqrt(pi)),0.0)
do 40 r=1,16
  h01(r)=h01(r)+aa*u(r)/w(r)**2
40 continue

aa=cplx(-l/(dk*sqrt(pi)),0.0)
bb=cplx(-l/sqrt(2.*pi),0.0)
do 60 r=1,16
  do 50 p=1,16
    if(27-r-p.le.0.or.27-r-p.gt.16) go to 50
    a=e-w(r)-w(p)-wu(27-r-p)
    h21(r,p,27-r-p)=(aa+bb*(u(r)/w(r)+u(p)/w(p)))/a
50 continue
60 continue

call hnorm

return
end

c* *****
subroutine vnrg(s,u)

c* finds the physical vacuum energy

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
real f,e,lf,le,s,u(16)
integer k,n,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

```

```

c*      calculate omega's for the given values of m and mu
do 10 n=1,16
  s=dk*float(n-9)
  w(n)=sqrt(m**2+s**2)
  wu(n)=sqrt(mu**2+s**2)
10      continue

      e=-0.0001
      call fg(u)
      le=e
      lf=f
      e=-1.0
      call fg(u)

20      s=(f-lf)/(e-le)
      le=e
      lf=f
      e=le-lf/s
      call fg(u)
      if(sqrt((e-le)**2).le.1.0e-11) go to 30
      if(sqrt(f*f).gt.1.0e-40) go to 20

30      s=e

      return
      end

c*      *****

      subroutine fg(u)

      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
      real b(16,17),u(16),a,bb,c,d,e,f,s
      integer spc,r,p

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
      common /b/ f,e

c*      first compute the function u
      bb=l**2*dk/(2.*pi)
      c=l**2/(sqrt(2.)*pi)
      d=l**2*dk/(8.*pi*mu*(e-mu))
      do 20 r=1,16
        b(r,17)=0.0
        s=0.0
        do 10 p=1,16
          b(r,p)=-d/w(p)**2
          if(r+p.le.9.or.r+p.gt.25) go to 10
          a=(e-w(r)-w(p)-wu(r+p-9))*w(p)*wu(r+p-9)
          b(r,17)=b(r,17)+c/a
          b(r,p)=b(r,p)-bb/(a*w(p))
          s=s-bb/(a*w(r))
10          continue
        b(r,r)=b(r,r)+e-2.*w(r)+s
20      continue

      call simeq(b,u)

```

```

c*      now calculate f
      bb=l**2*dk/(8.*pi)
      c=(l*dk)**2/(sqrt(32.)*pi)
      f=0.0
      do 40 r=1,16
      do 30 p=1,16
      if(r+p.le.9.or.r+p.gt.25) go to 30
      a=(e-w(r)-w(p)-wu(r+p-9))*w(r)*w(p)*wu(r+p-9)
      f=f+bb/a+c*u(r)/(w(r)*a)
30      continue
40      continue

      f=f-e

      return
      end

c*      *****

      subroutine simeq(b,u)

c*      solves simultaneous equations

      real b(16,17),u(16)
      integer m,n,l

      do 30 m=1,15
      do 20 n=m,15
      do 10 l=m,17
      b(n+1,l)=b(n+1,l)-(b(n+1,m)/b(m,m))*b(m,l)
10      continue
20      continue
30      continue

      do 60 m=1,15
      m1=17-m
      do 50 n=2,m1
      do 40 l=m1,17
      b(n-1,l)=b(n-1,l)-(b(n-1,m1)/b(m1,m1))*b(m1,l)
40      continue
50      continue
60      continue

      do 70 m=1,16
      u(m)=b(m,17)/b(m,m)
70      continue

      return
      end

```

```

c* *****
subroutine hnorm

real nrm
real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
complex h30(16,16,16),h21(16,16,16),h12(16,16,16)
integer r,p,q,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12

c* first find the present norm
nrm=conjg(h00)*h00

do 30 r=1,16
nrm=nrm+conjg(h10(r))*h10(r)*dk/(2.*w(r))
nrm=nrm+conjg(h01(r))*h01(r)*dk/(2.*wu(r))
do 20 p=1,16
nrm=nrm+conjg(h20(r,p))*h20(r,p)*dk**2/(4.*w(r)*w(p))
nrm=nrm+conjg(h11(r,p))*h11(r,p)*dk**2/(4.*w(r)*wu(p))
nrm=nrm+conjg(h02(r,p))*h02(r,p)*dk**2/(4.*wu(r)*wu(p))
do 10 q=1,16
nrm=nrm+conjg(h30(r,p,q))*h30(r,p,q)*dk**3/(8.*w(r)*w(p)*w(q))
nrm=nrm+conjg(h21(r,p,q))*h21(r,p,q)*dk**3/(8.*w(r)*w(p)*wu(q))
nrm=nrm+conjg(h12(r,p,q))*h12(r,p,q)*dk**3/(8.*wu(r)*wu(p)*wu(q))
10 continue
20 continue
30 continue

nrm=1.0/sqrt(nrm)

c* then normalize wavefunction
h00=h00*nrm
do 60 r=1,16
h10(r)=h10(r)*nrm
h01(r)=h01(r)*nrm
do 50 p=1,16
h20(r,p)=h20(r,p)*nrm
h11(r,p)=h11(r,p)*nrm
h02(r,p)=h02(r,p)*nrm
do 40 q=1,16
h30(r,p,q)=h30(r,p,q)*nrm
h21(r,p,q)=h21(r,p,q)*nrm
h12(r,p,q)=h12(r,p,q)*nrm
40 continue
50 continue
60 continue

return
end

```

```

c* *****

subroutine nulh

complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
complex h30(16,16,16),h21(16,16,16),h12(16,16,16)
integer r,p,q

common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12

h00=cplx(0.0,0.0)
do 30 r=1,16
h10(r)=cplx(0.0,0.0)
h01(r)=cplx(0.0,0.0)
do 20 p=1,16
h20(r,p)=cplx(0.0,0.0)
h11(r,p)=cplx(0.0,0.0)
h02(r,p)=cplx(0.0,0.0)
do 10 q=1,16
h30(r,p,q)=cplx(0.0,0.0)
h21(r,p,q)=cplx(0.0,0.0)
h12(r,p,q)=cplx(0.0,0.0)
10 continue
20 continue
30 continue

return
end

c* *****

subroutine push(z)

c* advance the system by dt in time. if z=0 that is all.
c* if z=1 also calculate the energy and report it

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
real r2,r4,r8,r12,r16,r32
complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
complex h30(16,16,16),h21(16,16,16),h12(16,16,16),nrg
integer z,r,p,q,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12
common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12

c* initialize h to zero
call nulh

c* first calculate interaction terms in hamiltonian
do 120 r=1,16
do 110 p=1,16
if(r-p.le.-9.or.r-p.gt.7) go to 40
h10(r)=h10(r)+p11(p,r-p+9)/(w(p)*wu(r-p+9))
h01(r)=h01(r)+p20(p,r-p+9)/(w(p)*w(r-p+9))
40 if(r+p.le.9.or.r+p.gt.25) go to 50
h20(r,p)=h20(r,p)+p01(r+p-9)*sqrt(2.)/(dk*wu(r+p-9))
h11(r,p)=h11(r,p)+sqrt(2.)*p10(r+p-9)/(dk*w(r+p-9))
50 if(r+p.lt.11.or.r+p.ge.27) go to 60
h00=h00+p21(r,p,27-r-p)/(w(r)*w(p)*wu(27-r-p))
60 do 100 q=1,16
if(r+p+q.ne.27) go to 70

```

```

70  h21(r,p,q)=h21(r,p,q)+p00*sqrt(2.)/dk
    if(r-q.le.-9.or.r-q.gt.7) go to 80
    h02(r,p)=h02(r,p)+p21(q,r-q+9,p)/(w(q)*w(r-q+9))
    h11(r,p)=h11(r,p)+p12(r-q+9,p,q)/(wu(q)*w(r-q+9))
    h02(p,r)=h02(p,r)+p21(q,r-q+9,p)/(w(r-q+9)*w(q))
    h11(p,r)=h11(p,r)+sqrt(1.5)*p30(p,q,r-q+9)/(w(q)*w(r-q+9))
80  if(p+q.le.9.or.p+q.gt.25) go to 90
    h30(r,p,q)=h30(r,p,q)+p11(r,p+q-9)/wu(p+q-9)
    h21(r,p,q)=h21(r,p,q)+p20(r,p+q-9)/w(p+q-9)
    h30(q,p,r)=h30(q,p,r)+p11(r,p+q-9)/wu(p+q-9)
    h21(q,p,r)=h21(q,p,r)+p02(r,p+q-9)/wu(p+q-9)
    h12(q,p,r)=h12(q,p,r)+p11(p+q-9,r)/w(p+q-9)
    h30(p,r,q)=h30(p,r,q)+p11(r,p+q-9)/wu(p+q-9)
    h12(p,r,q)=h12(p,r,q)+p11(p+q-9,r)/w(p+q-9)
    h21(p,r,q)=h21(p,r,q)+p20(r,p+q-9)/w(p+q-9)
90  if(r-q.le.-9.or.r-q.gt.7) go to 100
    h20(r,p)=h20(r,p)+p21(p,r-q+9,q)/(wu(q)*w(r-q+9))
    h20(p,r)=h20(p,r)+p21(p,r-q+9,q)/(wu(q)*w(r-q+9))
100 continue
110 continue
120 continue

c*  constants for use in the following
    r2=1./sqrt(2.*pi)
    r4=1./sqrt(4.*pi)
    r8=1./sqrt(8.*pi)
    r12=1./sqrt(12.*pi)
    r16=1./sqrt(16.*pi)
    r32=1./sqrt(32.*pi)

c*  and now calculate the kinetic terms
c*  and put proper constants in the interaction terms
    h00=-l*h00*dk**2/(8.*sqrt(pi))
    do 150 r=1,16
        h10(r)=w(r)*p10(r)-l*h10(r)*dk*r8
        h01(r)=wu(r)*p01(r)-l*h01(r)*dk*r16
    do 140 p=1,16
        h20(r,p)=(w(r)+w(p))*p20(r,p)-l*h20(r,p)*dk*r8
        h11(r,p)=(w(r)+wu(p))*p11(r,p)-l*h11(r,p)*dk*r4
        h02(r,p)=(wu(r)+wu(p))*p02(r,p)-l*h02(r,p)*dk*r32
    do 130 q=1,16
        h30(r,p,q)=(w(r)+w(p)+w(q))*p30(r,p,q)-l*h30(r,p,q)*r12
        h12(r,p,q)=(w(r)+wu(p)+wu(q))*p12(r,p,q)-l*h12(r,p,q)*r4
        h21(r,p,q)=(w(r)+w(p)+wu(q))*p21(r,p,q)-l*h21(r,p,q)*r2
130 continue
140 continue
150 continue

    if(z.eq.0) go to 190

c*  calculate the energy of the system
    nrg=conjg(p00)*h00
    do 180 r=1,16
        nrg=nrg+conjg(p10(r))*h10(r)*dk/(2.*w(r))
        nrg=nrg+conjg(p01(r))*h01(r)*dk/(2.*wu(r))
    do 170 p=1,16
        nrg=nrg+conjg(p20(r,p))*h20(r,p)*dk**2/(4.*w(r)*w(p))
        nrg=nrg+conjg(p11(r,p))*h11(r,p)*dk**2/(4.*w(r)*wu(p))
        nrg=nrg+conjg(p02(r,p))*h02(r,p)*dk**2/(4.*wu(r)*wu(p))
    do 160 q=1,16
        nrg=nrg+conjg(p30(r,p,q))*h30(r,p,q)*dk**3/(8.*w(r)*w(p)*w(q))
        nrg=nrg+conjg(p21(r,p,q))*h21(r,p,q)*dk**3/(8.*w(r)*w(p)*wu(q))
        nrg=nrg+conjg(p12(r,p,q))*h12(r,p,q)*dk**3/(8.*w(r)*wu(p)*wu(q))

```

```

160 continue
170 continue
180 continue

write(6,1000) t,nrg

c* now compute new wavefunctions
190 p00=p00-cmplx(0.0,dt)*h00
do 220 r=1,16
p10(r)=p10(r)-cmplx(0.0,dt)*h10(r)
p01(r)=p01(r)-cmplx(0.0,dt)*h01(r)
do 210 p=1,16
p20(r,p)=p20(r,p)-cmplx(0.0,dt)*h20(r,p)
p11(r,p)=p11(r,p)-cmplx(0.0,dt)*h11(r,p)
p02(r,p)=p02(r,p)-cmplx(0.0,dt)*h02(r,p)
do 200 q=1,16
p30(r,p,q)=p30(r,p,q)-cmplx(0.0,dt)*h30(r,p,q)
p21(r,p,q)=p21(r,p,q)-cmplx(0.0,dt)*h21(r,p,q)
p12(r,p,q)=p12(r,p,q)-cmplx(0.0,dt)*h12(r,p,q)
200 continue
210 continue
220 continue

c* update the time
t=t+dt

1000 format(4ht = ,e13.6,5h E = ,e13.6,e13.6/)

return
end

c* *****

subroutine knorm(z)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu(16),t
real k,i00,i10,i01,i20,i11,i02,i30,i21,i12,im(16),imu(16)
complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
integer r,p,q,spc,z

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

call norm

c* first calculate particle probabilities
i00=conjg(p00)*p00
i01=0.0
i10=0.0
i20=0.0
i11=0.0
i02=0.0
i30=0.0
i21=0.0
i12=0.0
i03=0.0
do 30 r=1,16
i01=i01+conjg(p01(r))*p01(r)*dk/(2.*wu(r))
i10=i10+conjg(p10(r))*p10(r)*dk/(2.*w(r))
do 20 p=1,16
i20=i20+conjg(p20(r,p))*p20(r,p)*dk**2/(4.*w(r)*w(p))
i11=i11+conjg(p11(r,p))*p11(r,p)*dk**2/(4.*w(r)*w(p))
i02=i02+conjg(p02(r,p))*p02(r,p)*dk**2/(4.*wu(r)*w(p))

```

```

do 10 q=1,16
  i30=i30+conjg(p30(r,p,q))*p30(r,p,q)*dk**3/(8.*w(r)*w(p)*w(q))
  i21=i21+conjg(p21(r,p,q))*p21(r,p,q)*dk**3/(8.*w(r)*w(p)*wu(q))
  i12=i12+conjg(p12(r,p,q))*p12(r,p,q)*dk**3/(8.*w(r)*wu(p)*wu(q))
10  continue
20  continue
30  continue

c*    and report them
      write(6,1001)
      write(6,2001) i00,i10,i01
      write(6,2002) i20,i11,i02
      write(6,2003) i30,i21,i12
      write(6,1000)

      if(z.eq.0) return

c*    now find the particle densities as a function of coordinate
c*    im(n) is density for m-particle
c*    imu(n) is density for mu-particle
do 40 r=1,16
  im(r)=0.0
  imu(r)=0.0
40  continue

do 70 r=1,16
  im(r)=im(r)+conjg(p10(r))*p10(r)
  imu(r)=imu(r)+conjg(p01(r))*p01(r)
do 60 p=1,16
  im(r)=im(r)+conjg(p20(r,p))*p20(r,p)*dk/(2.*w(p))
  im(r)=im(r)+conjg(p20(p,r))*p20(p,r)*dk/(2.*w(p))
  im(r)=im(r)+conjg(p11(r,p))*p11(r,p)*dk/(2.*wu(p))
  imu(r)=imu(r)+conjg(p02(r,p))*p02(r,p)*dk/(2.*wu(p))
  imu(r)=imu(r)+conjg(p02(p,r))*p02(p,r)*dk/(2.*wu(p))
  imu(r)=imu(r)+conjg(p11(p,r))*p11(p,r)*dk/(2.*w(p))
do 50 q=1,16
  im(r)=im(r)+conjg(p30(r,p,q))*p30(r,p,q)*dk**2/(4.*w(p)*w(q))
  im(r)=im(r)+conjg(p30(p,r,q))*p30(p,r,q)*dk**2/(4.*w(p)*w(q))
  im(r)=im(r)+conjg(p30(p,q,r))*p30(p,q,r)*dk**2/(4.*w(p)*w(q))
  im(r)=im(r)+conjg(p21(r,p,q))*p21(r,p,q)*dk**2/(4.*w(p)*wu(q))
  im(r)=im(r)+conjg(p21(p,r,q))*p21(p,r,q)*dk**2/(4.*w(p)*wu(q))
  im(r)=im(r)+conjg(p12(r,p,q))*p12(r,p,q)*dk**2/(4.*wu(p)*wu(q))
  imu(r)=imu(r)+conjg(p21(p,q,r))*p21(p,q,r)*dk**2/(4.*w(p)*w(q))
  imu(r)=imu(r)+conjg(p12(p,q,r))*p12(p,q,r)*dk**2/(4.*w(p)*wu(q))
  imu(r)=imu(r)+conjg(p12(p,r,q))*p12(p,r,q)*dk**2/(4.*w(p)*wu(q))
50  continue
60  continue
70  continue

c*    and print values out as a function of r
      write(6,1002)
do 80 r=1,16
  k=float(r-9)*dk
  write(6,1000) k,im(r),imu(r)
80  continue
      write(6,1000)

1000 format(e13.6,e13.6,e13.6,e13.6/)
1001 format(23hParticle probabilities:/)
1002 format(19hParticle densities:/)
2001 format(5hP00= ,e13.6,6h P10 =,e13.6,6h P01 =,e13.6/)
2002 format(5hP20= ,e13.6,6h P11 =,e13.6,6h P02 =,e13.6/)
2003 format(5hP30= ,e13.6,6h P21 =,e13.6,6h P12 =,e13.6/)
end

```



```

c* *****

subroutine init

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
complex wfft(16)
integer r,p,q,spc

common /ftms/ wfft
common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c* initialize all standard variables

spc=2
pi=acos(-1.0)
dk=2.0*pi/16.0
dx=2.0*pi/(16.0*dk)
mr=1.0
t=0.0

do 10 r=1,16
10 wfft(r)=cexp(cmplx(0.0,2.0*pi*float(r)/16.0))
continue

return
end

c* *****

subroutine ftpsi

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
integer spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c* fourier transform the wavefunction psi to the other space
c* spc=1 means it is represented in x space
c* spc=2 means it is represented in k space

c* if in x space, go to k space
if(spc.eq.1) call ftxtk

c* if in k space, go to x space
if(spc.eq.2) call ftktx

c* and change spc to reflect the new representation
spc=(2-spc)+1

return
end

c* *****

subroutine ftxtk

real fac,v(16),vu(16)
real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
complex u(16),vv(16),ww(16)
integer r,p,q,spc

```

```

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

c*   in x space, go to k space
      fac=sqrt(dx/dk)

      do 5 r=1,16
        v(r)=sqrt(2.*w(r))
        vu(r)=sqrt(2.*wu(r))
5      continue

c*   p00 doesn't need to be transformed
c*   first 1-d functions
      call fft(spc,p10)
      call fft(spc,p01)

      do 10 r=1,16
        p10(r)=p10(r)*fac*v(r)
        p01(r)=p01(r)*fac*vu(r)
10     continue

c*   then 2-d functions
      do 40 r=1,16
        do 20 p=1,16
          u(p)=p20(r,p)
          vv(p)=p11(r,p)
          ww(p)=p02(r,p)
20      continue
          call fft(spc,u)
          call fft(spc,vv)
          call fft(spc,ww)
          do 30 p=1,16
            p20(r,p)=u(p)*fac*v(p)
            p11(r,p)=vv(p)*fac*vu(p)
            p02(r,p)=ww(p)*fac*vu(p)
30      continue
40      continue

      do 70 r=1,16
        do 50 p=1,16
          u(p)=p20(p,r)
          vv(p)=p11(p,r)
          ww(p)=p02(p,r)
50      continue
          call fft(spc,u)
          call fft(spc,vv)
          call fft(spc,ww)
          do 60 p=1,16
            p20(p,r)=u(p)*fac*v(p)
            p11(p,r)=vv(p)*fac*vu(p)
            p02(p,r)=ww(p)*fac*vu(p)
60      continue
70      continue

c*   then the 3-d functions
      do 110 r=1,16
        do 100 p=1,16
          do 80 q=1,16
            u(q)=p30(r,p,q)
            vv(q)=p21(r,p,q)
            ww(q)=p12(r,p,q)
80      continue
          call fft(spc,u)

```

```

      call fft(spc,vv)
      call fft(spc,ww)
      do 90 q=1,16
      p30(r,p,q)=u(q)*fac*v(q)
      p21(r,p,q)=vv(q)*fac*vu(q)
      p12(r,p,q)=ww(q)*fac*vu(q)
90    continue
100   continue
110   continue

      do 150 r=1,16
      do 140 p=1,16
      do 120 q=1,16
      u(q)=p30(r,q,p)
      vv(q)=p21(r,q,p)
      ww(q)=p12(r,q,p)
120  continue
      call fft(spc,u)
      call fft(spc,vv)
      call fft(spc,ww)
      do 130 q=1,16
      p30(r,q,p)=u(q)*fac*v(q)
      p21(r,q,p)=vv(q)*fac*v(q)
      p12(r,q,p)=ww(q)*fac*vu(q)
130  continue
140  continue
150  continue

      do 190 r=1,16
      do 180 p=1,16
      do 160 q=1,16
      u(q)=p30(q,r,p)
      vv(q)=p21(q,r,p)
      ww(q)=p12(q,r,p)
160  continue
      call fft(spc,u)
      call fft(spc,vv)
      call fft(spc,ww)
      do 170 q=1,16
      p30(q,r,p)=u(q)*fac*v(q)
      p21(q,r,p)=vv(q)*fac*v(q)
      p12(q,r,p)=ww(q)*fac*v(q)
170  continue
180  continue
190  continue

      return
      end

c* *****

      subroutine ftktx

      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
      real fac,v(16),vu(16)
      complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
      complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
      complex u(16),vv(16),ww(16)
      integer r,p,q,spc

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
      common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

```

```

c*      in k space, go to x space
        fac=sqrt(dk/dx)

        do 5 r=1,16
          v(r)=1./sqrt(2.*w(r))
          vu(r)=1./sqrt(2.*wu(r))
5         continue

c*      p00 doesn't need to be transformed
c*      first 1-d functions
        do 10 r=1,16
          p10(r)=p10(r)*fac*v(r)
          p01(r)=p01(r)*fac*vu(r)
10        continue

        call fft(spc,p10)
        call fft(spc,p01)

c*      then 2-d functions
        do 40 r=1,16
          do 20 p=1,16
            u(p)=p20(r,p)*fac*v(p)
            vv(p)=p11(r,p)*fac*vu(p)
            ww(p)=p02(r,p)*fac*vu(p)
20          continue
            call fft(spc,u)
            call fft(spc,vv)
            call fft(spc,ww)
            do 30 p=1,16
              p20(r,p)=u(p)
              p11(r,p)=vv(p)
              p02(r,p)=ww(p)
30            continue
40          continue

        do 70 r=1,16
          do 50 p=1,16
            u(p)=p20(p,r)*fac*v(p)
            vv(p)=p11(p,r)*fac*v(p)
            ww(p)=p02(p,r)*fac*vu(p)
50          continue
            call fft(spc,u)
            call fft(spc,vv)
            call fft(spc,ww)
            do 60 p=1,16
              p20(p,r)=u(p)
              p11(p,r)=vv(p)
              p02(p,r)=ww(p)
60          continue
70        continue

c*      then the 3-d functions
        do 110 r=1,16
          do 100 p=1,16
            do 80 q=1,16
              u(q)=p30(r,p,q)*fac*v(q)
              vv(q)=p21(r,p,q)*fac*vu(q)
              ww(q)=p12(r,p,q)*fac*vu(q)
80            continue
            call fft(spc,u)
            call fft(spc,vv)
            call fft(spc,ww)

```

```

      do 90 q=1,16
      p30(r,p,q)=u(q)
      p21(r,p,q)=vv(q)
      p12(r,p,q)=ww(q)
90    continue
100    continue
110    continue

      do 150 r=1,16
      do 140 p=1,16
      do 120 q=1,16
      u(q)=p30(r,q,p)*fac*v(q)
      vv(q)=p21(r,q,p)*fac*v(q)
      ww(q)=p12(r,q,p)*fac*vu(q)
120    continue
      call fft(spc,u)
      call fft(spc,vv)
      call fft(spc,ww)
      do 130 q=1,16
      p30(r,q,p)=u(q)
      p21(r,q,p)=vv(q)
      p12(r,q,p)=ww(q)
130    continue
140    continue
150    continue

      do 190 r=1,16
      do 180 p=1,16
      do 160 q=1,16
      u(q)=p30(q,r,p)*fac*v(q)
      vv(q)=p21(q,r,p)*fac*v(q)
      ww(q)=p12(q,r,p)*fac*v(q)
160    continue
      call fft(spc,u)
      call fft(spc,vv)
      call fft(spc,ww)
      do 170 q=1,16
      p30(q,r,p)=u(q)
      p21(q,r,p)=vv(q)
      p12(q,r,p)=ww(q)
170    continue
180    continue
190    continue

      return
      end

c* *****

      subroutine fft(tc,ftm)

      integer n,tc,nv2,nm1,j,i,k,m,ind,l
      complex ftm(16),wfft(16),temp
      common /ftms/ wfft

c*      initialize wfft to exp(-ikx) in init

      if(tc.eq.1) go to 30
      do 10 n=1,16
      ftm(n)=conjg(ftm(n))/sqrt(16.0)
10    continue

```

c* rearrange data points so k=0 first

```

do 20 n=1,8
temp=ftm(n)
ftm(n)=ftm(n+16/2)
ftm(n+16/2)=temp
20 continue
go to 35

30 do 33 n=1,16
ftm(n)=ftm(n)/sqrt(16.0)
33 continue

```

c* shuffle data points

```

35 nv2=16/2
nm1=16-1
j=1
do 70 i=1,nm1
if(i.ge.j) go to 40
temp=ftm(j)
ftm(j)=ftm(i)
ftm(i)=temp
40 k=nv2
50 if(k.ge.j) go to 60
j=j-k
k=k/2
go to 50
60 j=j+k
70 continue

```

c* fourier transform data

```

k=1
j=16/2
80 m=1
90 ind=j
m=m+k+1
100 if(ind.eq.16/2) go to 110
ftm(m)=ftm(m)*wfft(ind)
ind=ind+j
m=m+1
go to 100
110 if(m.ne.16+1) go to 90
i=1
120 l=0
m=i+k
130 temp=ftm(i)+ftm(m)
ftm(m)=ftm(i)-ftm(m)
ftm(i)=temp
l=l+1
i=i+1
m=m+1
if(l.ne.k) go to 130
if(m.eq.16+1) go to 140
i=m
go to 120
140 if(k.eq.16/2) go to 150
k=k*2
j=j/2
go to 80

```

```

c*      end of transform

150     if(tc.eq.2) go to 170
        do 160 n=1,8
            temp=ftm(n)
            ftm(n)=ftm(n+16/2)
            ftm(n+16/2)=temp
160     continue
        go to 190

170     do 180 n=1,16
            ftm(n)=conjg(ftm(n))
180     continue

190     return
        end

c*      *****

subroutine xnorm(z)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu(16),t,v(16),vu(16)
real k,i00,i10,i01,i20,i11,i02,i30,i21,i12,im(16),imu(16)
complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
integer r,p,q,spc,z

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

c*      first calculate particle probabilities
        i00=conjg(p00)*p00
        i01=0.0
        i10=0.0
        i20=0.0
        i11=0.0
        i02=0.0
        i30=0.0
        i21=0.0
        i12=0.0
        i03=0.0
        do 30 r=1,16
            i01=i01+conjg(p01(r))*p01(r)*dx
            i10=i10+conjg(p10(r))*p10(r)*dx
        do 20 p=1,16
            i20=i20+conjg(p20(r,p))*p20(r,p)*dx**2
            i11=i11+conjg(p11(r,p))*p11(r,p)*dx**2
            i02=i02+conjg(p02(r,p))*p02(r,p)*dx**2
        do 10 q=1,16
            i30=i30+conjg(p30(r,p,q))*p30(r,p,q)*dx**3
            i21=i21+conjg(p21(r,p,q))*p21(r,p,q)*dx**3
            i12=i12+conjg(p12(r,p,q))*p12(r,p,q)*dx**3
10     continue
20     continue
30     continue

c*      and report them
        write(6,1001)
        write(6,2001) i00,i10,i01
        write(6,2002) i20,i11,i02
        write(6,2003) i30,i21,i12
        write(6,1000)

```

```

      if(z.eq.0) return

c*      now find the particle densities as a function of coordinate
c*      im(n) is density for m-particle
c*      imu(n) is density for mu-particle
      do 40 r=1,16
      im(r)=0.0
      imu(r)=0.0
40      continue

      do 70 r=1,16
      im(r)=im(r)+conjg(p10(r))*p10(r)
      imu(r)=imu(r)+conjg(p01(r))*p01(r)
      do 60 p=1,16
      im(r)=im(r)+conjg(p20(r,p))*p20(r,p)*dx
      im(r)=im(r)+conjg(p20(p,r))*p20(p,r)*dx
      im(r)=im(r)+conjg(p11(r,p))*p11(r,p)*dx
      imu(r)=imu(r)+conjg(p02(r,p))*p02(r,p)*dx
      imu(r)=imu(r)+conjg(p02(p,r))*p02(p,r)*dx
      imu(r)=imu(r)+conjg(p11(p,r))*p11(p,r)*dx
      do 50 q=1,16
      im(r)=im(r)+conjg(p30(r,p,q))*p30(r,p,q)*dx**2
      im(r)=im(r)+conjg(p30(p,r,q))*p30(p,r,q)*dx**2
      im(r)=im(r)+conjg(p30(p,q,r))*p30(p,q,r)*dx**2
      im(r)=im(r)+conjg(p21(r,p,q))*p21(r,p,q)*dx**2
      im(r)=im(r)+conjg(p21(p,r,q))*p21(p,r,q)*dx**2
      im(r)=im(r)+conjg(p12(r,p,q))*p12(r,p,q)*dx**2
      imu(r)=imu(r)+conjg(p21(p,q,r))*p21(p,q,r)*dx**2
      imu(r)=imu(r)+conjg(p12(p,q,r))*p12(p,q,r)*dx**2
      imu(r)=imu(r)+conjg(p12(p,r,q))*p12(p,r,q)*dx**2
50      continue
60      continue
70      continue

c*      and print values out as a function of r
      write(6,1002)
      do 80 r=1,16
      k=float(r-9)*dx
      write(6,1000) k,im(r),imu(r)
80      continue
      write(6,1000)

1000  format(e13.6,e13.6,e13.6,e13.6/)
1001  format(23hParticle probabilities:/)
1002  format(19hParticle densities:/)
2001  format(5hp00= ,e13.6,6h P10 =,e13.6,6h P01 =,e13.6/)
2002  format(5hp20= ,e13.6,6h P11 =,e13.6,6h P02 =,e13.6/)
2003  format(5hp30= ,e13.6,6h P21 =,e13.6,6h P12 =,e13.6/)

      end

c*      *****

      subroutine setm(f)

      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
      real a,b,c,d,e,nrm,u(16)
      real t10(16),t11(16,16),t12(16,16,16),t30(16,16,16)
      complex f(16)
      complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
      complex h30(16,16,16),h21(16,16,16),h12(16,16,16)
      integer r,p,q,n,spc

```



```

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12

c*   now set up the wavefunction - use h for the wavefctn for now
      call nulh

c*   calculate wavefunction
      do 200 n=1,16

c*   get proper value of the energy and the u fctn for k-state n
      call mnrq(n,e,u)

c*   set up one particle wavefunction and zero others
      do 50 r=1,16
        t10(r)=0.0
      do 40 p=1,16
        t11(r,p)=0.0
      do 30 q=1,16
        t12(r,p,q)=0.0
        t30(r,p,q)=0.0
30    continue
40    continue
50    continue
      t10(n)=1.0

c*   then set up the two particle part
      do 70 r=1,16
        if(n-r.le.-9.or.n-r.gt.7) go to 70
        t11(n-r+9,r)=t10(n)*u(r)
70    continue

c*   then the three particle parts
      b=-1/sqrt(4.0*pi)
      do 100 r=1,16
        do 90 q=1,16
          if(n-r-q.le.-18.or.n-r-q.gt.-2) go to 80
          c=0.0
          d=0.0
          a=b/(e-w(n-r-q+18)-wu(r)-wu(q))
          if(n-r.gt.-9.and.n-r.le.7) c=u(r)/w(n-r+9)
          if(n-q.gt.-9.and.n-q.le.7) d=u(q)/w(n-q+9)
          t12(n-r-q+18,r,q)=a*(c+d)
80    if(n-r.le.-9.or.n-r.gt.7) go to 90
          if(r-q.le.-9.or.r-q.gt.7) go to 90
          a=b/((e-w(n-r+9)-w(q)-w(r-q+9))*sqrt(3.0))
          t30(n-r+9,r-q+9,q)=t30(n-r+9,r-q+9,q)+a*u(r)/wu(r)
          c=0.0
          d=0.0
          if(n-q.gt.-9.and.n-q.le.7) c=u(n-q+9)/wu(n-q+9)
          if(n-r+q.gt.0.and.n-r+q.le.16) d=u(n-r+q)/wu(n-r+q)
          t30(n-r+9,r-q+9,q)=t30(n-r+9,r-q+9,q)+a*(c+d)
90    continue
100   continue

c*   now normalize this state
      nrm=0.0
      do 130 r=1,16
        nrm=nrm+t10(r)**2*dk/(2.*w(r))
      do 120 p=1,16
        nrm=nrm+t11(r,p)**2*dk**2/(4.*w(r)*wu(p))
      do 110 q=1,16
        nrm=nrm+t12(r,p,q)**2*dk**3/(8.*w(r)*wu(p)*wu(q))
        nrm=nrm+t30(r,p,q)**2*dk**3/(8.*w(r)*wu(p)*wu(q))

```

```

110 continue
120 continue
130 continue
    nrm=1.0/sqrt(nrm)
    do 160 r=1,16
        t10(r)=t10(r)*nrm
    do 150 p=1,16
        t11(r,p)=t11(r,p)*nrm
    do 140 q=1,16
        t12(r,p,q)=t12(r,p,q)*nrm
        t30(r,p,q)=t30(r,p,q)*nrm
140 continue
150 continue
160 continue

c* now add the single k-state u to the total wavefunction h
c* with the proper phase given by f(n)
    do 190 r=1,16
        h10(r)=h10(r)+f(n)*t10(r)
    do 180 p=1,16
        h11(r,p)=h11(r,p)+f(n)*t11(r,p)
    do 170 q=1,16
        h12(r,p,q)=h12(r,p,q)+f(n)*t12(r,p,q)
        h30(r,p,q)=h30(r,p,q)+f(n)*t30(r,p,q)
170 continue
180 continue
190 continue

200 continue

c* and normalize the total state
    call hnorm

    return
end

c* *****

subroutine mnrg(k,s,u)

c* finds the physical m-particle energy level

    real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
    real f,e,lf,le,s,u(16)
    integer k,n,r,spc

    common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
    common /b/ f,e

c* set up the constants w,wu
    do 10 r=1,16
        w(r)=sqrt(m**2+(float(r-9)*dk)**2)
        wu(r)=sqrt(mu**2+(float(r-9)*dk)**2)
10 continue

    e=0.01
    call fm(k,u)
    le=e
    lf=f
    e=1.0
    call fm(k,u)

```

```

20      s=(f-lf)/(e-le)
         le=e
         lf=f
         e=le-lf/s
         call fm(k,u)
         if(sqrt((e-le)**2).le.1.0e-11) go to 30
         if(sqrt(f*f).gt.1.0e-40) go to 20

30      s=e

         return
         end

c*      *****

subroutine fm(k,u)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
real f,e,c,d,b(16,17),u(16),a
integer spc,r,q,k

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /b/ f,e

do 20 r=1,16
do 10 q=1,17
b(r,q)=0.0
10      continue
20      continue

c=l**2*dk/(8.0*pi)
do 70 r=1,16
b(r,r)=b(r,r)+e-wu(r)
if(k-r.le.-9.or.k-r.gt.7) go to 25
b(r,17)=-l/(sqrt(2.0*pi))*w(k)
b(r,r)=b(r,r)-w(k-r+9)
25      do 60 q=1,16
if(k-r.le.-9.or.k-r.gt.7) go to 50
if(r-q.le.-9.or.r-q.gt.7) go to 40
d=e-w(k-r+9)-w(q)-w(r-q+9)
b(r,r)=b(r,r)-c/(d*w(r-q+9)*w(q)*wu(r))
if(k-r+q.le.0.or.k-r+q.gt.16) go to 30
d=c/(w(r-q+9)*wu(k-r+q)*w(q))
b(r,k-r+q)=b(r,k-r+q)-d/(e-w(k-r+9)-w(q)-w(r-q+9))
30      if(k-q.le.-9.or.k-q.gt.7) go to 40
d=c/(w(q)*wu(k-q+9)*w(r-q+9))
b(r,k-q+9)=b(r,k-q+9)-d/(e-w(k-r+9)-w(q)-w(r-q+9))
40      if(k-r-q.le.-18.or.k-r-q.gt.-2) go to 60
d=e-w(k-r-q+18)-wu(r)-wu(q)
b(r,r)=b(r,r)-2.0*c/(d*w(k-r+9)*w(k-r-q+18)*wu(q))
50      if(k-r-q.le.-18.or.k-r-q.gt.-2) go to 60
if(k-q.le.-9.or.k-q.gt.7) go to 60
d=w(k-r-q+18)*wu(q)*w(k-q+9)
b(r,q)=b(r,q)-2.0*c/(d*(e-w(k-r-q+18)-wu(r)-wu(q)))
60      continue
70      continue

call simeq(b,u)

```

```

      f=0.0
      do 80 r=1,16
      if(k-r.le.-9.or.k-r.gt.7) go to 80
      f=f+u(r)/(w(r)*wu(k-r+9))
80    continue

      f=-f*l*dk/sqrt(8.0*pi)-e+w(k)

      return
      end

c*  *****

      subroutine norm

      real nrm
      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu(16),t
      complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
      complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
      integer r,p,q,spc

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
      common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

c*  first find the present norm
      nrm=conjg(p00)*p00

      do 30 r=1,16
      nrm=nrm+conjg(p10(r))*p10(r)*dk/(2.*w(r))
      nrm=nrm+conjg(p01(r))*p01(r)*dk/(2.*wu(r))
      do 20 p=1,16
      nrm=nrm+conjg(p20(r,p))*p20(r,p)*dk**2/(4.*w(r)*w(p))
      nrm=nrm+conjg(p11(r,p))*p11(r,p)*dk**2/(4.*w(r)*wu(p))
      nrm=nrm+conjg(p02(r,p))*p02(r,p)*dk**2/(4.*wu(r)*wu(p))
      do 10 q=1,16
      nrm=nrm+conjg(p30(r,p,q))*p30(r,p,q)*dk**3/(8.*w(r)*w(p)*w(q))
      nrm=nrm+conjg(p21(r,p,q))*p21(r,p,q)*dk**3/(8.*w(r)*w(p)*wu(q))
      nrm=nrm+conjg(p12(r,p,q))*p12(r,p,q)*dk**3/(8.*w(r)*wu(p)*wu(q))
10    continue
20    continue
30    continue

      nrm=1.0/sqrt(nrm)

c*  then normalize wavefunction
      p00=p00*nrm
      do 60 r=1,16
      p10(r)=p10(r)*nrm
      p01(r)=p01(r)*nrm
      do 50 p=1,16
      p20(r,p)=p20(r,p)*nrm
      p11(r,p)=p11(r,p)*nrm
      p02(r,p)=p02(r,p)*nrm
      do 40 q=1,16
      p30(r,p,q)=p30(r,p,q)*nrm
      p21(r,p,q)=p21(r,p,q)*nrm
      p12(r,p,q)=p12(r,p,q)*nrm
40    continue
50    continue
60    continue

      return
      end

```

```

c* *****
c*
c*      subroutine setmu(f)
c*
c*      this sets up a single physical mu-particle state with momentum
c*      distribution given by the complex function f.
c*      the state is set up in the common block h since this routine
c*      is generally called by 'prod' which shifts it to the block
c*      p while setting up multi-particle states.
c*
c*      local variables
c*      real a,b,c,e,nrm,u(16)
c*      complex f(16)
c*      integer r,p,q,n,j
c*
c*      common variables
c*      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
c*      real t01(16),t02(16,16),t20(16,16),t21(16,16,16)
c*      complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
c*      complex x h30(16,16,16),h21(16,16,16),h12(16,16,16)
c*      integer spc
c*
c*      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
c*      common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12
c*
c*      set the total wavefunction to zero initially
c*      call nulh
c*
c*      calculate wavefunction by superposition of eigenstates
c*      n=eigenstate parameter throughout, k=dk*(n-9)
c*      do 230 n=1,16
c*
c*      first zero the temporary wavefunction, which will be the eigenstate
c*      do 50 r=1,16
c*          t01(r)=0.0
c*          do 40 p=1,16
c*              t20(r,p)=0.0
c*              t02(r,p)=0.0
c*              do 30 q=1,16
c*                  t21(r,p,q)=0.0
c*              30 continue
c*          40 continue
c*      50 continue
c*
c*      get proper value of the energy and the u fctn for k-state n
c*      call munrg(n,e,u)
c*
c*      set up the one particle part
c*      t01(n)=1.0
c*
c*      then the first two particle part
c*      do 70 r=1,16
c*          if(n-r.le.-9.or.n-r.gt.7) go to 70
c*          t20(r,n-r+9)=u(r)
c*      70 continue
c*
c*      then the three particle part
c*      a=-1/sqrt(2.0*pi)
c*      do 90 r=1,16
c*          do 80 q=1,16
c*              j=n-r-q+18
c*              if(j.le.0.or.j.gt.16) go to 80
c*              b=0.0

```

```

c=0.0
if(n-q.gt.-9.and.n-q.le.7) b=u(q)/w(n-q+9)
if(n-r.gt.-9.and.n-r.le.7) c=u(r)/w(n-r+9)
t21(r,q,j)=a*(b+c)/(e-w(r)-w(q)-wu(j))
80  continue
90  continue

c*  and the final 2 particle part
a=-l*dk/sqrt(32.0*pi)
do 120 r=1,16
if(n-r.le.-9.or.n-r.gt.7) go to 120
do 110 q=1,16
j=n-r-q+18
b=0.0
c=0.0
if(r-q.le.-9.and.r-q.gt.7) go to 105
b=t21(q,r-q+9,n-r+9)/(w(q)*w(r-q+9))
105 if(j.gt.0.and.j.le.16) c=t21(q,j,r)/(w(q)*w(j))
t02(r,n-r+9)=t02(r,n-r+9)+a*(b+c)/(e-wu(n-r+9)-wu(r))
110 continue
120 continue

c*  now normalize this state
nrm=0.0
do 160 r=1,16
nrm=nrm+t01(r)**2*dk/(2.*wu(r))
do 150 p=1,16
nrm=nrm+t20(r,p)**2*dk**2/(4.*w(r)*w(p))
nrm=nrm+t02(r,p)**2*dk**2/(4.*wu(r)*wu(p))
do 140 q=1,16
nrm=nrm+t21(r,p,q)**2*dk**3/(8.*w(r)*w(p)*wu(q))
140 continue
150 continue
160 continue
nrm=1.0/sqrt(nrm)
do 190 r=1,16
t01(r)=t01(r)*nrm
do 180 p=1,16
t20(r,p)=t20(r,p)*nrm
t02(r,p)=t02(r,p)*nrm
do 170 q=1,16
t21(r,p,q)=t21(r,p,q)*nrm
170 continue
180 continue
190 continue

c*  now add the single k-state u to the total wavefunction h
c*  with the proper phase and amplitude given by f(n)
do 220 r=1,16
h01(r)=h01(r)+f(n)*t01(r)
do 210 p=1,16
h20(r,p)=h20(r,p)+f(n)*t20(r,p)
h02(r,p)=h02(r,p)+f(n)*t02(r,p)
do 200 q=1,16
h21(r,p,q)=h21(r,p,q)+f(n)*t21(r,p,q)
200 continue
210 continue
220 continue

c*  and go do next momentum eigenstate
230 continue

```

```

c*      finally, normalize the total state
      call hnorm

      return
      end

c*      *****

      subroutine munrg(k,s,u)

c*      finds the physical mu-particle energy level

      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
      real f,e,lf,le,s,u(16)
      integer k,n,r,spc

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
      common /b/ f,e

c*      set up the constants w,wu
      do 10 r=1,16
      w(r)=sqrt(m**2+(float(r-9)*dk)**2)
      wu(r)=sqrt(mu**2+(float(r-9)*dk)**2)
10      continue

      e=0.01
      call fmu(k,u)
      le=e
      lf=f
      e=1.0
      call fmu(k,u)

20      s=(f-lf)/(e-le)
      le=e
      lf=f
      e=le-lf/s
      call fmu(k,u)
      if(sqrt((e-le)**2).le.1.0e-11) go to 30
      if(sqrt(f*f).gt.1.0e-40) go to 20

30      s=e

      return
      end

c*      *****

      subroutine fmu(k,u)

      real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
      real u(16),a,b(16,17),c,d,e,f
      integer spc,r,k,j,q

      common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
      common /b/ f,e

      do 20 r=1,16
      do 10 q=1,17
      b(r,q)=0.0
10      continue
20      continue

```

```

c=-l/sqrt(4.0*pi)
d=-l**2*dk/(4.0*pi)
do 70 r=1,16
  b(r,r)=b(r,r)+e-w(r)
  if(k-r.le.-9.or.k-r.gt.7) go to 30
  b(r,r)=b(r,r)-w(k-r+9)
  b(r,17)=c/wu(k)
30  do 60 q=1,16
    j=k-r-q+18
    if(k-r.le.-9.or.k-r.gt.7) go to 50
    if(j.le.0.or.j.gt.16) go to 40
    b(r,r)=b(r,r)+d/((e-w(r)-w(j)-wu(q))*w(j)*w(k-r+9)*wu(q))
40  if(r-q.le.-9.or.r-q.gt.7) go to 50
    a=d/((e-wu(q)-w(k-r+9)-w(r-q+9))*wu(q)*w(r-q+9))
    b(r,k-r+9)=b(r,k-r+9)+a/w(r)
    if(k+q-r.le.0.or.k+q-r.gt.16) go to 50
    b(r,r-q+9)=b(r,r-q+9)+a/w(k+q-r)
50  if(q+r.le.9.or.q+r.gt.25) go to 60
    if(j.le.0.or.j.gt.16) go to 60
    a=(e-w(r)-w(j)-wu(q))*w(r+q-9)*w(j)*wu(q)
    b(r,j)=b(r,j)+d/a
60  continue
70  continue

call simeq(b,u)

f=0.0
c=-l*dk/sqrt(16.*pi)
do 80 r=1,16
  if(k-r.le.-9.or.k-r.gt.7) go to 80
  f=f+c*u(r)/(w(r)*w(k-r+9))
80  continue

f=f-e+wu(k)

return
end

c* *****
c*
c* subroutine renorm
c*
c* this routine finds the solution of a set of eqns of the form
c* ar(ao)=r, where ao and b are vectors.

real u(16),ao(3),an(3),ar(3),arn(3),d(3,3),r(3),sq2
real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
integer n,p,count,spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c* count is a loop index to see how many times the loop has
c* been used. initialize to zero
count=0

c* assign the physical values we want to end up with
r(1)=mr
r(2)=mur
r(3)=lr

```



```

c*      guess at what the unrenormalized values should be
c*      here we just guess that they are the same as the physical
do 10 n=1,3
  ao(n)=r(n)
10      continue

c*      get renormalized values for initial guess
call nrgs(ao(1),ao(2),ao(3),ar)

c*      loop begins here
c*      shift ao by a small amount in each direction to calculate derivatives s
30      do 50 n=1,3
  an(n)=ao(n)+(ao(n)/ar(n))*(r(n)-ar(n))
50      continue

      do 55 n=1,3
  do 52 p=1,3
    d(n,p)=0.
52      continue
55      continue
  call nrgs(an(1),ao(2),ao(3),arn)
  do 60 n=1,3
    if(arn(n).ne.ar(n)) d(n,1)=(arn(n)-ar(n))/(an(1)-ao(1))
60      continue
    if(arn(1).eq.ar(1).and.an(1).eq.ao(1)) d(1,1)=1.

  call nrgs(ao(1),an(2),ao(3),arn)
  do 70 n=1,3
    if(arn(n).ne.ar(n)) d(n,2)=(arn(n)-ar(n))/(an(2)-ao(2))
70      continue
    if(arn(2).eq.ar(2).and.an(2).eq.ao(2)) d(2,2)=1.

  call nrgs(ao(1),ao(2),an(3),arn)
  do 80 n=1,3
    if(arn(n).ne.ar(n)) d(n,3)=(arn(n)-ar(n))/(an(3)-ao(3))
80      continue
    if(arn(3).eq.ar(3).and.an(3).eq.ao(3)) d(3,3)=1.

c*      invert the tensor d(i,j)
call matinv(d)

c*      use d inv to make a new guess at ao
do 100 p=1,3
  do 90 n=1,3
    ao(p)=ao(p)+d(n,p)*(r(n)-ar(n))
90      continue
100     continue

c*      calculate a new value of a
call nrgs(ao(1),ao(2),ao(3),ar)

c*      see if a is small enough:
c*          if so, we're done
c*          if not, loop again, unless stuck
sq2=0.0
do 120 n=1,3
  sq2=sq2+(ar(n)-r(n))**2
120  continue
count=count+1
if(count.eq.10) go to 130
if(sq2.lt.1.0e-15) go to 130
go to 30

```

```

c*      if done, set bare values for output, read actual
c*      renormalized values out again
130     call nrgs(ao(1),ao(2),ao(3),ar)
        mr=ar(1)
        mur=ar(2)
        lr=ar(3)
        m=ao(1)
        mu=ao(2)
        l=ao(3)
        call nrgs(m,mu,l,ar)

        write(6,1001) m,mu,l
        write(6,1002) mr,mur,lr
        call vnrg(sq2,u)
        write(6,1003) sq2

1001    format(6h M = ,e14.7,7h Mu = ,e14.7,6h L = ,e14.7/)
1002    format(6h Mr = ,e14.7,7h Mur = ,e14.7,6h Lr = ,e14.7/)
1003    format(16hVacuum energy = ,e14.7/)

        return
        end

c*      *****

subroutine matinv(s)

real s(3,3),si(3,3),det
integer i,j

c*      first calculate determinant
det=s(1,1)*(s(2,2)*s(3,3)-s(2,3)*s(3,2))
det=det-s(2,1)*(s(1,2)*s(3,3)-s(1,3)*s(3,2))
det=det+s(3,1)*(s(1,2)*s(2,3)-s(1,3)*s(2,2))

c*      if det=0 matrix is singular - can't invert
if(det.le.1e-20) write(6,1000)
1000    format(15hSINGULAR MATRIX/)

c*      then the inverse matrix elements
si(1,1)=(s(2,2)*s(3,3)-s(2,3)*s(3,2))/det
si(1,2)=-(s(2,1)*s(3,3)-s(2,3)*s(3,1))/det
si(1,3)=(s(2,1)*s(3,2)-s(2,2)*s(3,1))/det
si(2,1)=-(s(1,2)*s(3,3)-s(1,3)*s(3,2))/det
si(2,2)=(s(1,1)*s(3,3)-s(1,3)*s(3,2))/det
si(2,3)=-(s(1,1)*s(3,2)-s(1,2)*s(3,1))/det
si(3,1)=(s(1,2)*s(2,3)-s(1,3)*s(2,2))/det
si(3,2)=-(s(1,1)*s(2,3)-s(1,3)*s(2,1))/det
si(3,3)=(s(1,1)*s(2,2)-s(1,2)*s(2,1))/det

c*      and then substitute si into s
do 20 i=1,3
do 10 j=1,3
s(i,j)=si(i,j)
10    continue
20    continue

        return
        end

```

```

c* *****

subroutine nrgs(me,mue,le,a)

real me,mue,le,a(3),u(16),e
real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
integer spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

m=me
mu=mue
l=le

call vnrg(e,u)
call mnrg(9,a(1),u)
a(1)=a(1)-e
call munrg(9,a(2),u)
a(2)=a(2)-e
call lamda(a(3))

return
end

c* *****

subroutine setup(n1,n2,n3,f1,f2,f3)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
complex f1(16),f2(16),f3(16)
complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
complex h30(16,16,16),h21(16,16,16),h12(16,16,16)
complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
integer r,p,q,spc,n1,n2,n3

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc
common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12
common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

c* set up a total wavefunction consisting of
c* particles/vacuum states. n=0-->vac n=1-->m n=2-->mu

c* get the first state
if(n1+n2+n3.eq.0) call setvac
if(n1+n2+n3.eq.0) go to 5
if(n1.eq.0) go to 35
if(n1.eq.1) call setm(f1)
if(n1.eq.2) call setmu(f1)

c* move it from h to p
5 p00=h00
do 30 r=1,16
p10(r)=h10(r)
p01(r)=h01(r)
do 20 p=1,16
p20(r,p)=h20(r,p)
p11(r,p)=h11(r,p)
p02(r,p)=h02(r,p)
do 10 q=1,16
p30(r,p,q)=h30(r,p,q)
p21(r,p,q)=h21(r,p,q)
p12(r,p,q)=h12(r,p,q)

```

```

10      continue
20      continue
30      continue

c*      now get second state
35      if(n2.eq.0) return
          if(n2.eq.1) call setm(f2)
          if(n2.eq.2) call setmu(f2)

c*      and take the tensor product of h and p and put it in p
          call prod

c*      now get the third state
40      if(n3.eq.0) return
          if(n3.eq.1) call setm(f3)
          if(n3.eq.2) call setmu(f3)

c*      and take the tensor product again
          call prod
          call norm

          return
          end

c*      *****

          subroutine prod

          complex h00,h01(16),h10(16),h20(16,16),h11(16,16),h02(16,16)
          complex h30(16,16,16),h21(16,16,16),h12(16,16,16)
          complex p00,p01(16),p10(16),p20(16,16),p11(16,16),p02(16,16)
          complex p30(16,16,16),p21(16,16,16),p12(16,16,16)
          complex tm
          integer r,p,q

          common /h/ h00,h01,h10,h20,h11,h02,h30,h21,h12
          common /wave/ p00,p01,p10,p20,p11,p02,p30,p21,p12

c*      compute the product of h and p

          do 30 r=1,16
          do 20 p=1,16
          do 10 q=1,16
              tm=p12(r,p,q)*h00+p00*h12(r,p,q)+h10(r)*p02(p,q)
              tm=tm+p10(r)*h02(p,q)+(p11(r,p)*h01(q)+p11(r,q)*h01(p))/2.0
              p12(r,p,q)=tm+(h11(r,p)*p01(q)+h11(r,q)*p01(p))/2.0
              tm=p21(r,p,q)*h00+p00*h21(r,p,q)+p20(r,p)*h01(q)
              tm=tm+h20(r,p)*p01(q)+(p11(r,q)*h10(p)+p11(p,q)*h10(r))/2.0
              p21(r,p,q)=tm+(h11(r,q)*p10(p)+h11(p,q)*p10(r))/2.0
              tm=p30(r,p,q)*h00+p00*h30(r,p,q)
              tm=tm+(p20(r,p)*h10(q)+p20(r,q)*h10(p)+p20(p,q)*h10(r))/3.0
              tm=tm+(h20(r,p)*p10(q)+h20(r,q)*p10(p)+h20(p,q)*p10(r))/3.0
              p30(r,p,q)=tm
10          continue
20          continue
30          continue

c*      now do 2-d part
          do 50 r=1,16
          do 40 p=1,16
              tm=p20(r,p)*h00+p00*h20(r,p)

```

```

p20(r,p)=tm+(p10(r)*h10(p)+p10(p)*h10(r))/2.0
tm=p11(r,p)*h00+p00*h11(r,p)+p01(p)*h10(r)
p11(r,p)=tm+p10(r)*h01(p)
tm=p02(r,p)*h00+p00*h02(r,p)
p02(r,p)=tm+(p01(r)*h01(p)+p01(p)*h01(r))/2.0
40  continue
50  continue

c*  then the 1-d part
do 60 r=1,16
p10(r)=p10(r)*h00+p00*h10(r)
p01(r)=p01(r)*h00+p00*h10(r)
60  continue

c*  and dont forget the 0-d part
p00=p00*h00

c*  normalize the state
call norm

return
end

c*  *****

subroutine lamda(lt)

real m,mu,l,mr,mur,lr,dk,dx,dt,pi,w(16),wu(16),t
real lt
integer spc

common /a/ m,mu,l,mr,mur,lr,dk,dx,dt,pi,w,wu,t,spc

c*  just a dummy for now
lt=l

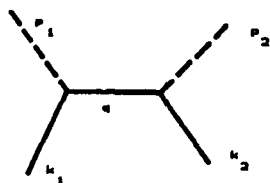
return
end

```

APPENDIX E

PERTURBATIVE SCATTERING CALCULATIONS

To calculate the probability of producing a pair of μ -particles in the scattering of a pair of m -particles, we must consider the diagram



Writing the S -matrix element from the Feynman rules derived in Chapter 3, we have

$$S^{p_1 p_2 \leftarrow k_1 k_2} = \frac{2i\lambda^2}{(2\pi)^2} \int (2\pi)^4 \delta^2(k_1 - q - p_1) \delta^2(k_2 + q - p_2) \frac{i}{q^2 - m^2 + i\epsilon} \frac{d^2 q}{(2\pi)^2}$$

or

$$S^{p_1 p_2 \leftarrow k_1 k_2} = -4i\lambda^2 \delta(k_1 - p_2 + k_2 - p_1) \delta(\omega_{k_1} - \omega_{p_2} + \omega_{k_2} - \omega_{p_1}) ((\omega_{k_2} - \omega_{p_2})^2 - (k_2 - p_2)^2 - m^2 + i\epsilon)^{-1}$$

Since we are doing a scattering experiment with wave packets instead of plane waves, this must be integrated over the packets. In particular,

$$\psi_{o_2}^{\text{final}}(p_1, p_2) = \int S^{p_1 p_2 \leftarrow k_1 k_2} \psi_{i_0}^{\text{init}}(k_1, k_2) \frac{dk_1 dk_2}{4\omega_{k_1} \omega_{k_2}}$$

or,

$$\psi_{o_2}^f(p_1, p_2) = i\lambda^2 \psi_{i_0}^i(p_1, p_2) (((\omega_{p_1} - \omega_{p_2})^2 - (p_1 - p_2)^2 - m^2)^{-1} - m^{-2}) (|p_1| \omega_{p_2} + |p_2| \omega_{p_1})^{-1}$$

The probability of finding a pair of μ -particles in any state is

$$P_{o_2}^{\text{final}} = \int \psi_{o_2}^{f*}(p_1, p_2) \psi_{o_2}^f(p_1, p_2) \frac{dp_1 dp_2}{4\omega_{p_1} \omega_{p_2}}$$

or

$$P_{02}^{final} = \lambda^4 \int |((\omega_{p_2} - \omega_{p_1})^2 - (\mathbf{p}_1 - \mathbf{p}_2)^2 - m^2)^{-1} - m^{-2}|^2 \frac{\psi_{20}^{i*}(\mathbf{p}_1, \mathbf{p}_2) \psi_{20}^i(\mathbf{p}_1, \mathbf{p}_2)}{|\mathbf{p}_1| \omega_{p_2} + |\mathbf{p}_2| \omega_{p_1}} \frac{d\mathbf{p}_1 d\mathbf{p}_2}{4\omega_{p_1} \omega_{p_2}}$$

This is integrated numerically by the Fortran program SCATTER for any given input wavefunction.

```

program scatter
real pi,ko,dk,prob,l,a,c,nrm,w(256),k,ki,kj
integer no,i,j,n1
real dx,x1,x2,ph,n(256)
complex b,f1(256),f2(256),psi(256,256)

call link('unit6=tty//')

pi=acos(-1.0)
ko=5.0*2.0*pi/16.0
no=32
dk=2.0*pi/16.0
dx=2*pi/(float(no)*dk)
prob=0
nrm=0
l=0.1
n1=no/2+1

x1=float(n1)*dx+5.0
x2=float(n1)*dx-5.0
do 10 i=1,no
k=float(i-n1)*dk
w(i)=sqrt(1+k*k)
f1(i)=cexp(cmplx(-(ko+k)**2,-(ko+k)*x1))
f1(i)=f1(i)+cexp(cmplx(-(ko+k)**2,-(ko+k)*(x1+ph)))
f1(i)=f1(i)+cexp(cmplx(-(ko+k)**2,-(ko+k)*(x1-ph)))
f2(i)=cexp(cmplx(-(ko-k)**2,(ko-k)*x2))
f2(i)=f2(i)+cexp(cmplx(-(ko-k)**2,(ko-k)*(x2+ph)))
f2(i)=f2(i)+cexp(cmplx(-(ko-k)**2,(ko-k)*(x2-ph)))
10 continue

do 20 i=1,no
do 15 j=1,no
psi(i,j)=(f1(i)*f2(j)+f1(j)*f2(i))*sqrt(w(i)*w(j))
nrm=nrm+psi(i,j)*conjg(psi(i,j))*dk**2/(4.0*w(i)*w(j))
15 continue
20 continue
nrm=1./sqrt(nrm)
do 27 i=1,no
do 26 j=1,no
psi(i,j)=psi(i,j)*nrm
26 continue
27 continue

do 40 i=1,no
ki=float(i-n1)*dk
do 30 j=1,no
kj=float(j-n1)*dk
if(ki.eq.0.and.kj.eq.0) go to 30
a=(1/((w(j)-w(i))**2-(kj-ki)**2-1)-1)**2
c=psi(i,j)*conjg(psi(i,j))
prob=prob+a*c/((w(i)*abs(kj)+w(j)*abs(ki))**2*w(i)*w(j))
30 continue
40 continue
nrm=0.0
do 60 i=1,no
n(i)=0.0
do 50 j=1,no
n(i)=n(i)+conjg(psi(i,j))*psi(i,j)*dk/(2.0*w(j))
n(i)=n(i)+conjg(psi(j,i))*psi(j,i)*dk/(2.0*w(j))
50 continue
nrm=nrm+n(i)*dk/(2.0*w(i))
60 continue

```



```
      write(6,100) nrm
      do 70 i=1,no
      k=float(i-n1)*dk
      write(6,110) k,n(i)
70    continue

      prob=l**4*prob*dk*dk/4.0

      write(6,100) prob
100    format(14HProbability = ,E16.9//)
110    format(e13.6,e13.6)

      call exit
      end
```

APPENDIX F

MASS PERTURBATION THEORY CALCULATIONS

The three programs listed here solve the eigenvalue problems in the mass perturbation model field theory discussed in Chapter 7. MPERT2 solves the three particle truncation equations, (182a), (183), (185a), and (186). MPERT3 solves the five particle truncation equations (182) and (185). FIGURE2 creates the data for the perturbation expansion depicted in Fig. 11 in the text.

```

program mpert2; {2 equation trivial system solver}
var
  fname      :string[25];
  fout       :text;
  k,p,n,n2,j :integer;
  dk,e,el,s,m,ev,em:real;
  sq2,sq6    :real;
  w,v,f,fl   :array[1..512] of real;

procedure zero;
begin
  for k:=1 to n do f[k]:=0;
end;

procedure next;
begin
  for k:=1 to n do begin fl[k]:=f[k]; f[k]:=0; end;
end;

procedure initial;
begin
  n2:=(n div 2)+1; dk:=2*3.1415926/n;
  sq2:=sqrt(2); sq6:=sqrt(6);
  zero;
  for k:=1 to n do begin
    w[k]:=sqrt(1+dk*dk*(k-n2)*(k-n2));
    v[k]:= (1+m*m/(2*w[k]*w[k]))*w[k];
  end;
end;

procedure getvac;
begin
  e:=0; zero;
  repeat
    el:=e;
    for k:=1 to n do f[k]:=m*m/(sq2*dk*(e-2*v[k]));
    e:=0;
    for k:=1 to n do e:=e+m*m*f[k]*dk/(4*sq2*w[k]*w[k]);
  until abs((e-el)/e)<0.00001;
  ev:=e;
end;

procedure getmass;
begin
  e:=1; zero;
  repeat
    el:=e; next; s:=0;
    for k:=1 to n do f[k]:=m*m/(sq6*dk*(e-v[n2]-2*v[k]));
    e:=0;
    for k:=1 to n do e:=e+f[k]*dk/(w[k]*w[k]);
    e:=v[n2]+m*m*sq6*(2*f[n2]*dk+e)/8;
  until abs((e-el)/e)<0.00001;
  em:=e;
end;

```

```
begin
  write('Number of grid points? '); readln(n);
  write('File to put data in? '); readln(fname);
  assign(fout,fname); rewrite(fout);
  for j:=1 to 30 do begin
    m:=j/10;
    initial;
    getvac;
    getmass;
    writeln('MU = ',m:4:6,' Mass = ',em-ev,' Theoretical mass = ',sqrt(1+m*m));
    writeln(fout,m:12:10,' ',em-ev);
  end;
  close(fout);
end.
```

```

program mpert3; {3 equation trivial system solver}
var
  fname      :string[25];
  fout       :text;
  k,p,n,n2,j :integer;
  dk,e,el,s,m,ev,em:real;
  sq2,sq6    :real;
  w,v,f,fl   :array[1..512] of real;

procedure zerof;
begin
  for k:=1 to n do f[k]:=0;
end;

procedure nextf;
begin
  for k:=1 to n do begin fl[k]:=f[k]; f[k]:=0; end;
end;

procedure initial;
begin
  n2:=(n div 2)+1; dk:=2*3.1415926/n; s:=0;
  sq2:=sqrt(2); sq6:=sqrt(6);
  zerof; nextf;
  for k:=1 to n do begin
    w[k]:=sqrt(1+dk*dk*(k-n2)*(k-n2));
    v[k]:=(1+m*m/(2*w[k]*w[k]))*w[k];
  end;
end;

procedure getvac;
begin
  e:=0; zerof;
  repeat
    el:=e; nextf; s:=0;
    for k:=1 to n do for p:=1 to n do
      f[k]:=f[k]+(fl[p]+fl[k])/(w[p]*w[p]*(e-2*v[k]-2*v[p]));
    for k:=1 to n do f[k]:=f[k]+2*fl[k]/(w[k]*w[k]*(e-4*v[k]));
    for k:=1 to n do f[k]:=(m*m/(sq2*dk)+sqr(m*m)*f[k]/8)/(e-2*v[k]);
    e:=0;
    for k:=1 to n do e:=e+m*f[k]*dk/(4*sq2*w[k]*w[k]);
    until abs((e-el)/e)<0.00001;
  ev:=e;
end;

procedure getmass;
begin
  e:=1; zerof;
  repeat
    el:=e; nextf; s:=0;
    for k:=1 to n do for p:=1 to n do
      f[k]:=f[k]+(fl[k]+fl[p])/(w[p]*w[p]*(e-v[n2]-2*v[k]-2*v[p]));
    for k:=1 to n do
      f[k]:=f[k]+4*fl[k]/(w[k]*w[k]*(e-v[n2]-4*v[k]))+2*(fl[n2]+fl[k])
      /(w[k]*w[k]*(e-3*v[n2]-2*v[k]));
    for k:=1 to n do f[k]:=(m*m/(sq6*dk)+sqr(m*m)*f[k]/8)/(e-v[n2]-2*v[k]);
    e:=0;
    for k:=1 to n do e:=e+f[k]*dk/(w[k]*w[k]);
    e:=v[n2]+m*m*sq6*(2*f[n2]*dk+e)/8;
    until abs((e-el)/e)<0.00001;
  em:=e;
end;

```

```
begin
  write('Number of grid points? '); readln(n);
  write('File to put data in? '); readln(fname);
  assign(fout,fname); rewrite(fout);
  for j:=1 to 30 do begin
    m:=j/10;
    initial;
    getvac;
    getmass;
    writeln('MU = ',m:4:6,' Mass = ',em-ev,' Theoretical mass = ',sqrt(1+m*m));
    writeln(fout,m:12:10,' ',em-ev);
  end;
  close(fout);
end.
```

```
program figure2;          (Perturbation series for figure 2)
var
  m,mm,a,b,c,d,e      :real;
  j                    :integer;
  fout                 :text;
begin
  assign(fout,'FIGURE2.DAT'); rewrite(fout);
  for j:=1 to 60 do begin
    m:=j/20; mm:=m*m;
    a:=1+mm/2;
    b:=a-sqr(mm)/8;
    c:=b+mm*sqr(mm)/16;
    d:=c-5*sqr(sqr(mm))/128;
    e:=d+25*mm*sqr(sqr(mm))/1280;
    writeln(fout,m:4:6,' ',a:6:8,' ',b:6:8,' ',c:6:8,' ',d:6:8,' ',e:6:8,' ',sqrt(1+mm):6:8);
  end;
  close(fout);
end.
```

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